# Mathematics - Probability and Statistics

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Fall 2025

# Syllabus

- **Session 1:** Why Study Statistics?
- **Session 2:** Graphical Representation
- **Session 3:** Descriptive Statistics
- **Session 4:** Boxplots & Normal Distribution
- **Session 5:** Basic Probability & Counting Rules
- **Session 6:** Conditional Probability & Independence
- **Session 7:** Total Probability & Bayes' Theorem
- **Session 8:** Discrete Random Variables
- **Session 9:** Continuous Random Variables
- **Session 10:** Joint Distributions & Correlation
- **Session 11: Exercices/Summary**
- **Session 12:** Central Limit Theorem & Simulation
- **Session 13:** Point Estimation & MLE
- **Session 14:** Confidence Intervals
- **Session 15:** Hypothesis Testing – Basics

# Graphical representation

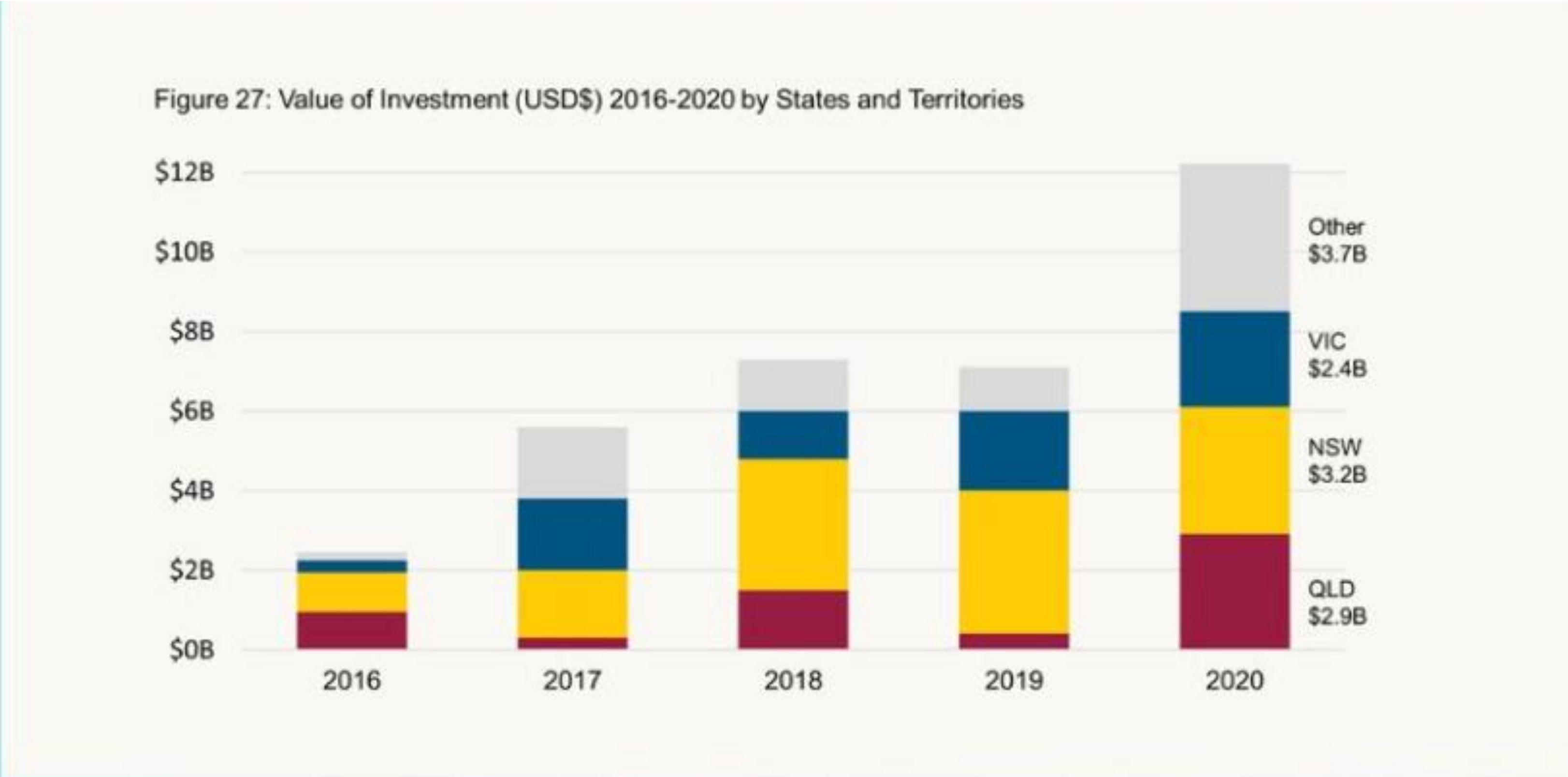
Session 2

# What is data visualisation?

**Data visualisation** is the process of **representing data** and information through visual elements like **charts, graphs, and maps**. It transforms complex datasets into clear, accessible visuals, making it easier **to identify patterns, trends, and outliers**. This helps people **make faster, more informed decisions** across various fields.

# Graphical representation

What is the message?



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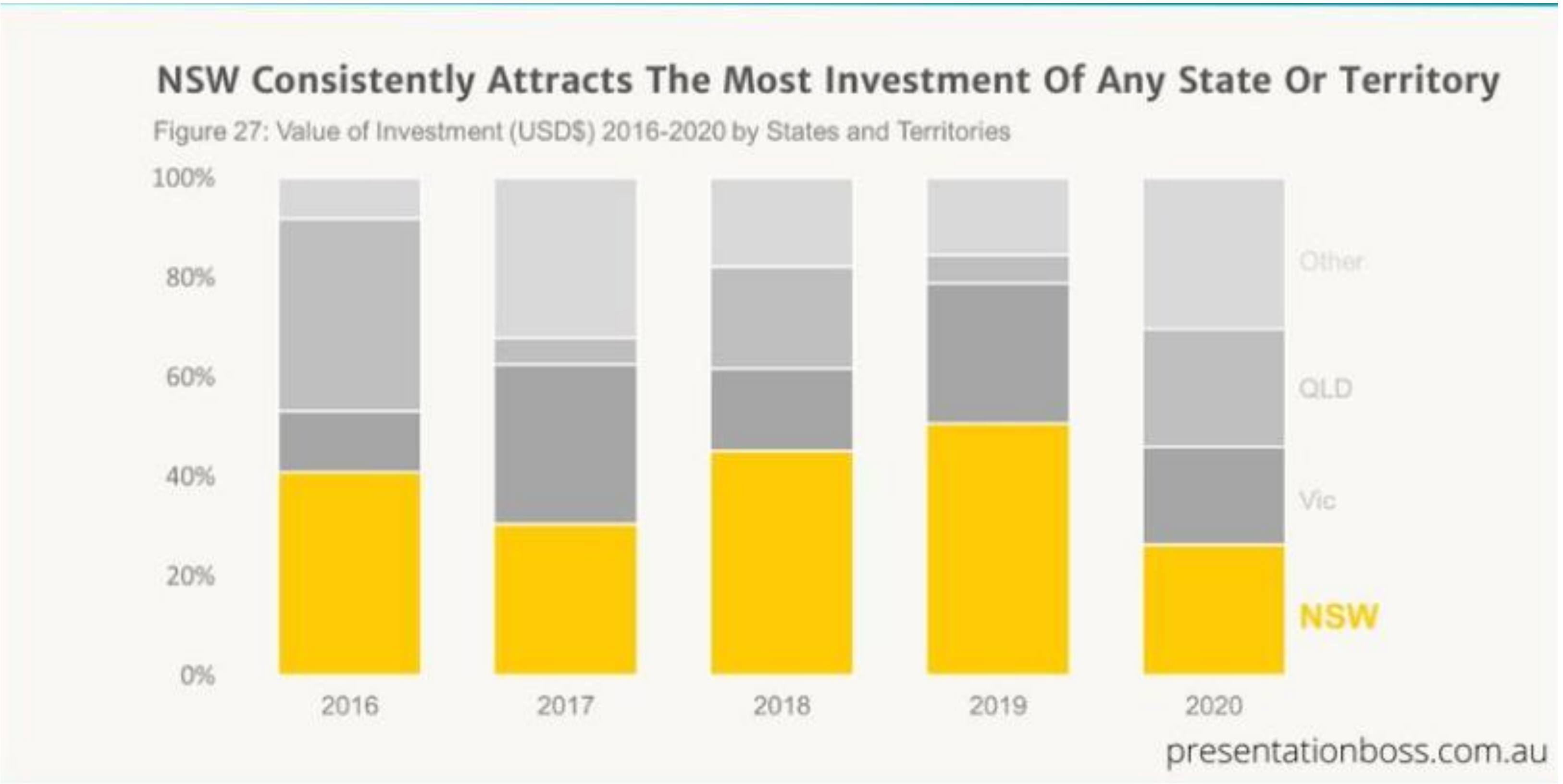


# Data to graph

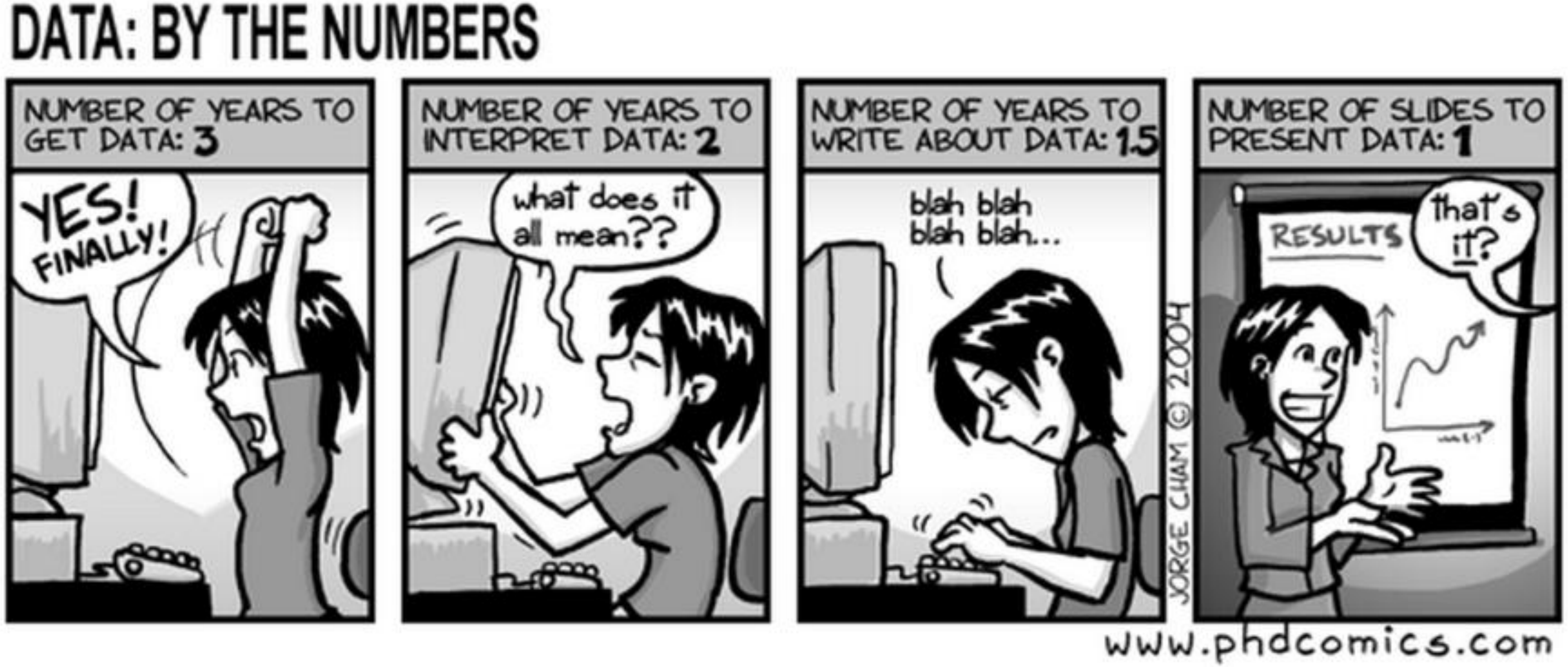



Figure 2.1: Data By the Numbers (by Jorge Cham)

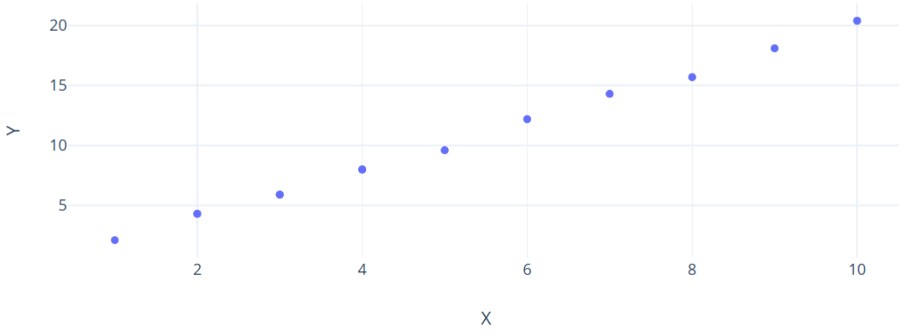
# Why data visualization is needed?

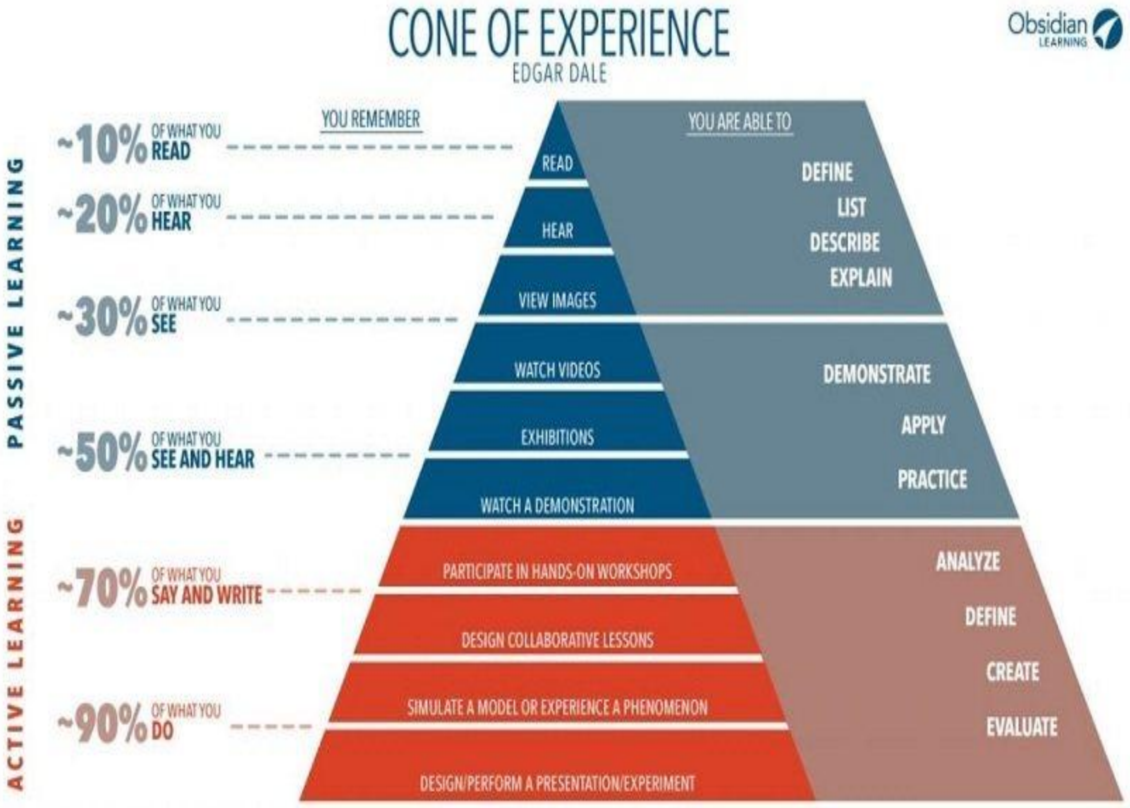
**Our brains are wired for visuals:**

- We process images **60,000× faster** than text

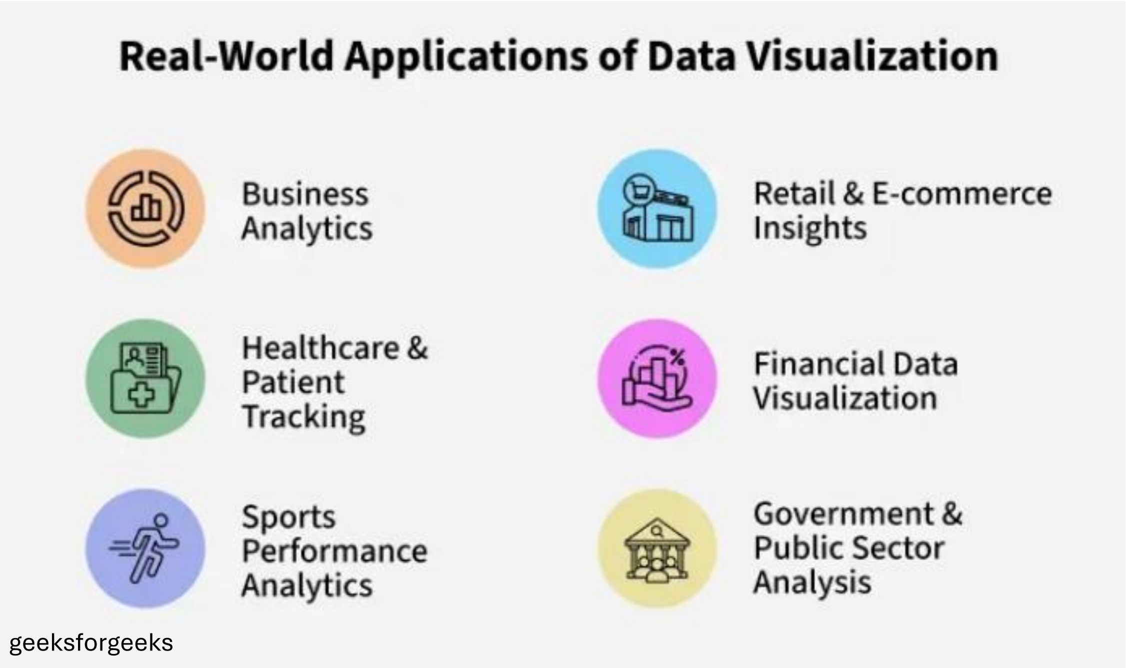
Trends, outliers, clusters, and gaps stand out instantly in a chart
Tables may hide insights that a simple plot can reveal

| | |
|---|---|
| 1 | 2.1 |
| 2 | 4.3 |
| 3 | 5.9 |
| 4 | 8.0 |
| 5 | 9.6 |
| 6 | 12.2 |
| 7 | 14.3 |
| 8 | 15.7 |
| 9 | 18.1 |



# Real world application




| | |
|---|---|
| Public Policy | Track climate change trends and carbon emissions |
| Business | Sales dashboards, customer behavior |
| Health | Epidemic tracking, patient diagnostics, drug effectiveness |
| Finance | Stock market trends, risk visualization, fraud detection |
| Human Rights | Visualizing refugee flows or conflict zones |
| Engineering | Sensor data monitoring, anomaly detection |
| Research | Experimental results, network analysis |
| Journalism | Data stories (e.g., elections, education access) |

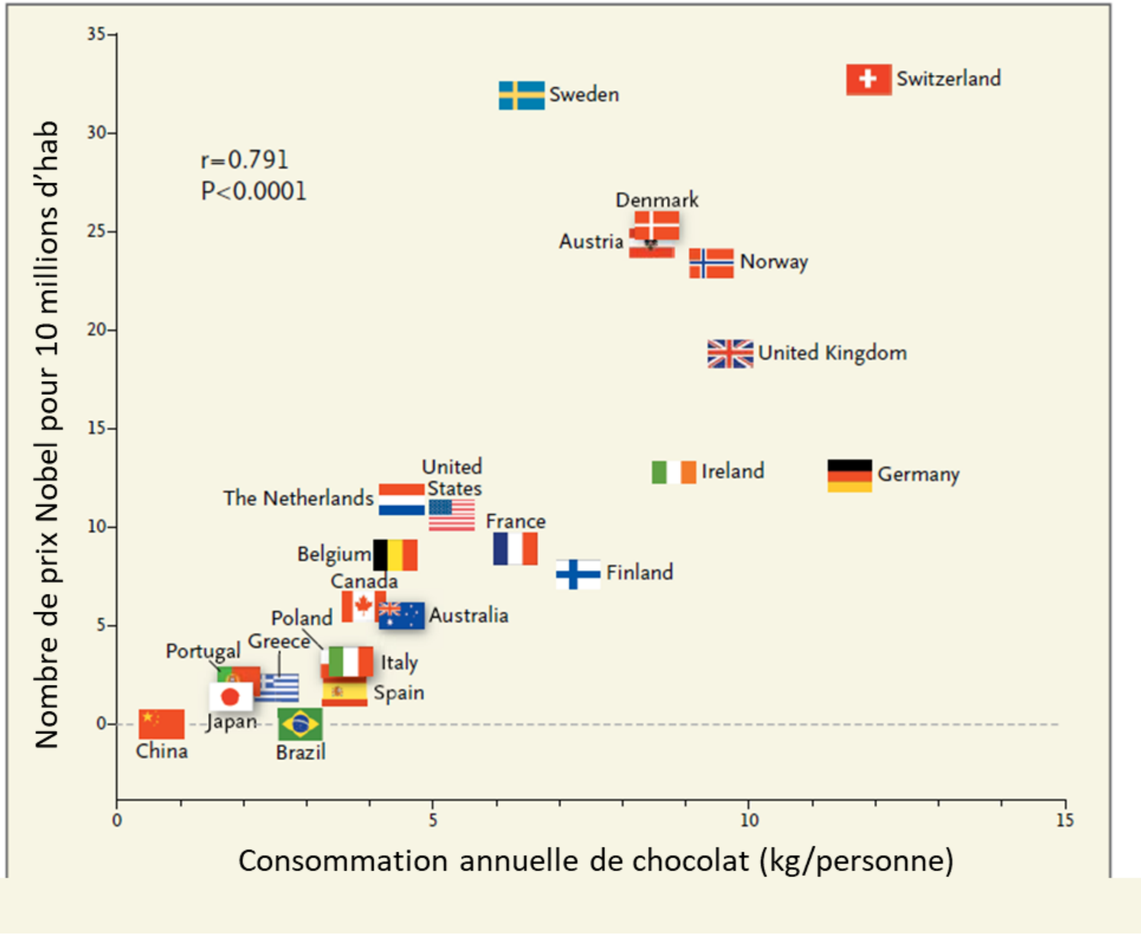
# Importance of data visualisation

- **Simplifies Complex Data:** Converts large and complicated datasets into charts and graphs that are easier to understand.
- **Highlights Patterns and Trends:** Makes it easier to spot relationships and trends that are hard to see in raw numbers.
- **Saves Time:** Enables quick interpretation of data, reducing the need to scan through lengthy tables.
- **Enhances Communication:** Helps share insights clearly, even with people who aren’t data experts.
- **Tells a Clear Story:** Guides the viewer through the data logically, supporting better understanding and decisions..

# Cons

**Visualization has real risks:**

- **Misleading design**: truncated axes, cherry-picked data, 3D effects
- **Oversimplification**: hides uncertainty or nuance
- **Cognitive bias**: viewers may interpret visuals emotionally
- **False authority**: “it looks professional, so it must be true…”
- **Accessibility issues**: poor color choices exclude color-blind users
- **Correlation shown as causation**



# When Visuals Mislead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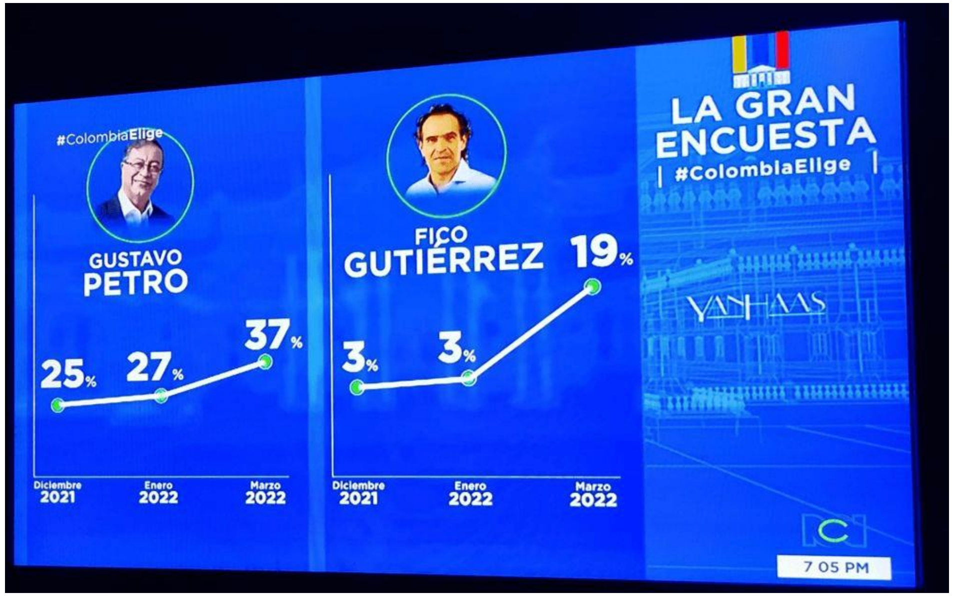

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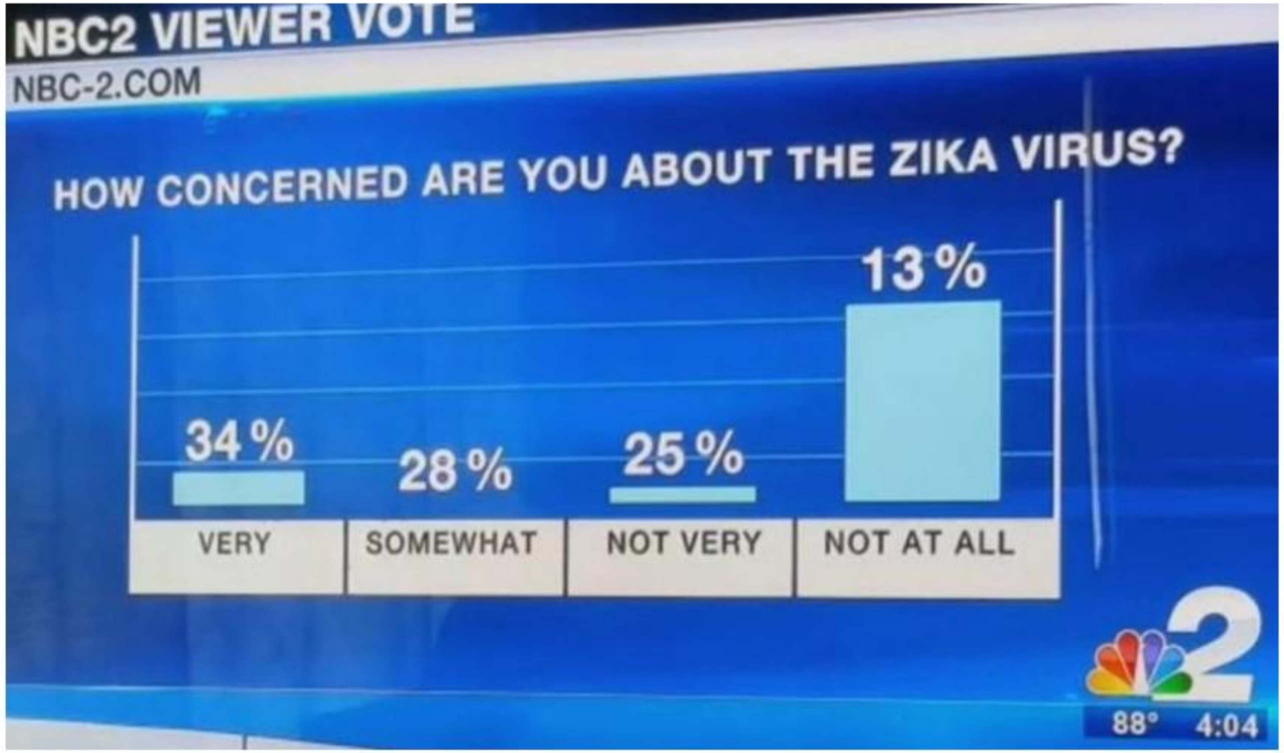
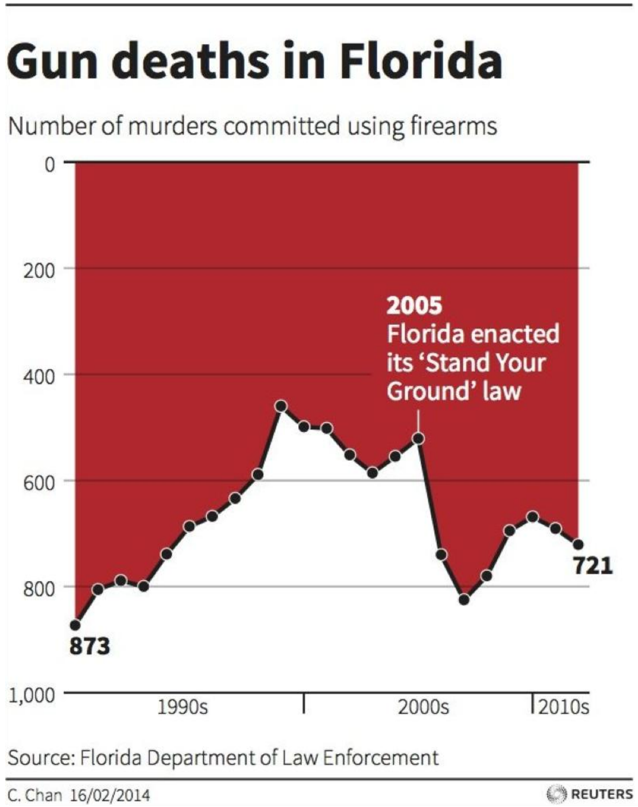


# Before You Plot — Structure Your Data

## You can’t visualize what you haven’t cleaned!

**What is Data Structuring?**
(Also called *data wrangling*, *data preparation*, or *data engineering*)

- Remove duplicates and errors (e.g. misspelling)
- Handle missing values (drop, fill, interpolate)
- Convert formats (e.g. dates, categories)
- Normalize or scale values
- Reshape data (pivot, melt) for plotting

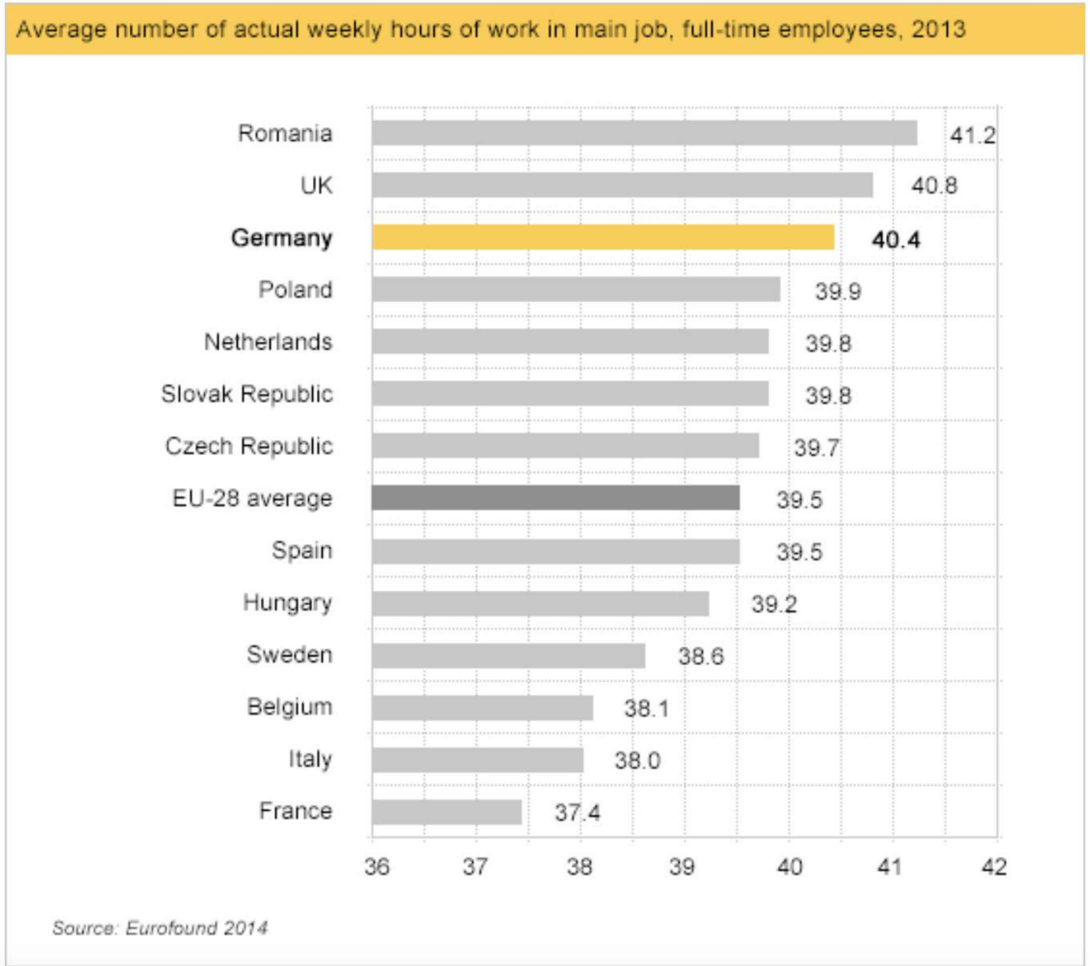
# Bar charts

**A bar graph plots data with the help of bars, which represent value on the y-axis and category on the x-axis. The main purpose of bar charts is to provide comparisons between categories**

A bar chart is used when you want to show a distribution of data points or perform a comparison of metric values across different subgroups of your data. From a bar chart, we can see which groups are highest or most common, and how other groups compare against the others.

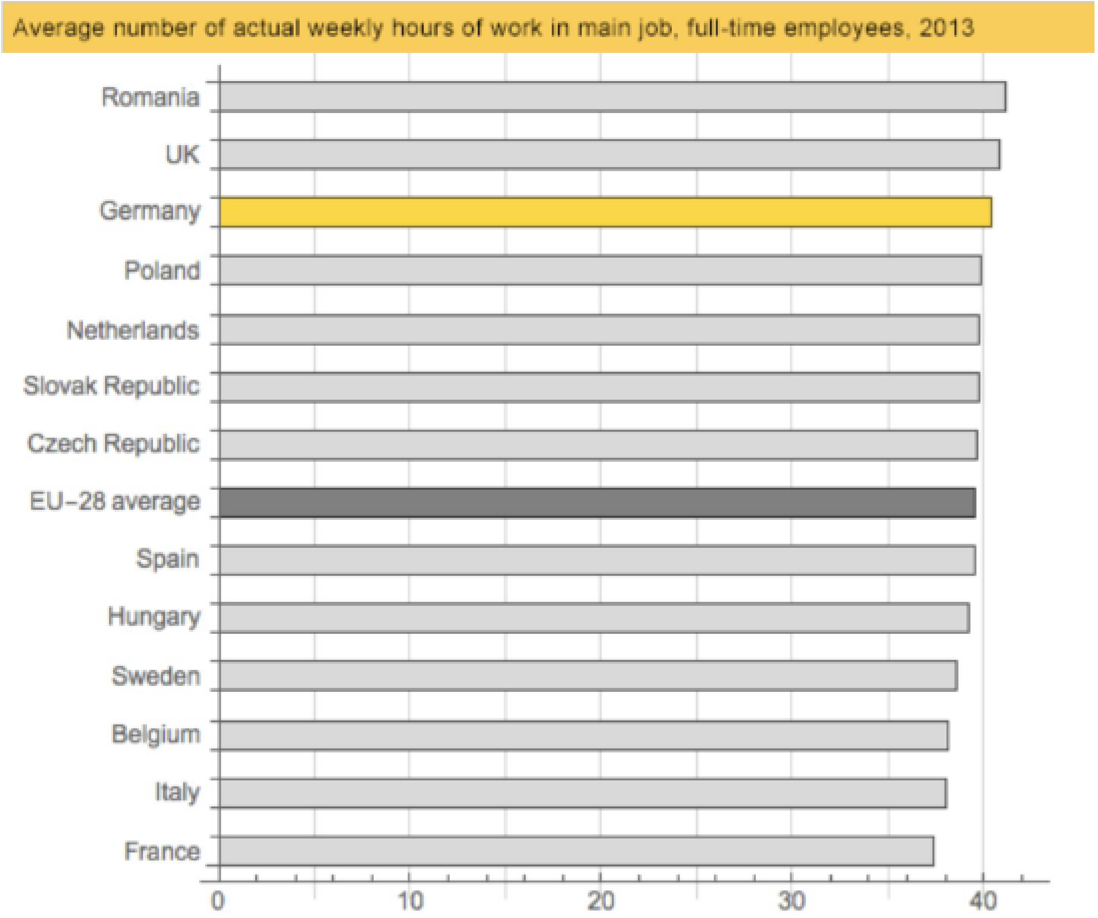
# Bar charts

What conclusions do you draw from this graph?

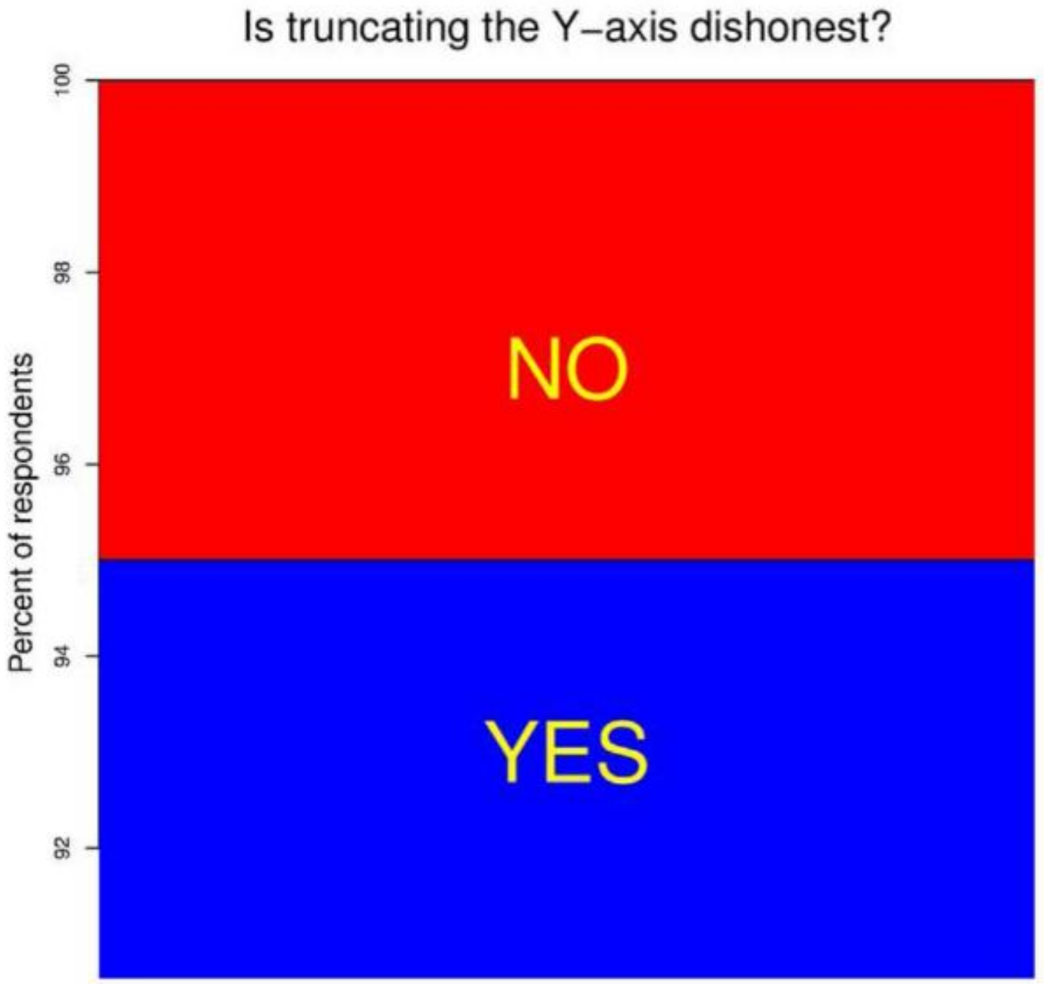


# Bar charts

And from this one?



# Graphical representation



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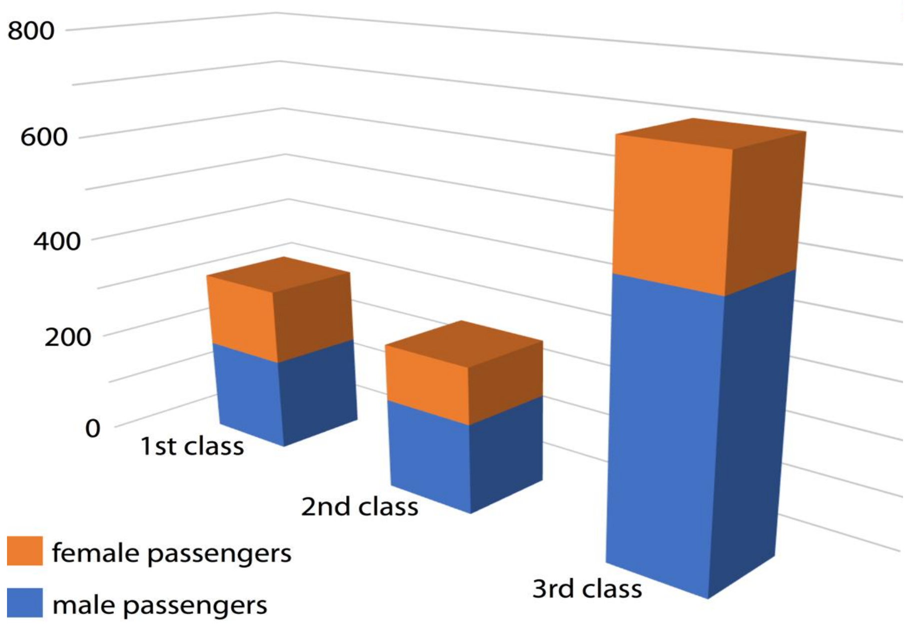



Fundamentals of Data Visualization

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1) **Always start y axis from zero: bar chart** emphasizes the absolute **magnitude of the values** associated with each category

2) **Do not use 3D:** Difficult to read the right values

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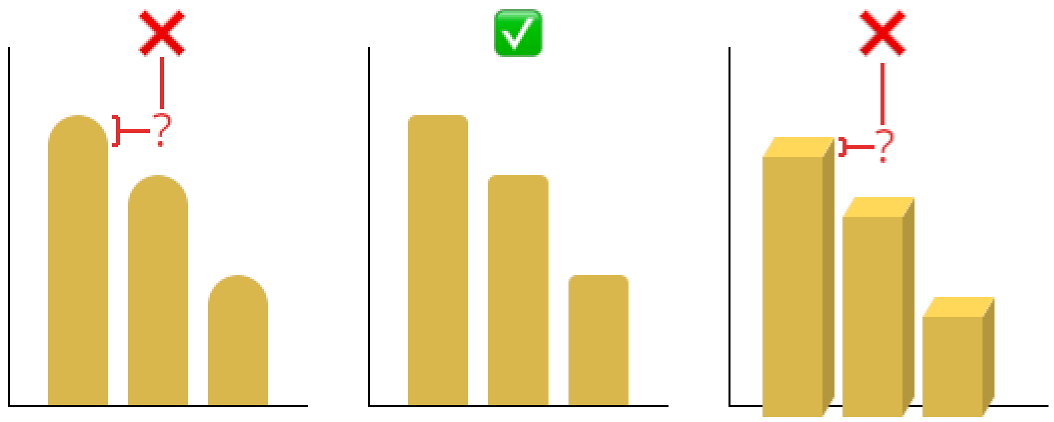

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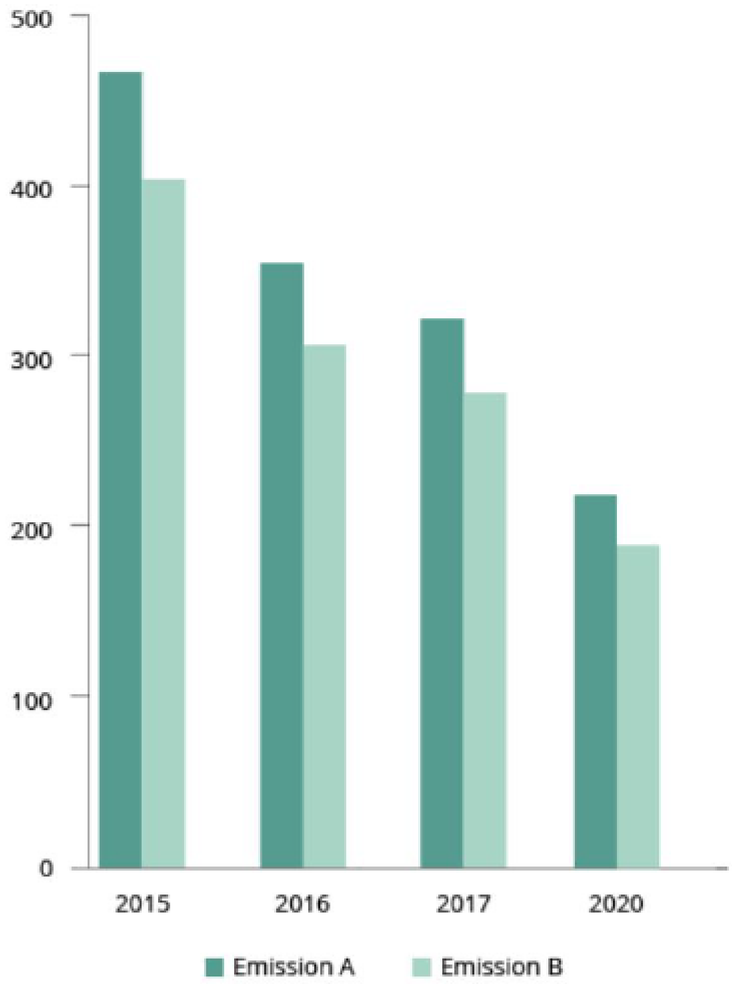
What do you see?



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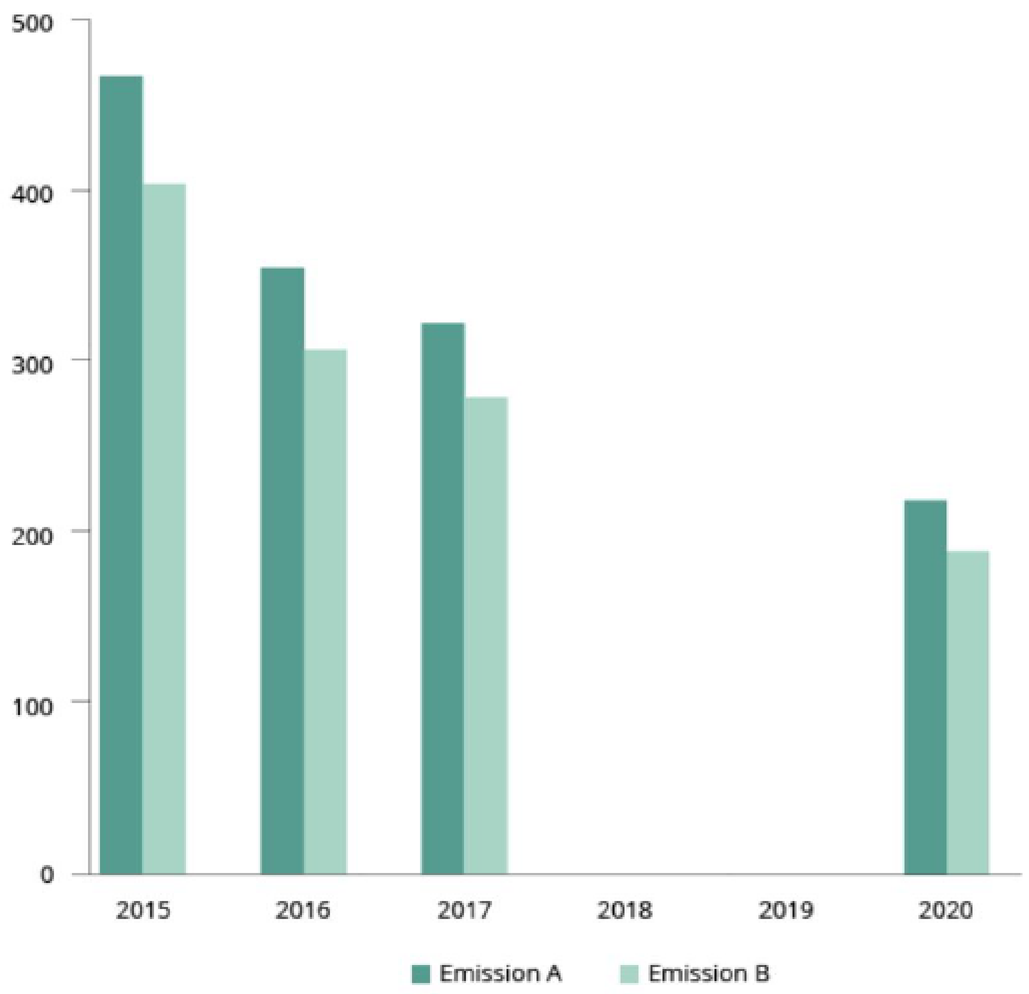


**Note:** No data were reported in 2018 and 2019

# Bar charts

You have 5 seconds to find the position of France in the ranking

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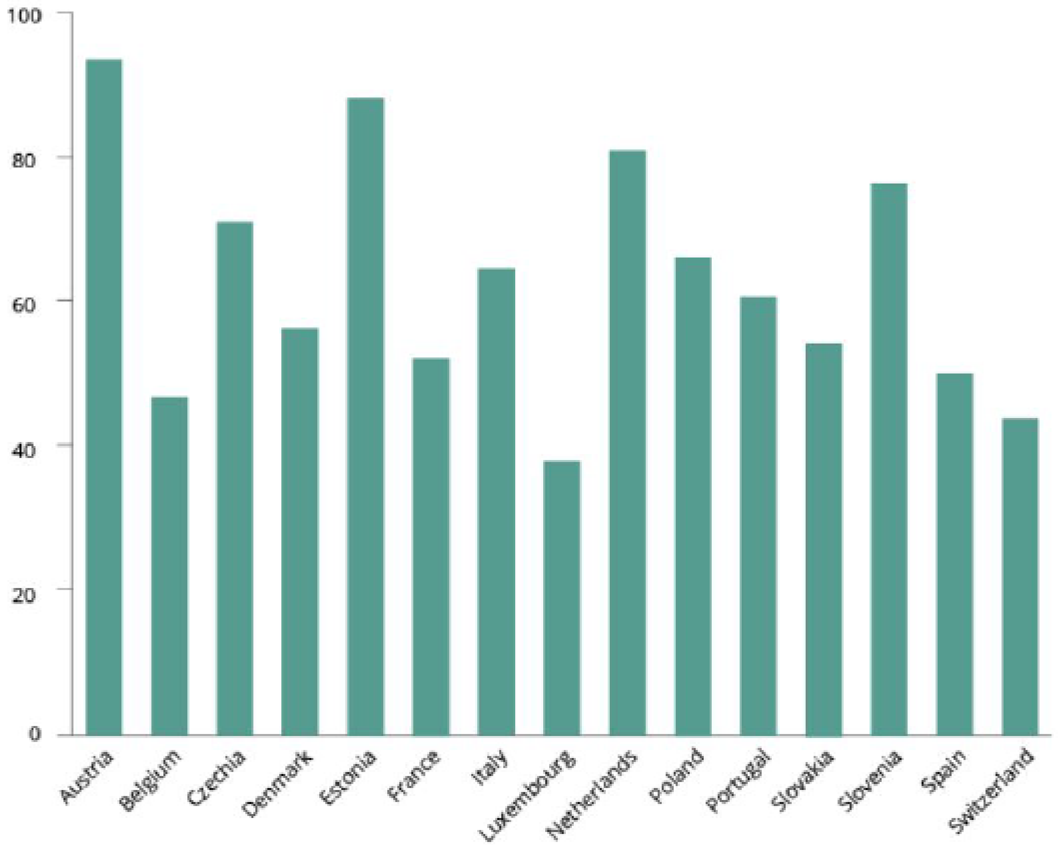

100
80
60
40
20
0
Austria
Belgium
Czechia
Denmark
Estonia
France
Italy
Luxembourg
Netherlands
Poland
Portugal
Slovakia
Slovenia
Spain
Switzerland

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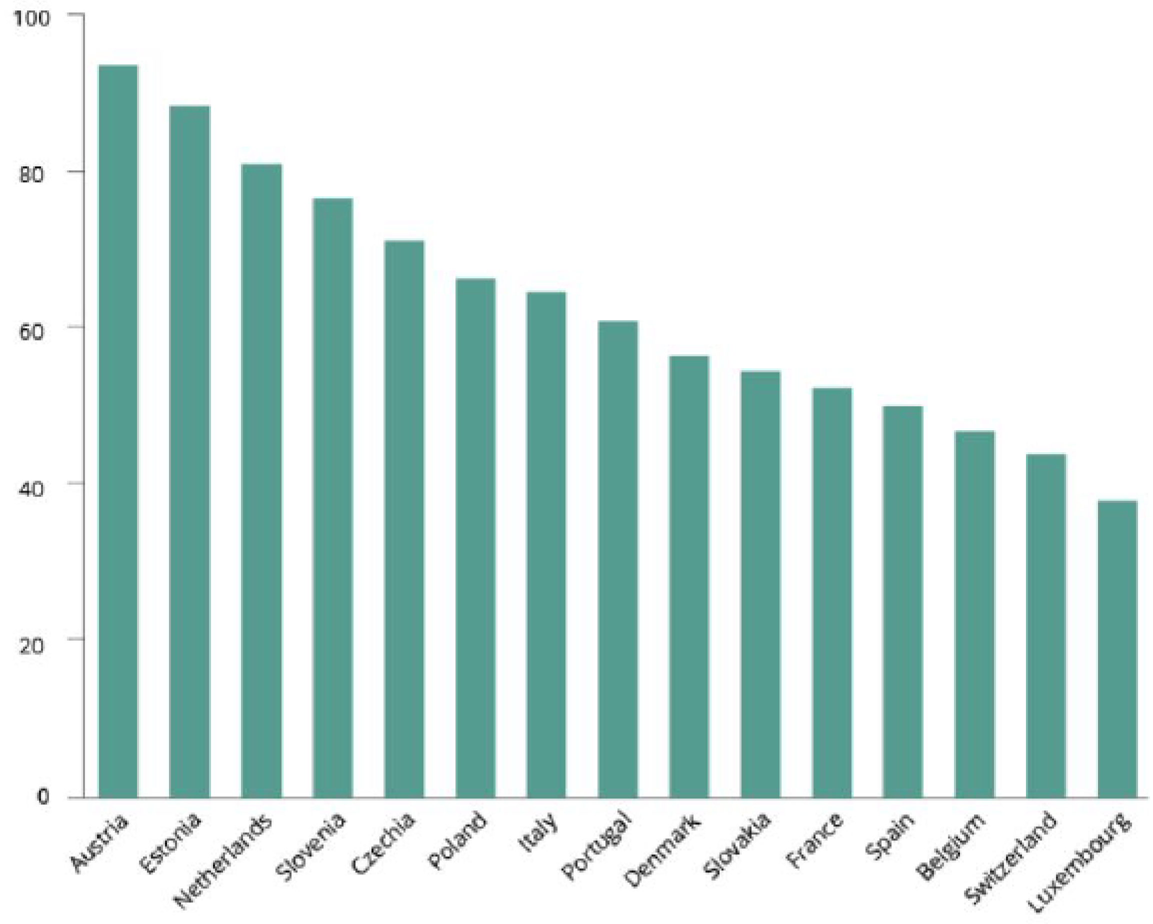
100
80
60
40
20
0
Austria
Estonia
Netherlands
Slovenia
Czechia
Poland
Italy
Portugal
Denmark
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France
Spain
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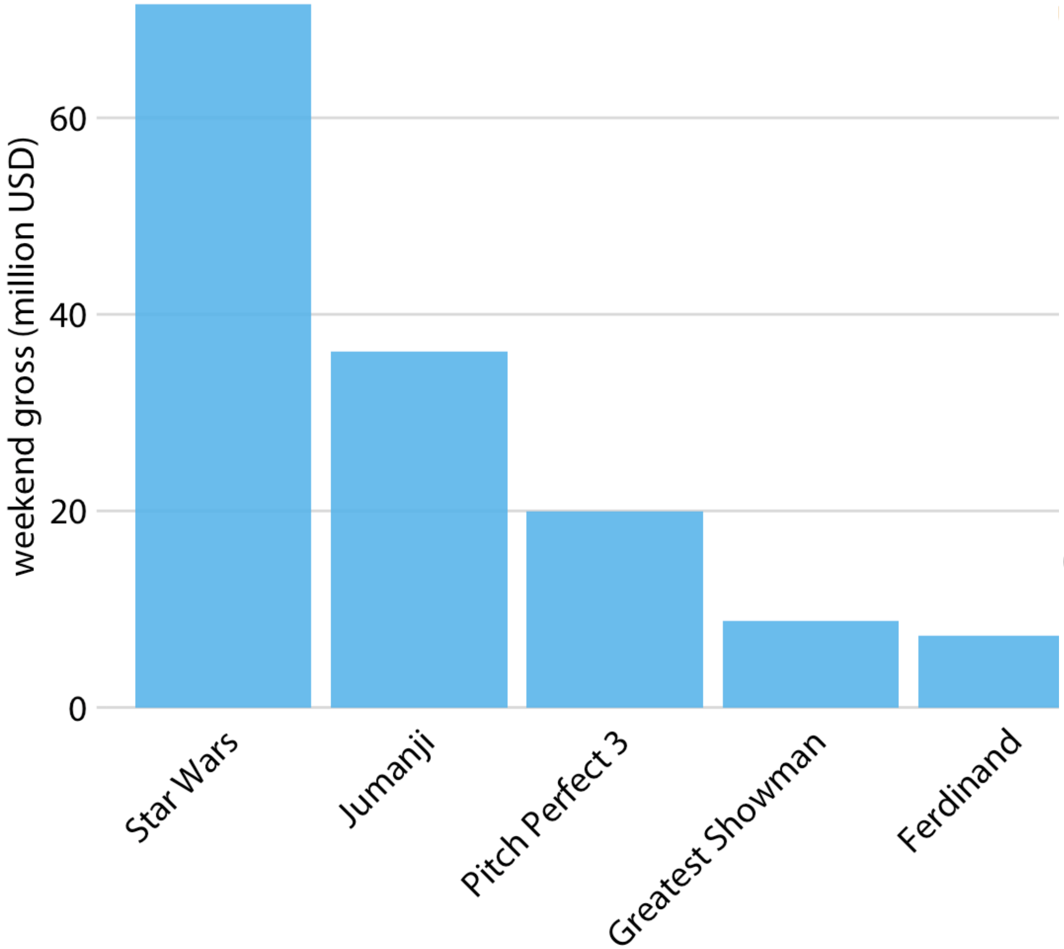
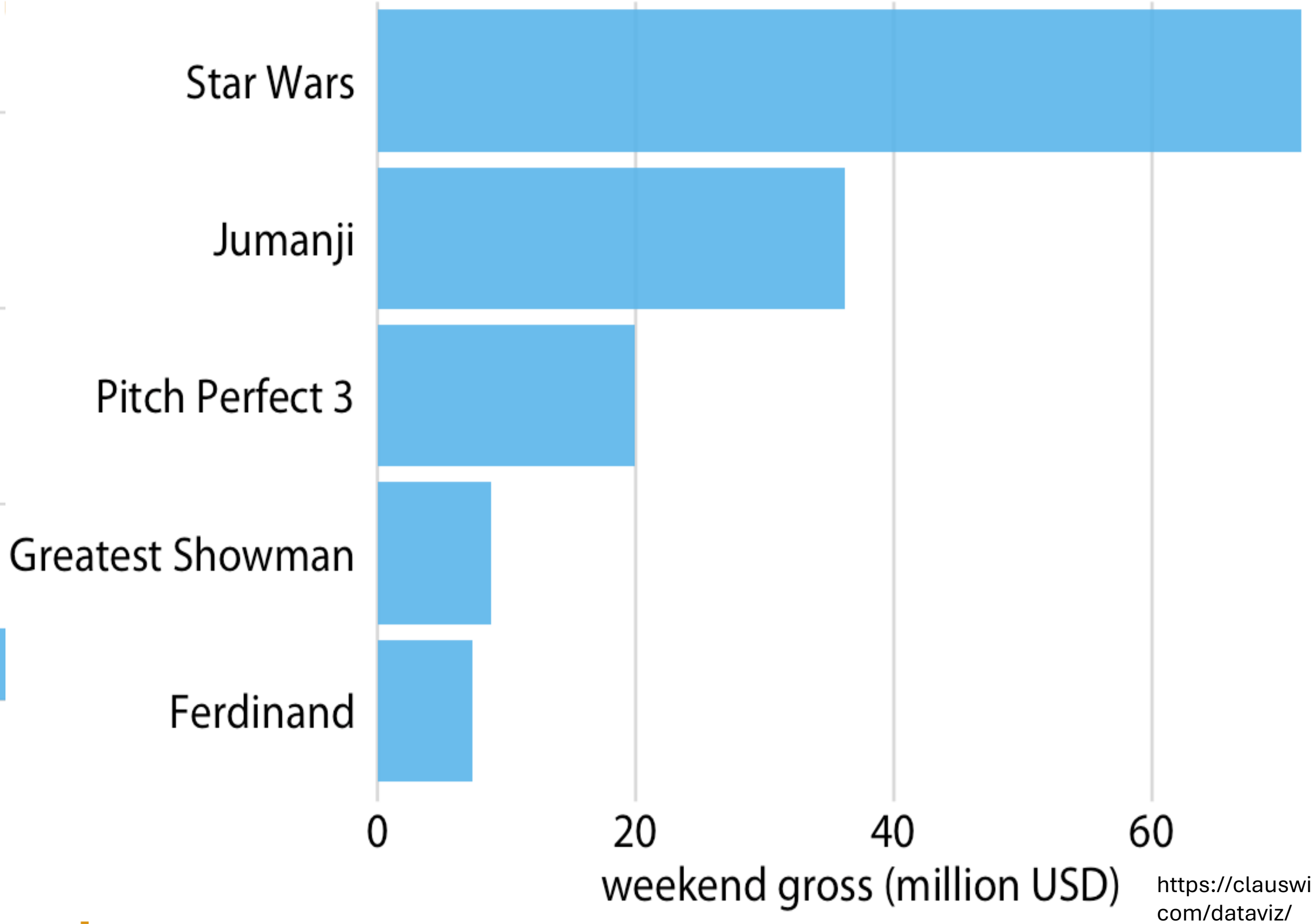
Switzerland
Luxembourg

# Bar charts




https://clauswilke.com/dataviz/

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4) **Sort your data:** Allow to make the comparison faster

5) **Horizontal orientation for long label**

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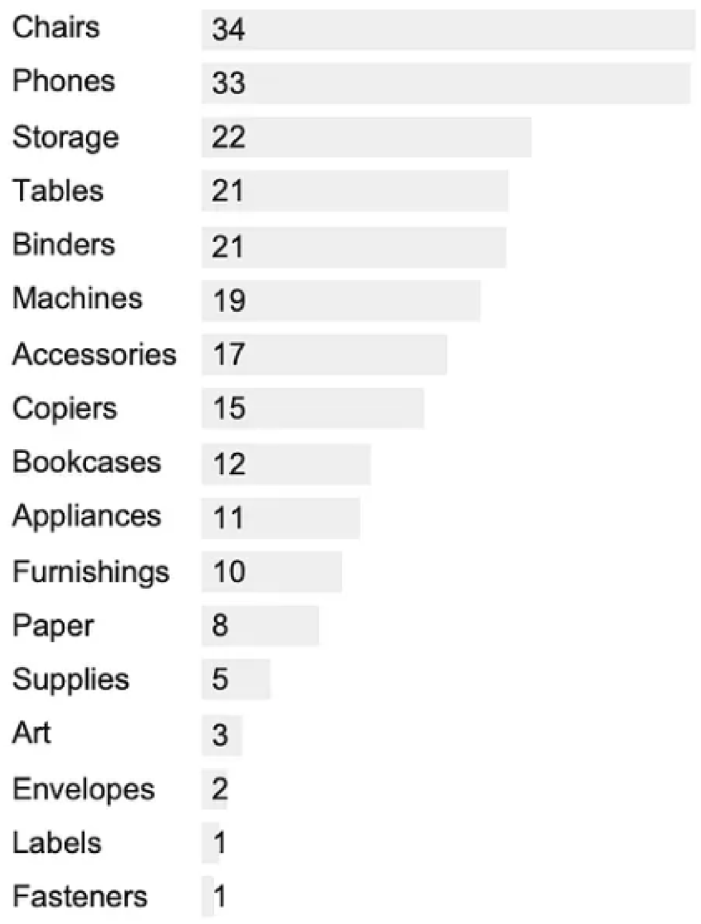


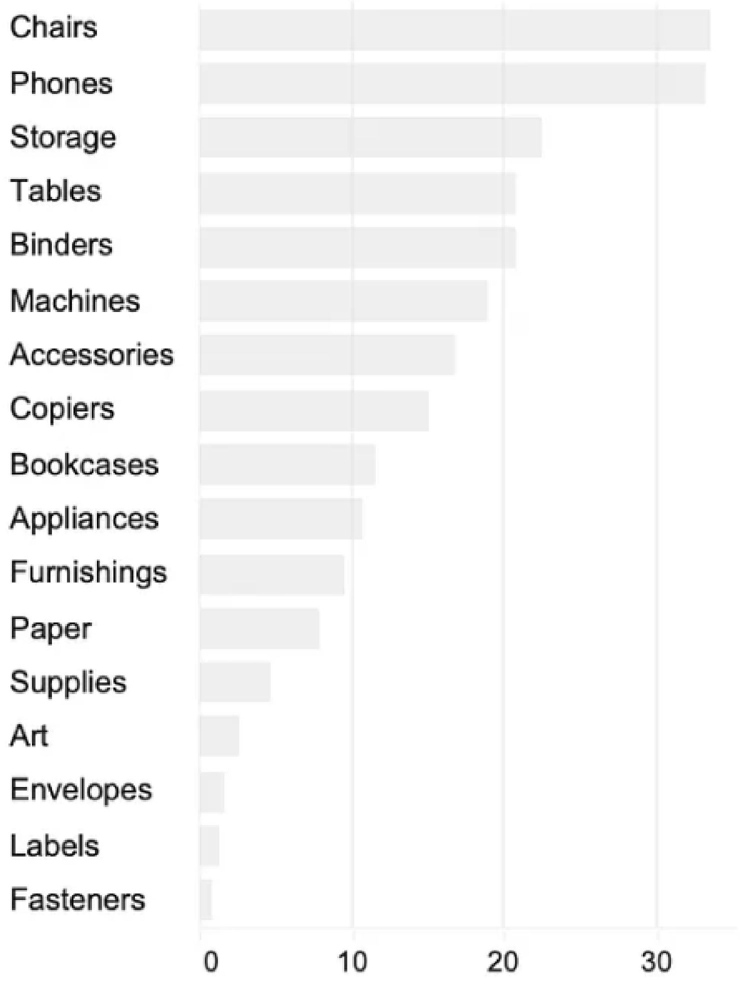


[Bar Charts Best Practices | by Anastasiya Kuznetsova | Medium]

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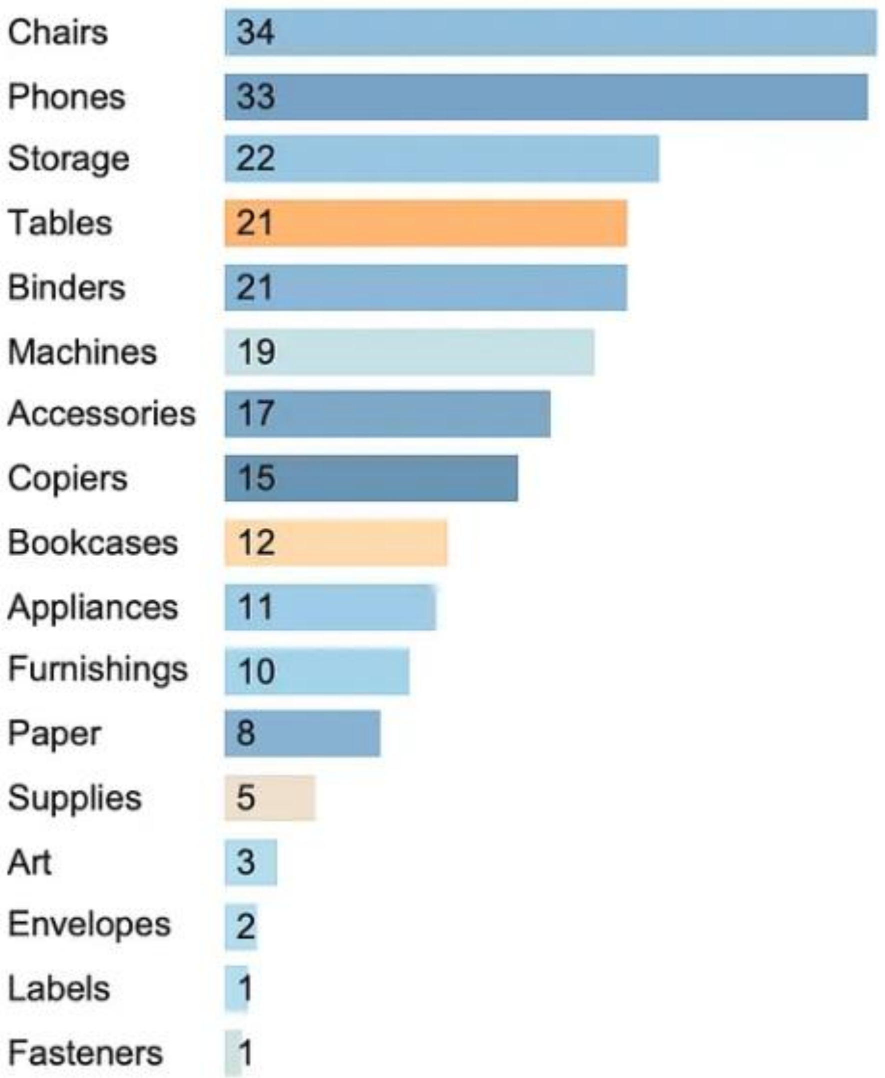


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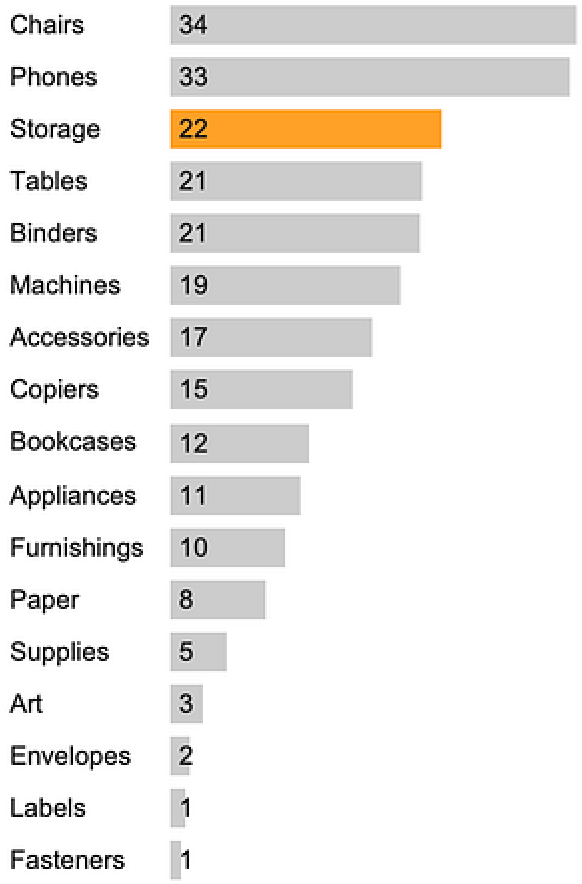
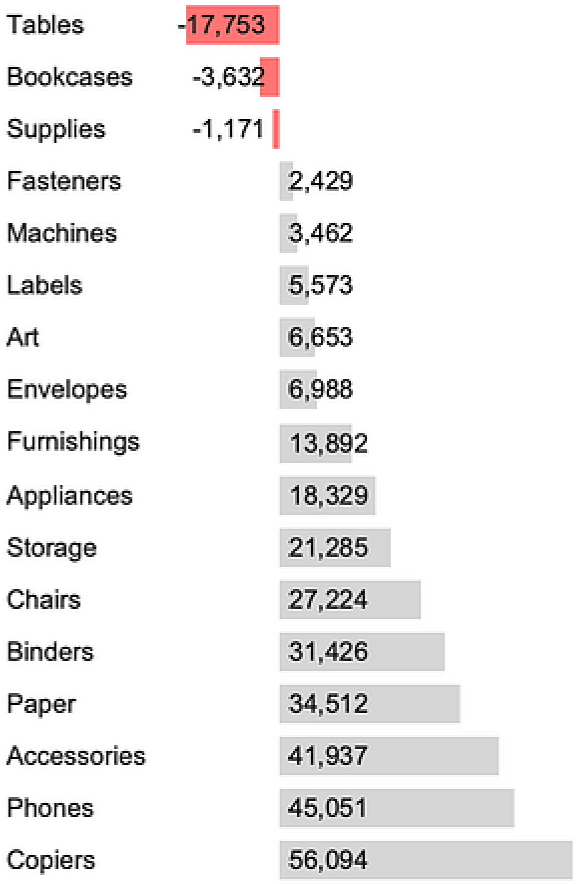
# Bar charts

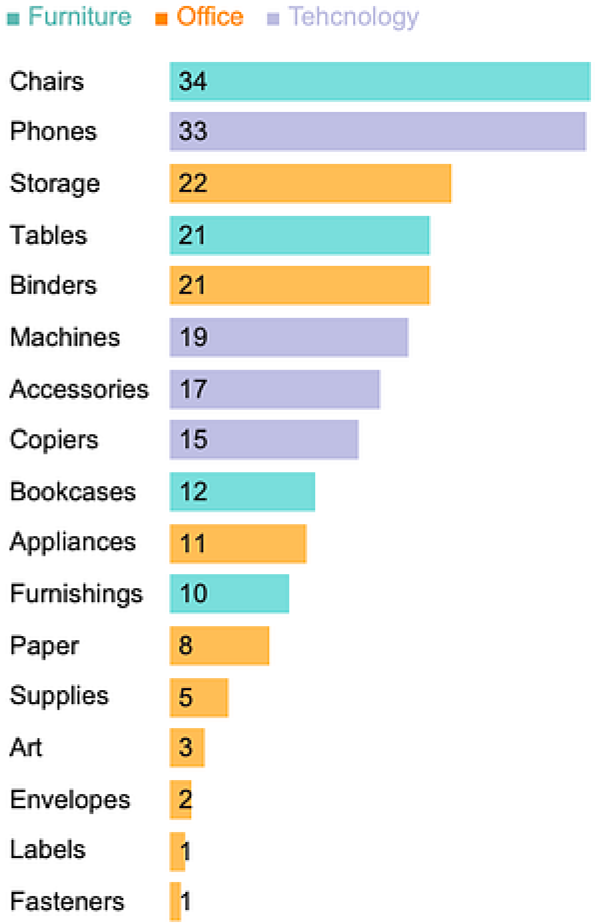
✅ **Highlight Storage**




✅ **Alert: Not generating profit**




✅ **Distinguish**

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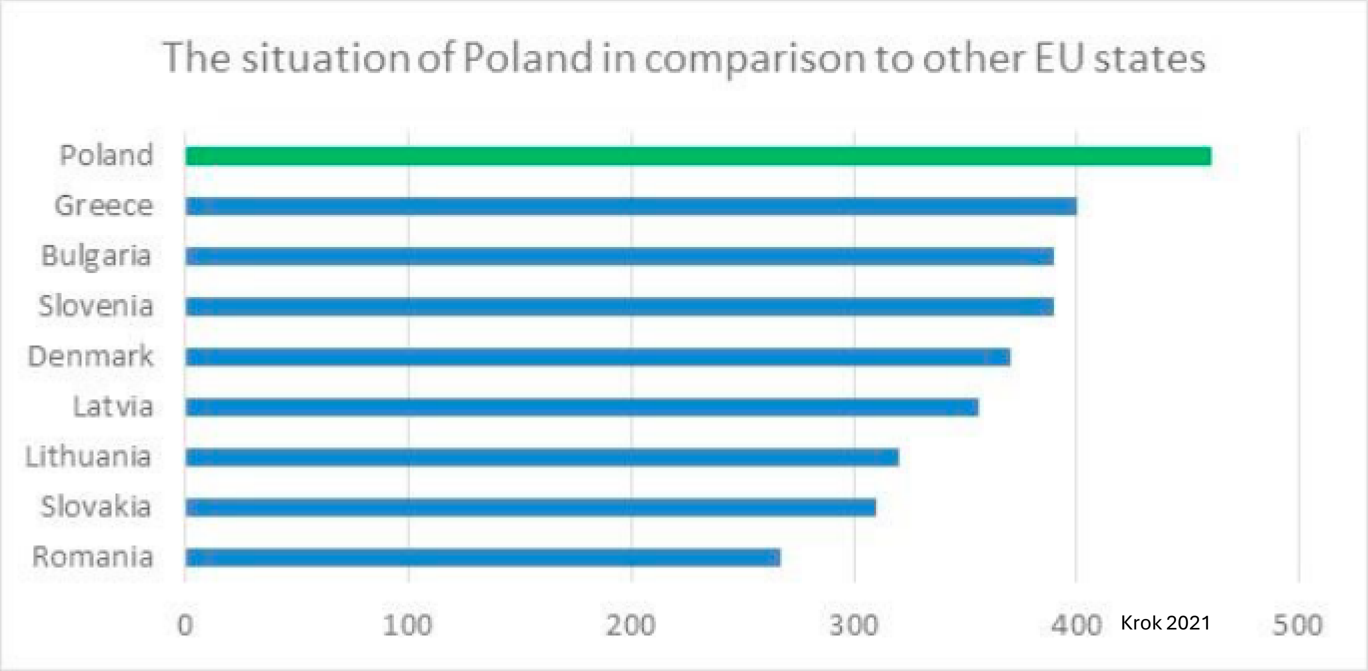
What do you conclude?



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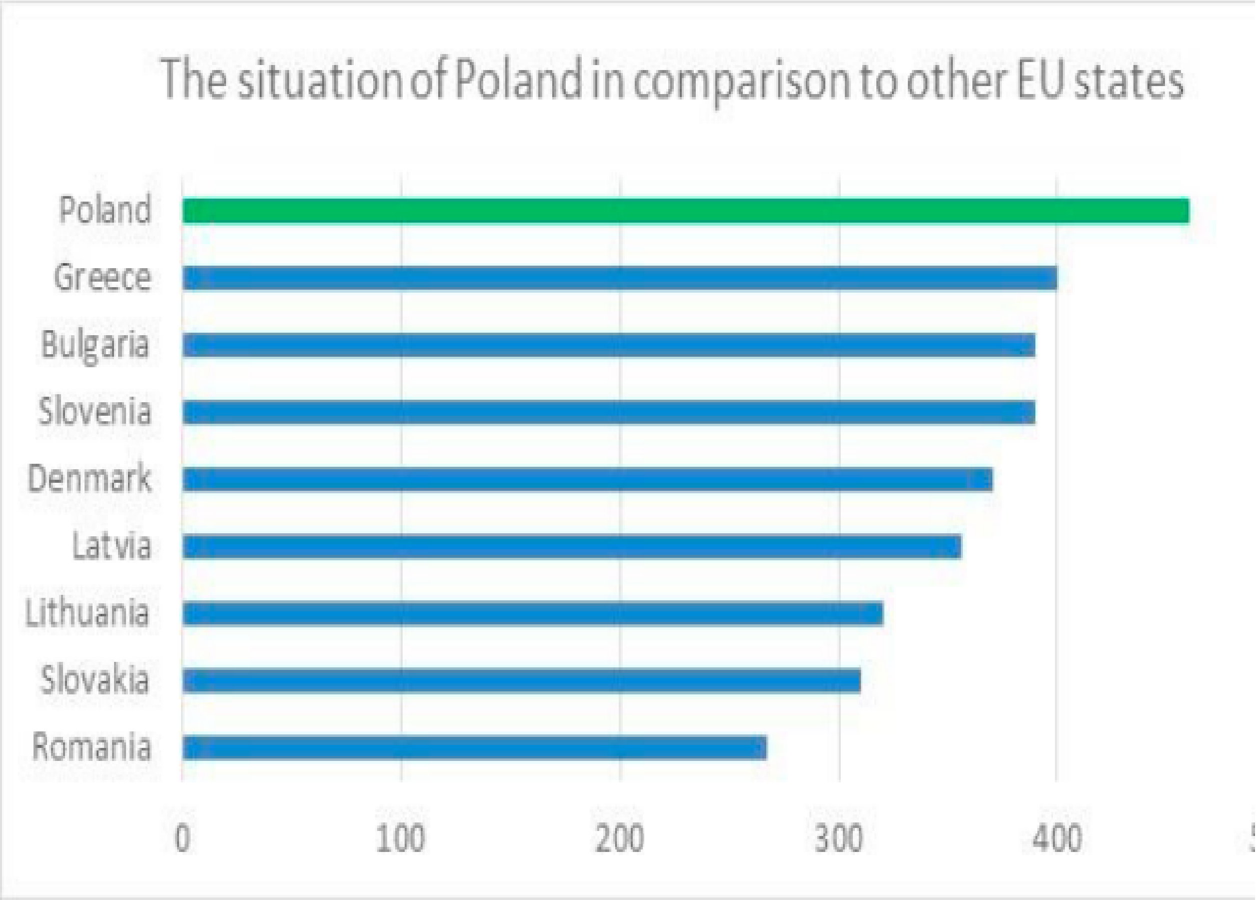
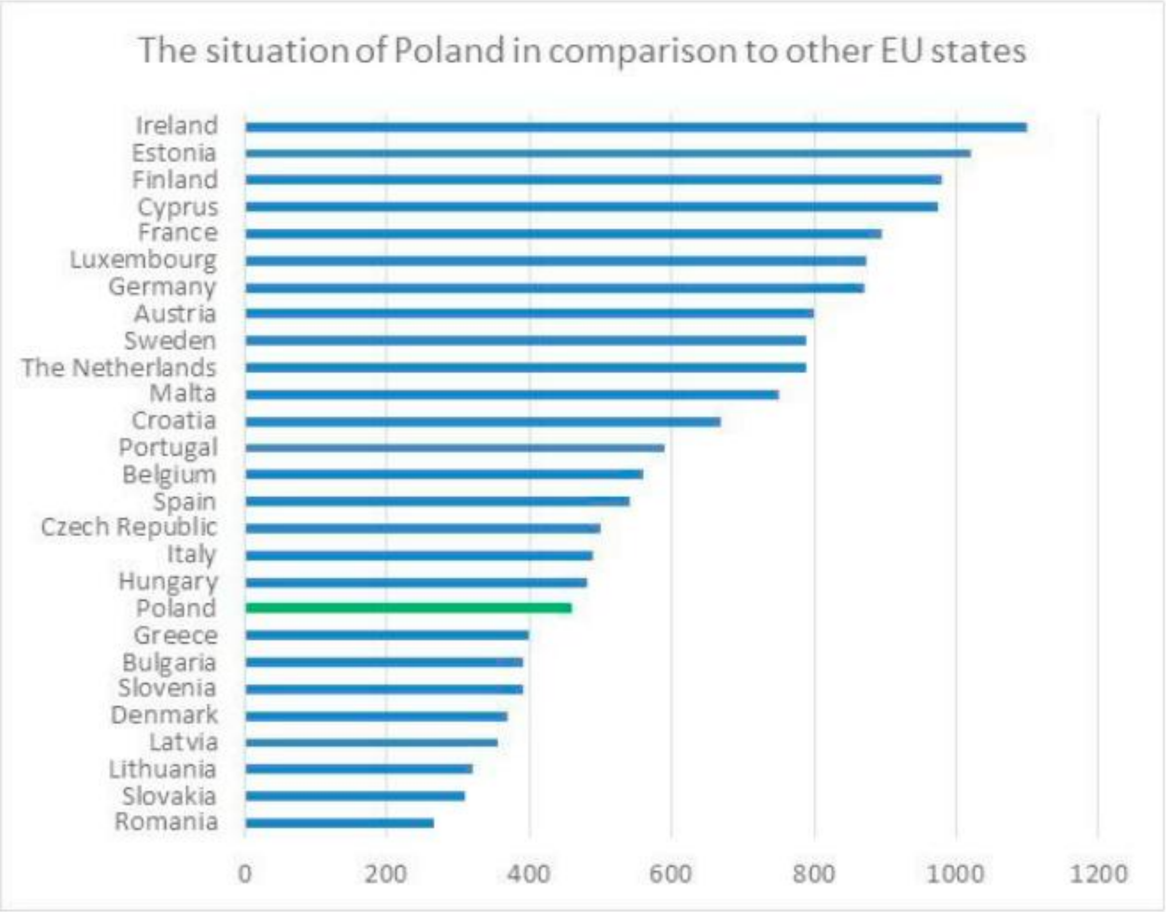

The situation of Poland in comparison to other EU states
Poland
Greece
Bulgaria
Slovenia
Denmark
Latvia
Lithuania
Slovakia
Romania
0
100
200
300
400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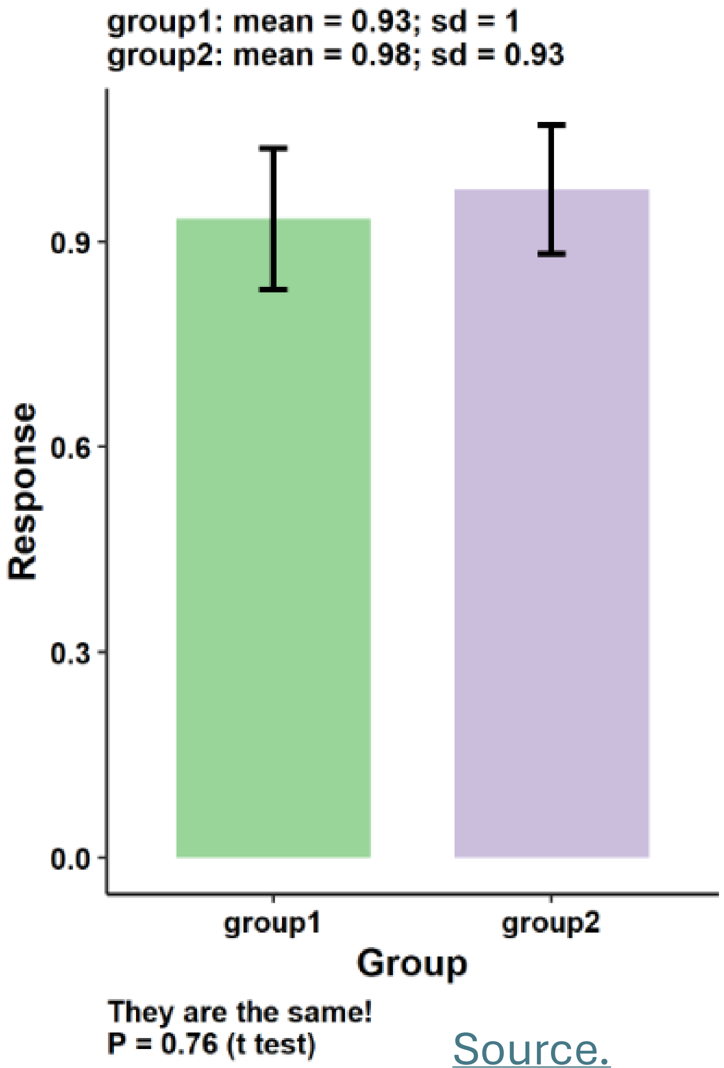


Source.

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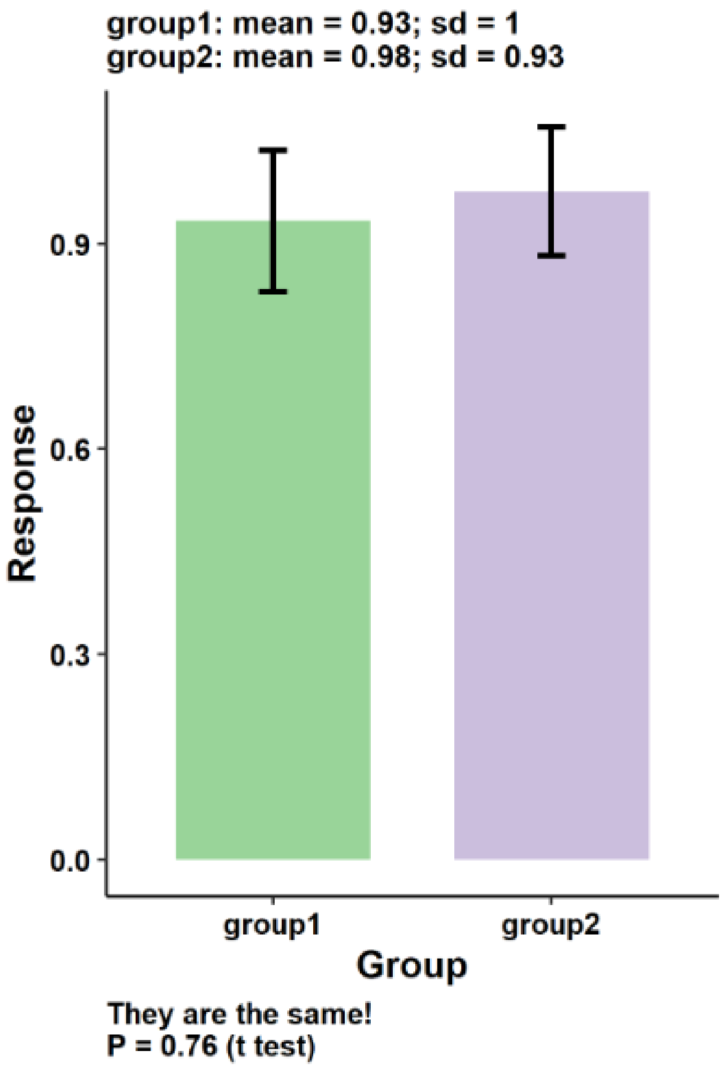


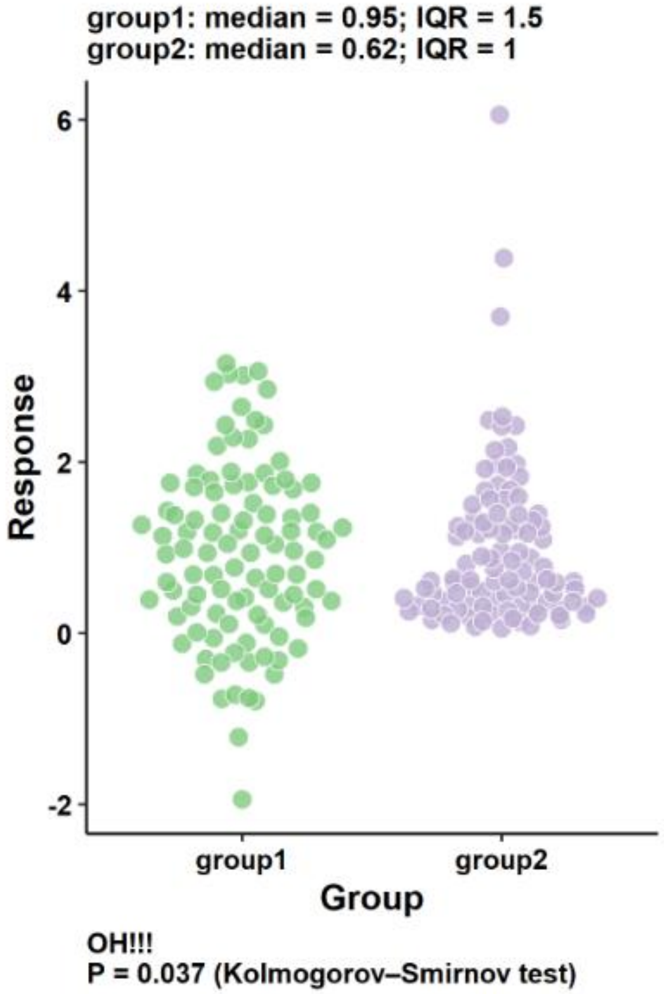


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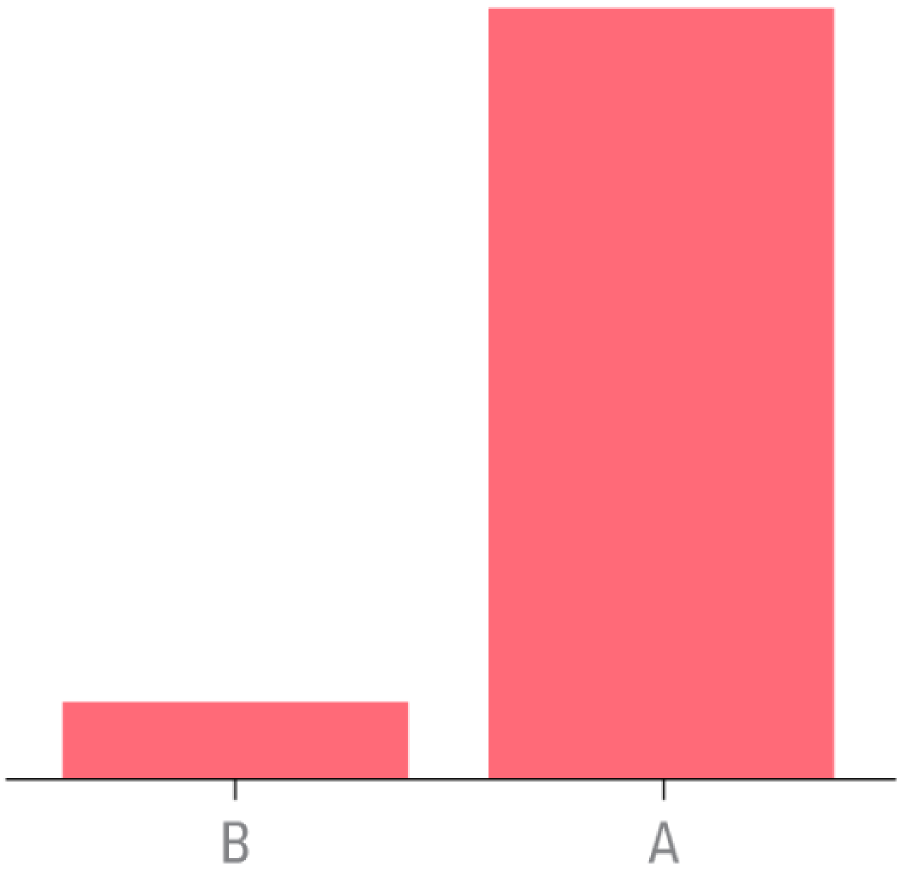
9) **Do not make Bar Plots for Means Separation**

# Bar charts




The value of A is how many times as large as the value of B?

9

8

10

5

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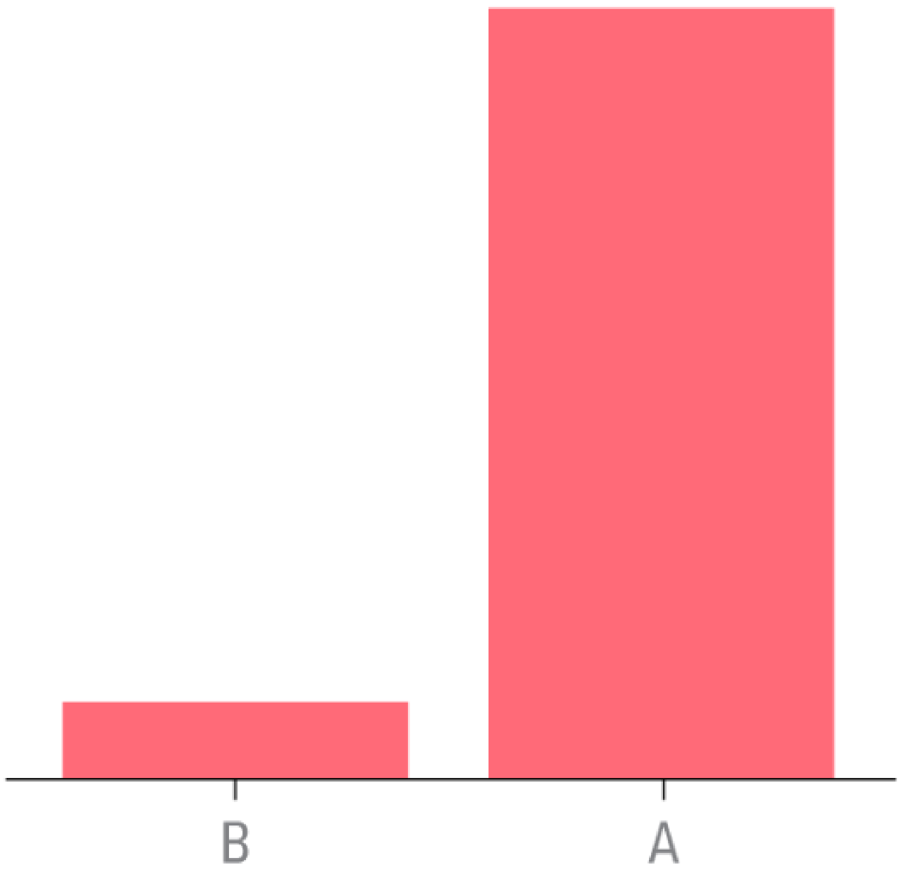


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# Histogram charts

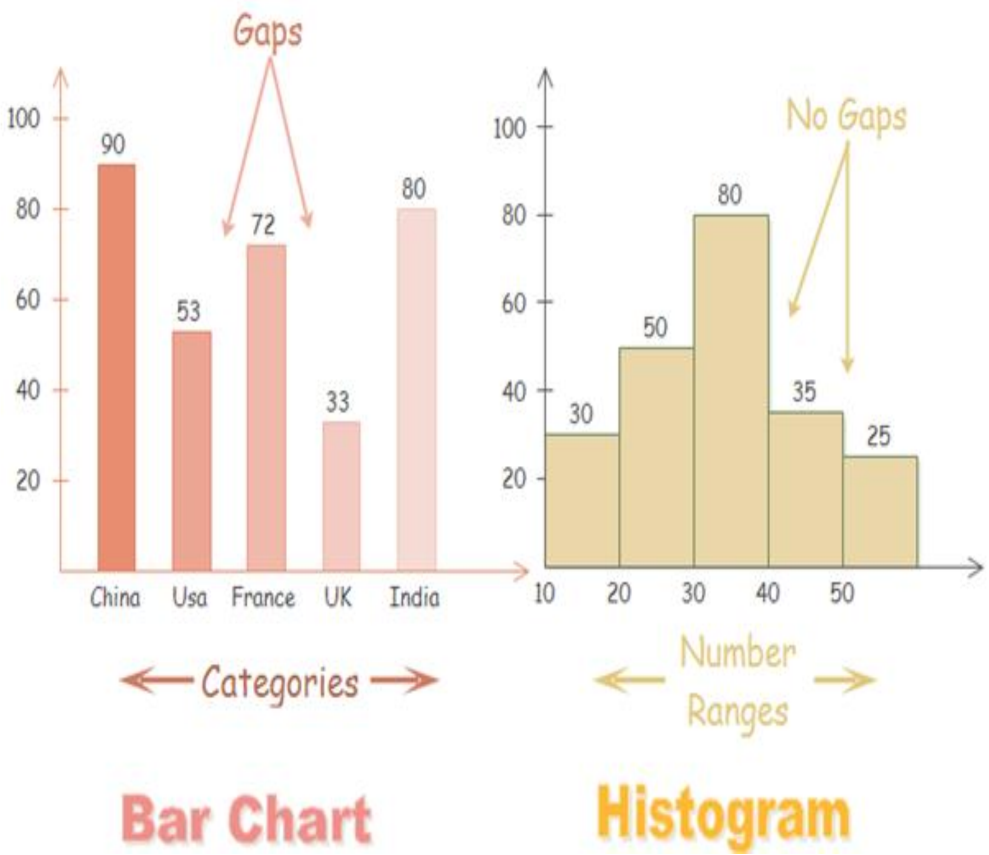
- We frequently encounter the situation where we would like to understand how a particular variable is distributed in a dataset.

- **A histogram displays the distribution of numerical data by grouping values into intervals (bins) and showing their frequency as bars. It helps reveal the shape, spread and patterns in the data.**

- Histogram use **Non-discrete quantitative data** (discrete numerical or categorical data for bar chart)

**Useful for:**
- To visualize variability
- To identify outliers
- To identify multimodal distributions

Example from edraw



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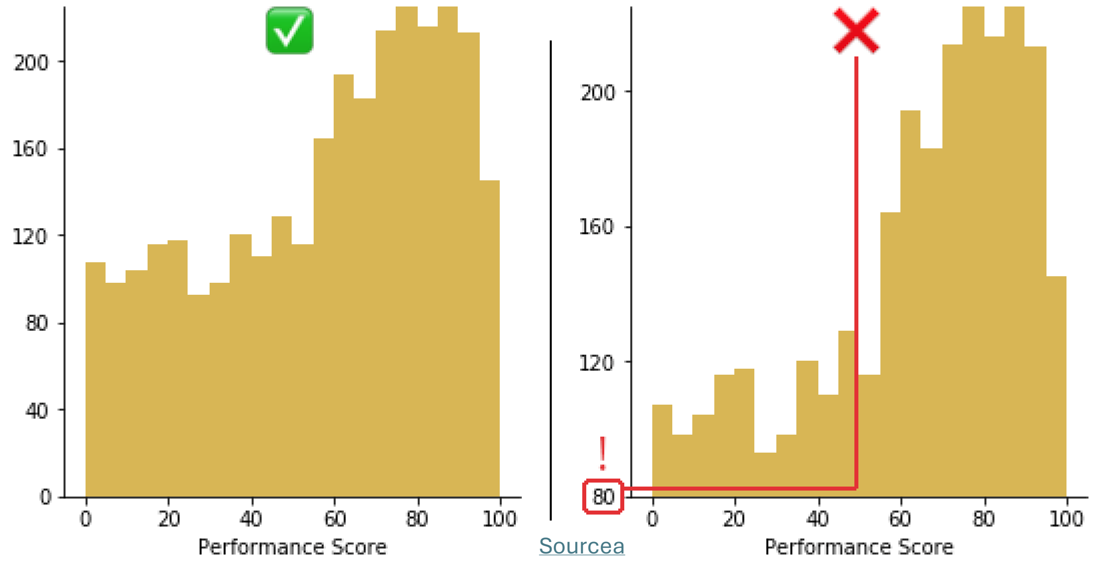
**1) The base value must always be zero**



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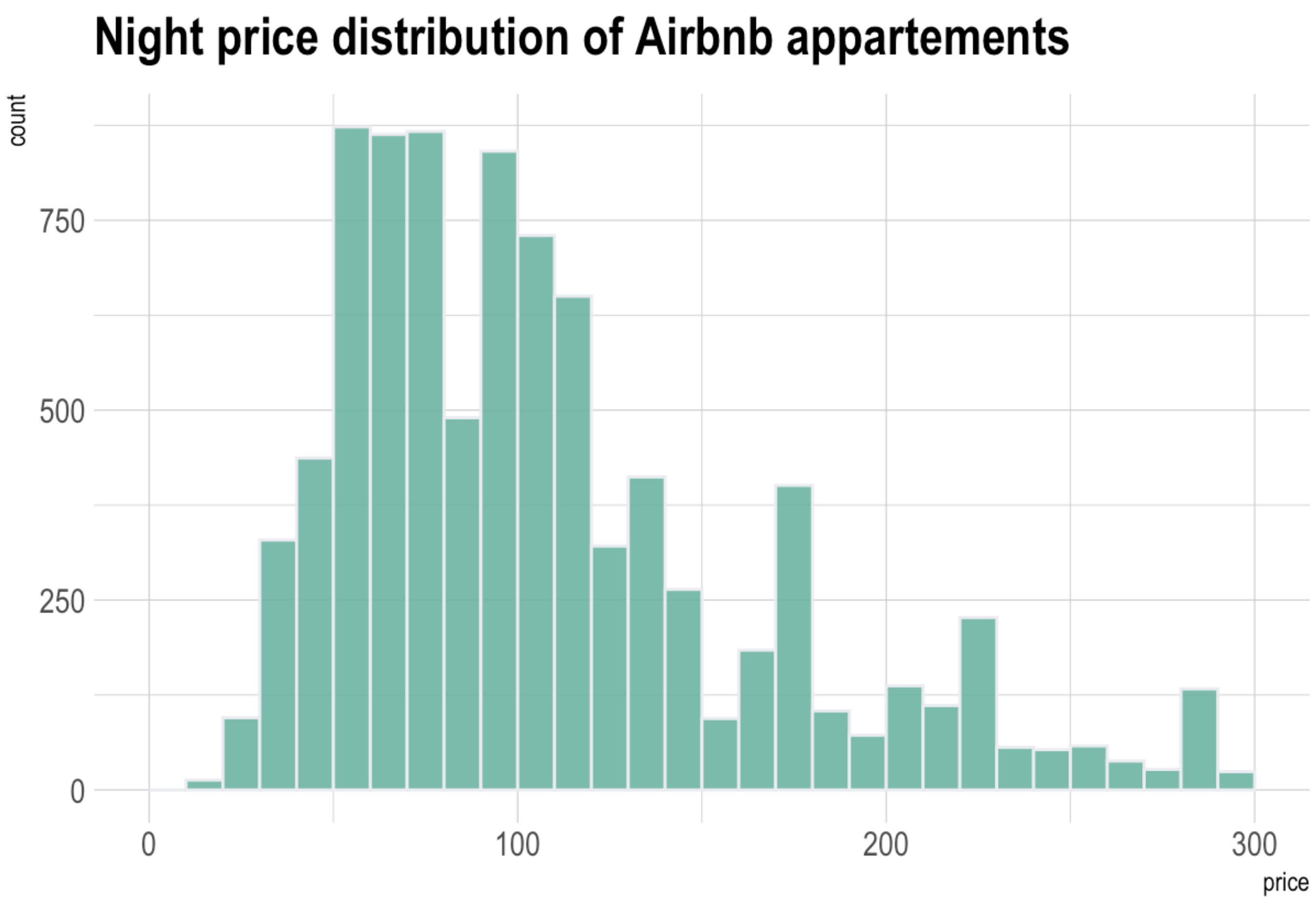

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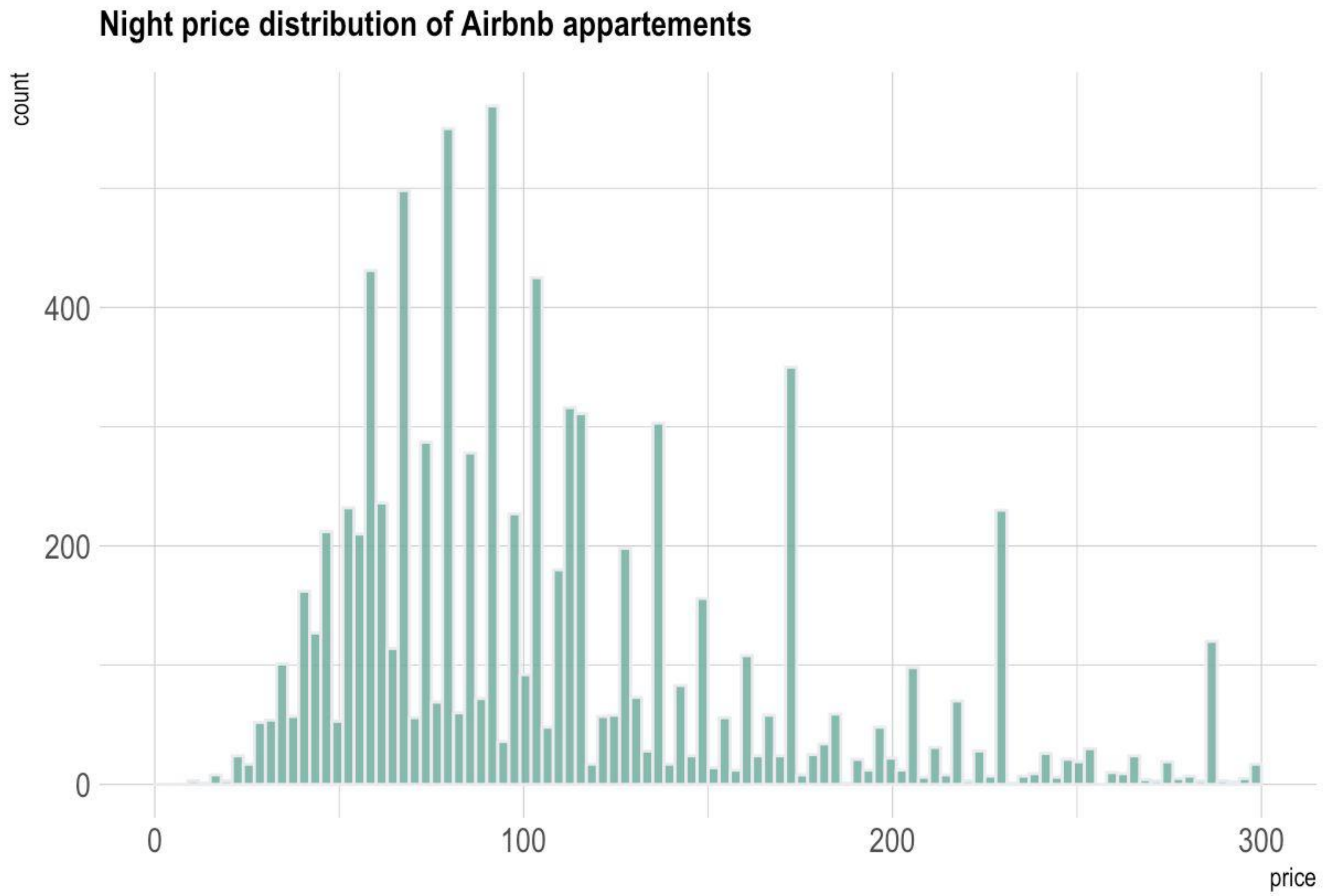

# Histogram charts

**1) The base value must always be zero**

**2) Try several bin size!!:** If bin width is too small, then the histogram becomes overly peaky and visually busy and the main trends in the data may be obscured. If the bin width is too large, then smaller features in the distribution of the data can be hidden.

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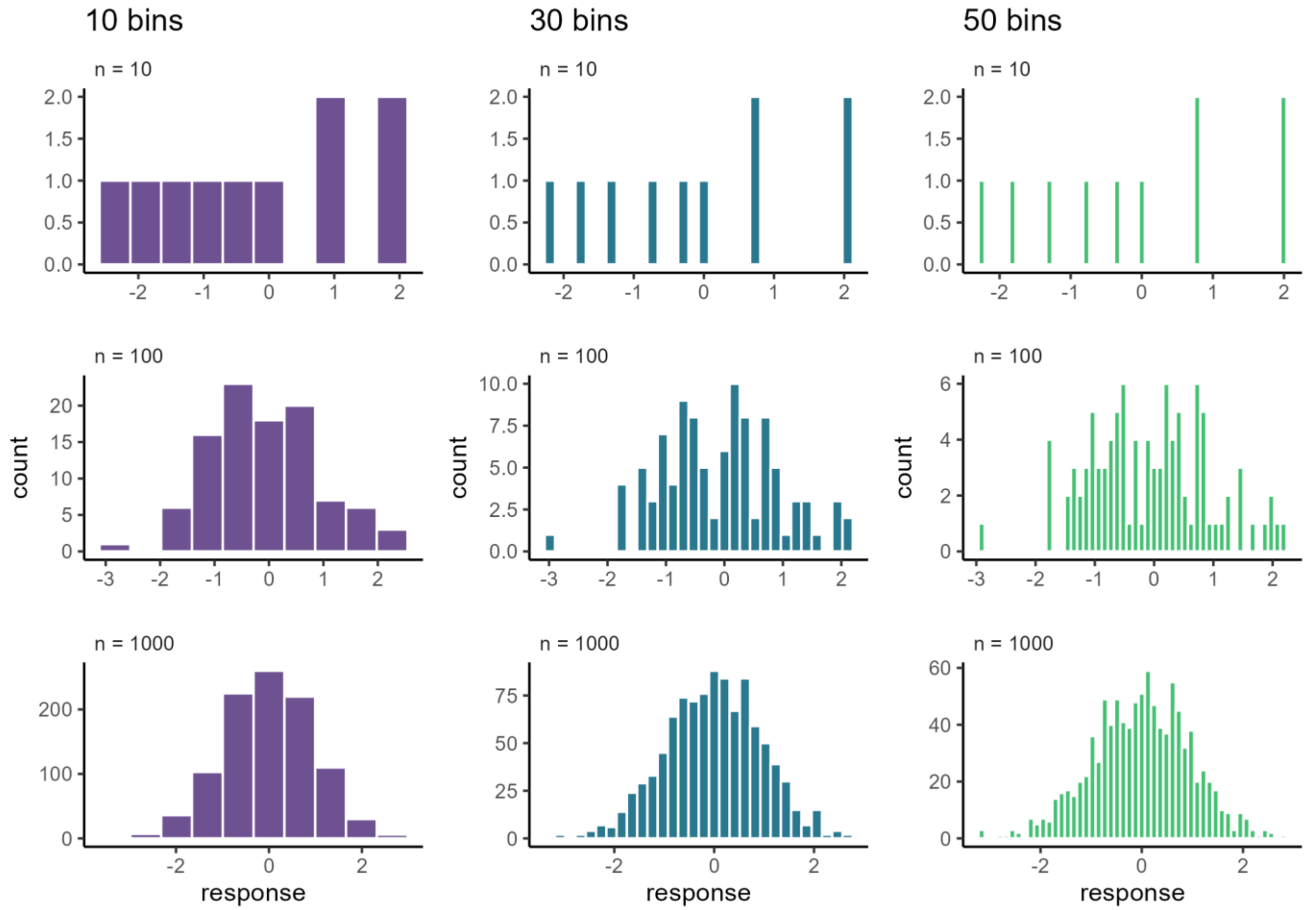
**3) Do use weird colors**

# Histogram charts

Random number from the same distribution




Wow, the appearance does change with different bin numbers.

Source.

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**4) Never use histogram on a small dataset!!**

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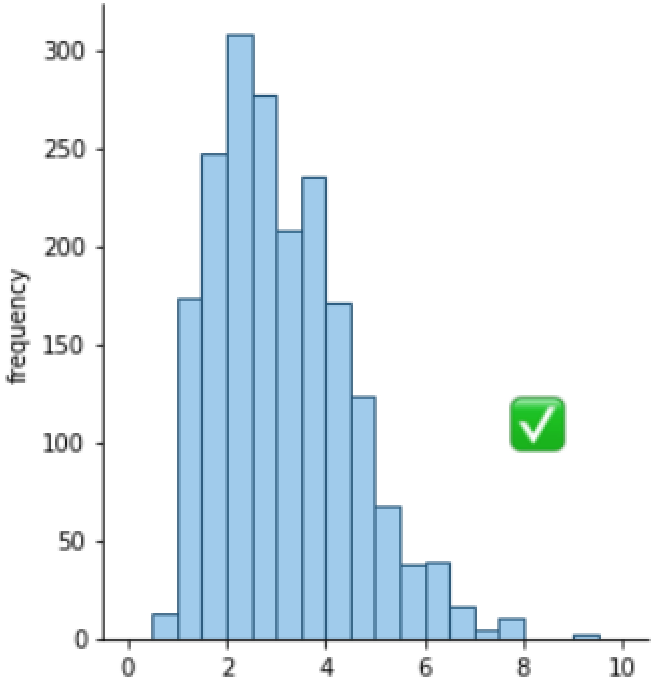
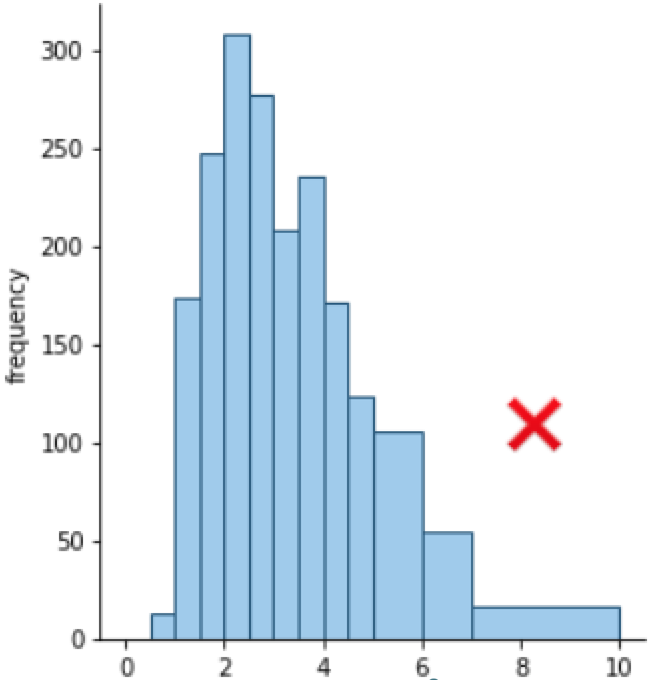
**5) Use the same bin width**

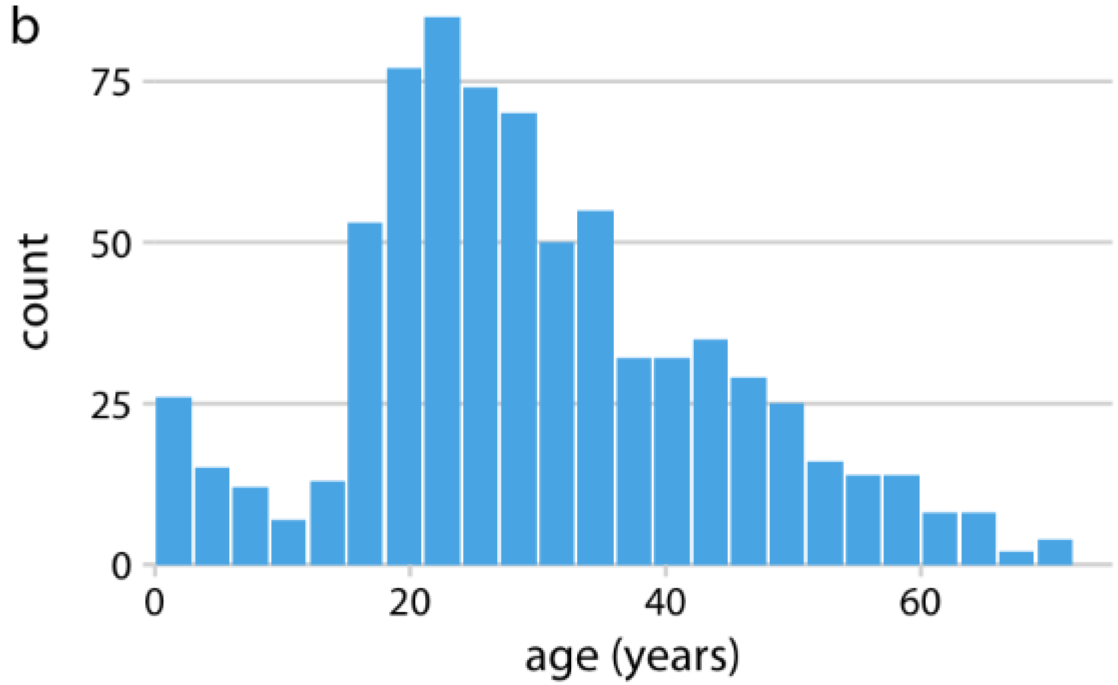


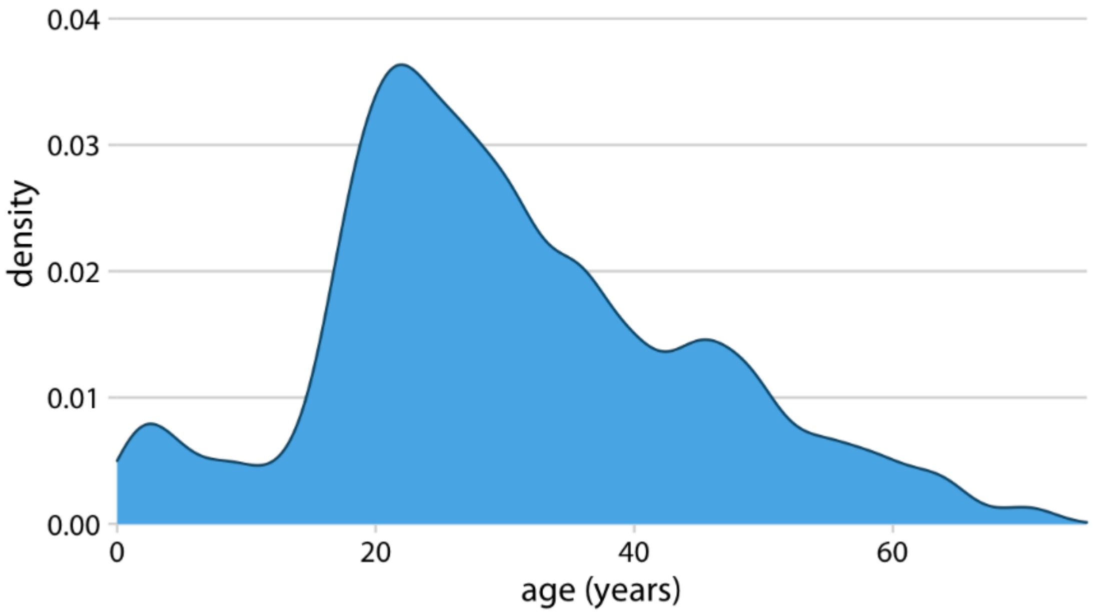




Sourcea

# Density curve

In a density plot, we attempt to visualize the underlying probability distribution of the data by drawing an appropriate continuous curve




https://clauswilke.com/dataviz/

# Pie charts

**A pie chart shows how a total amount is divided between levels of a categorical variable as a circle divided into radial slices.**

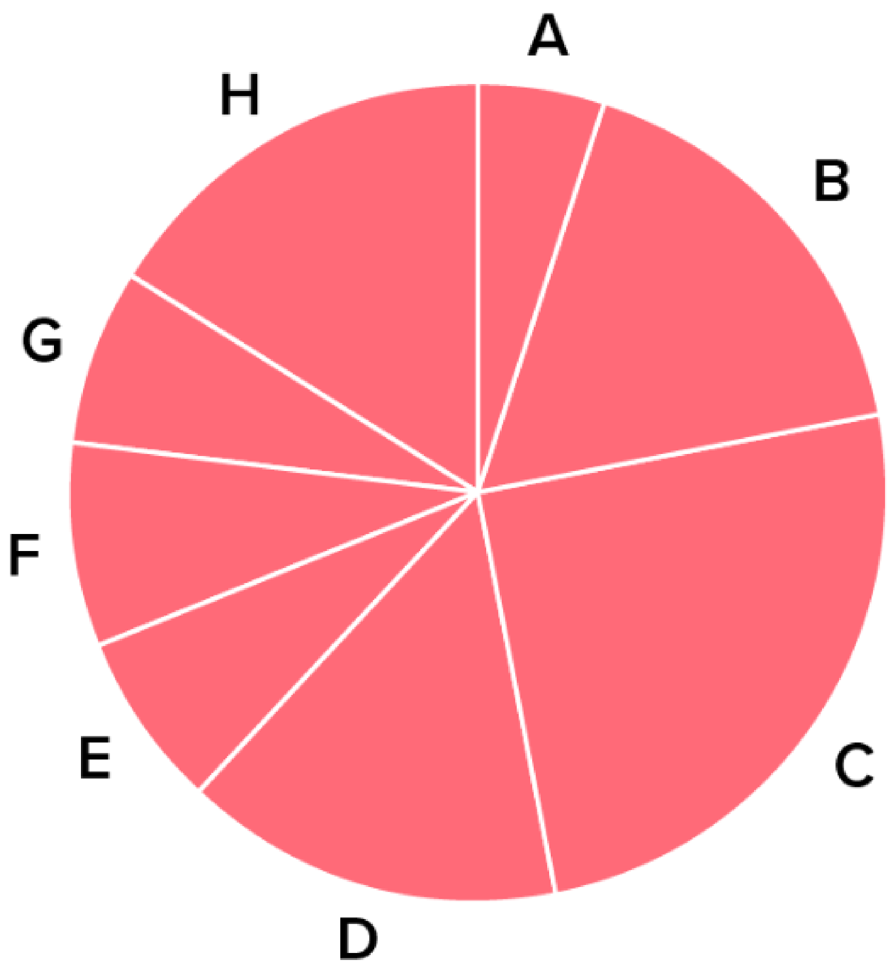
Your primary objective in a pie chart should be to compare each group's contribution to the whole, as opposed to comparing groups to each other

HOWEVER, TRY TO AVOID THEM

Table 10.1: Pros and cons of common approaches to visualizing proportions: pie charts, stacked bars, and side-by-side bars.

| | Pie chart | Stacked bars | Side-by-side bars |
|---|---|---|---|
| Clearly visualizes the data as proportions of a whole | ✓ | ✓ | × |
| Allows easy visual comparison of the relative proportions | × | × | ✓ |
| Visually emphasizes simple fractions, such as 1/2, 1/3, 1/4 | ✓ | × | × |
| Looks visually appealing even for very small datasets | ✓ | × | ✓ |
| Works well when the whole is broken into many pieces | × | × | ✓ |
| Works well for the visualization of many sets of proportions or time series of proportions | × | ✓ | × |

# Pie charts




Which is the third largest segment in the pie chart?

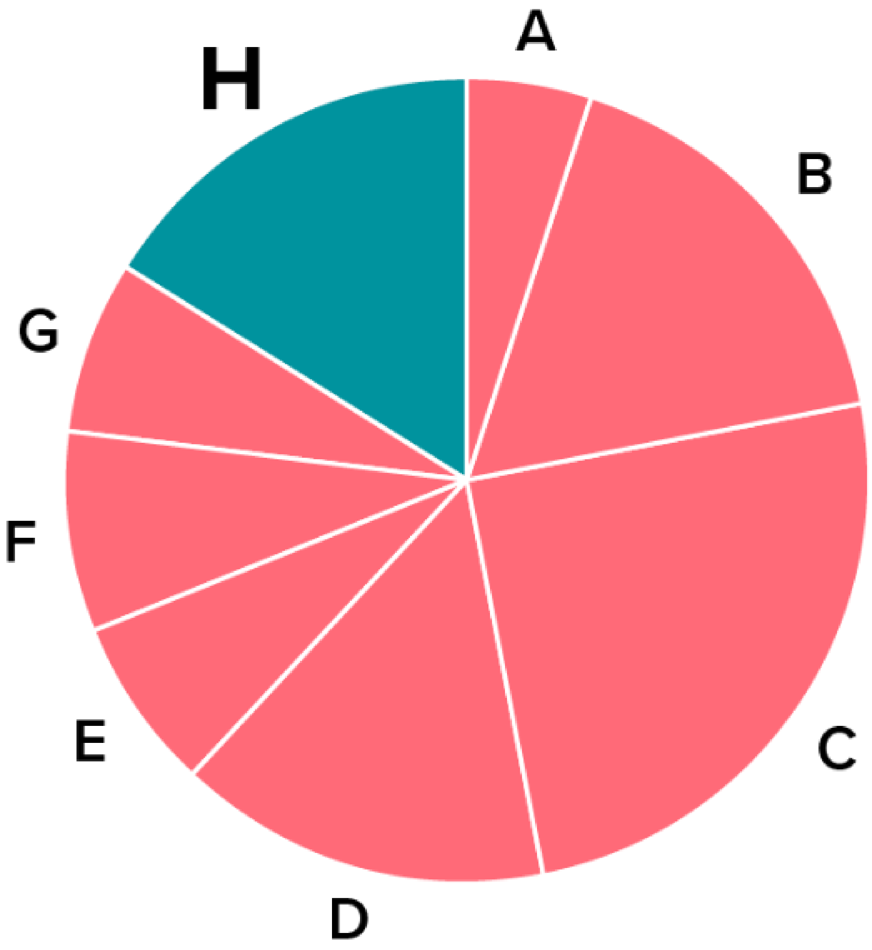
A

H

B

D

# Pie charts




Which is the third largest segment in the pie chart?

A

H

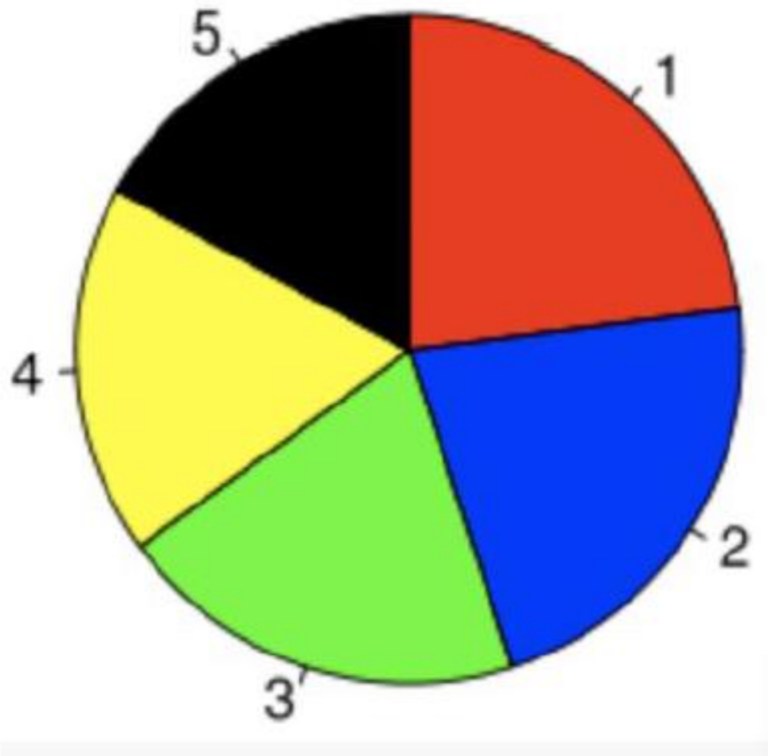
B

D

# Pie charts

You have 5 seconds to find the largest part

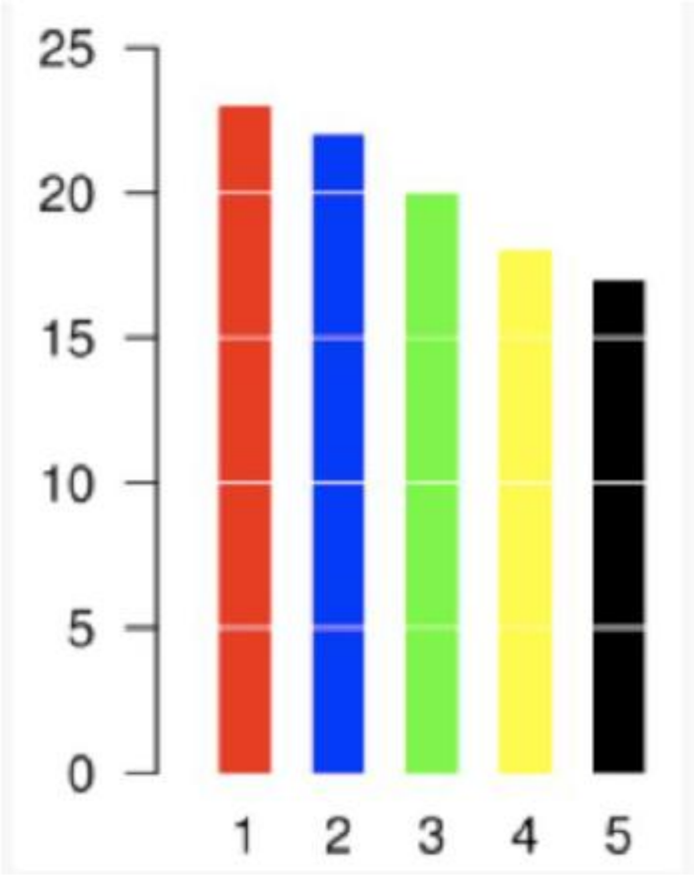
# Pie charts



# Pie charts

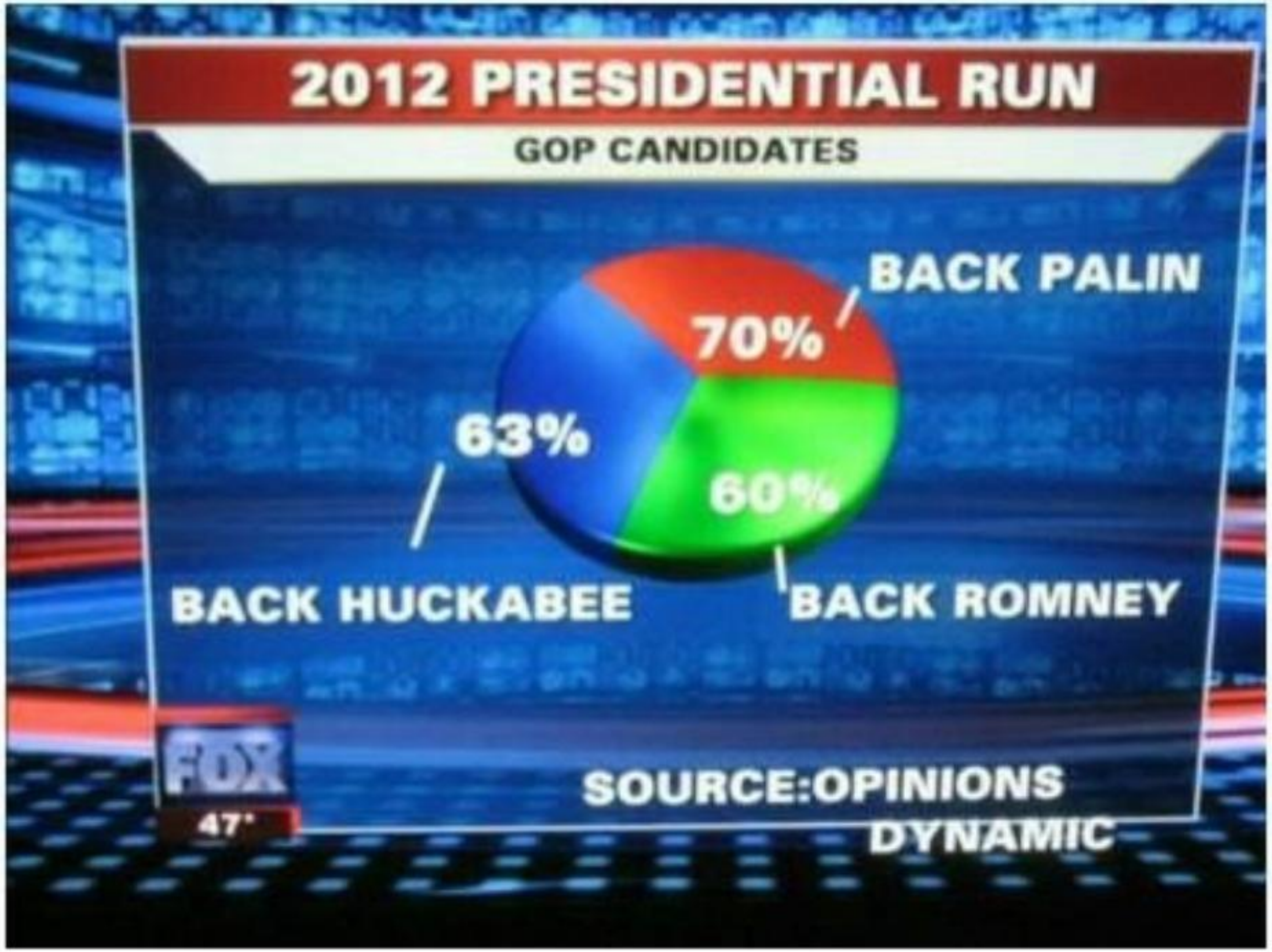
You have 5 seconds to find the largest part

# Pie charts



# Pie charts

Fox News Election 2012 Presidential Run GOP Candidates

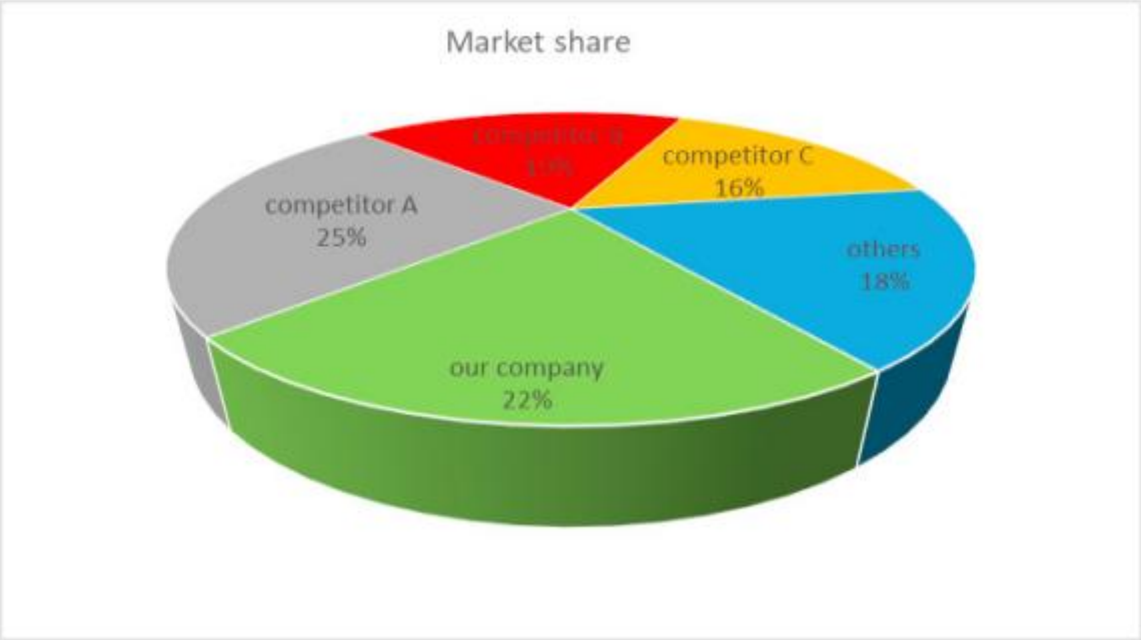


# Pie charts

**1) The parts of a pie chart must always be equal to 100%.**

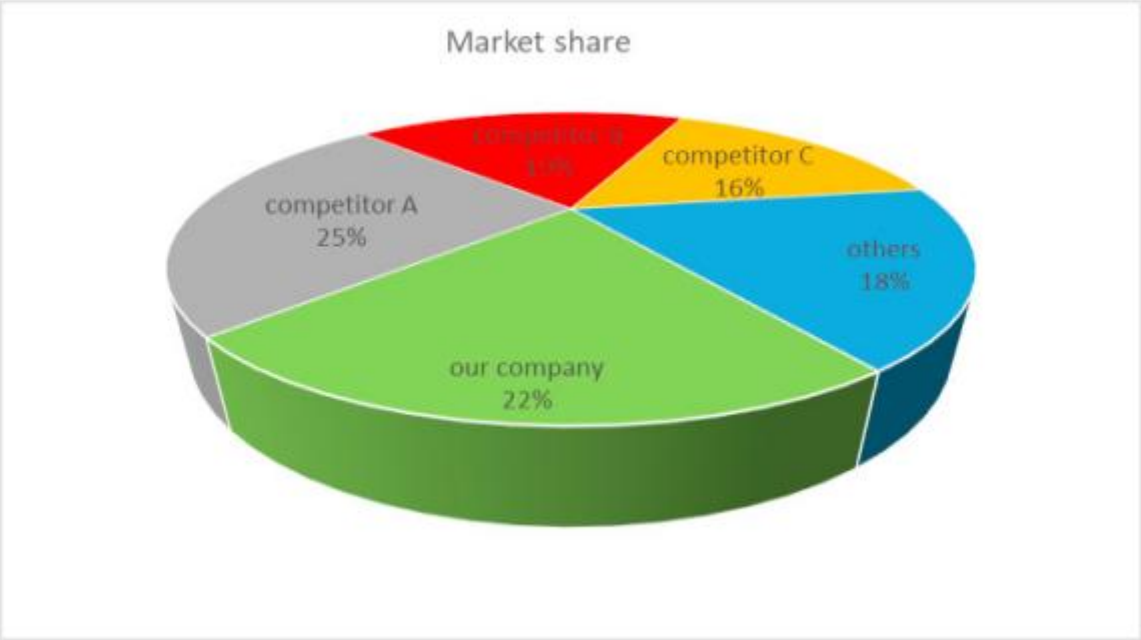
# Pie charts

What do you conclude?



# Pie charts

What do you conclude?



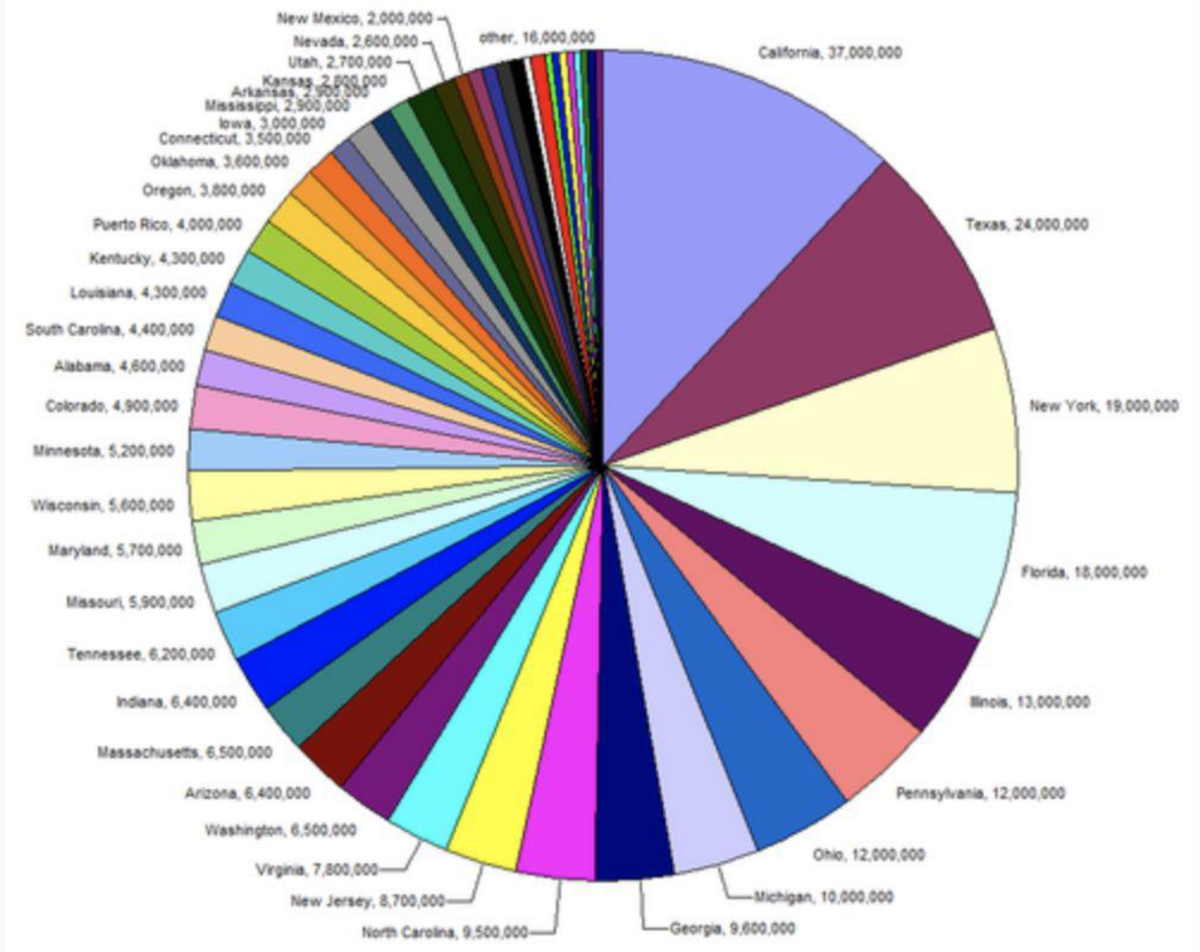
# Pie charts

**1) The parts of a pie chart must always be equal to 100%.**

**2) Do not use 3D:** Difficult to read the right values

# Pie charts

What do you conclude?



# Pie charts

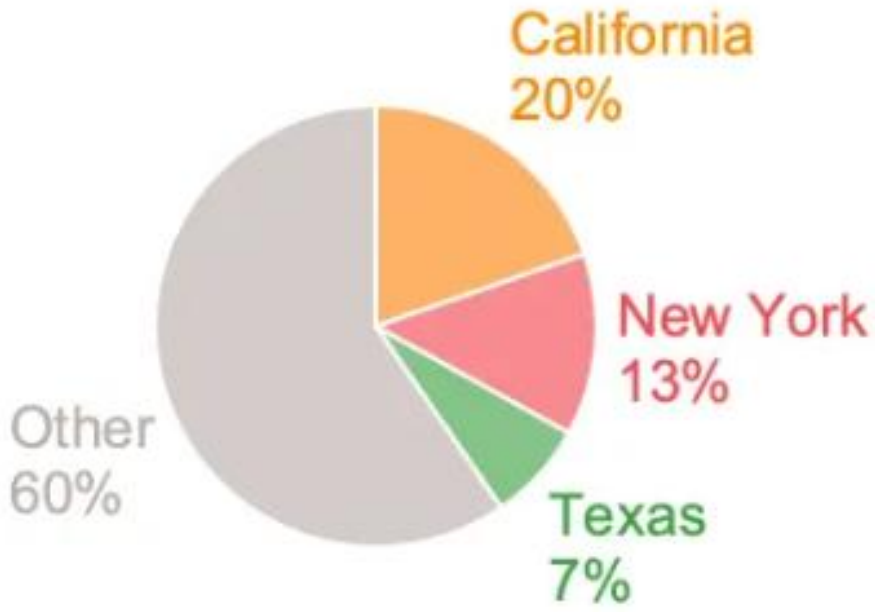
1) **The parts of a pie chart must always be equal to 100%.**

2) **Do not use 3D:** Difficult to read the right values

3) **No more than 4-5 items**

# Pie charts

What do you see?



# Pie charts

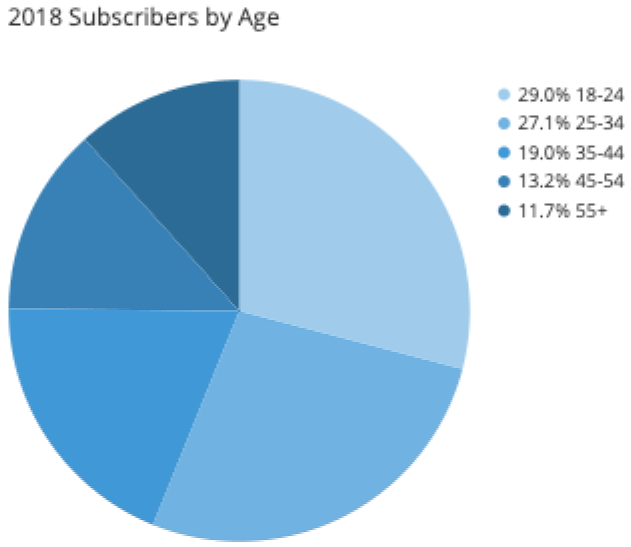
1) **The parts of a pie chart must always be equal to 100%.**

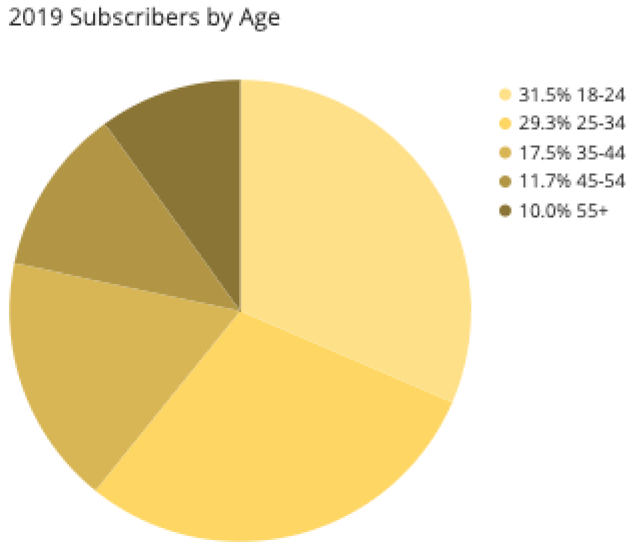
2) **Do not use 3D:** Difficult to read the right values

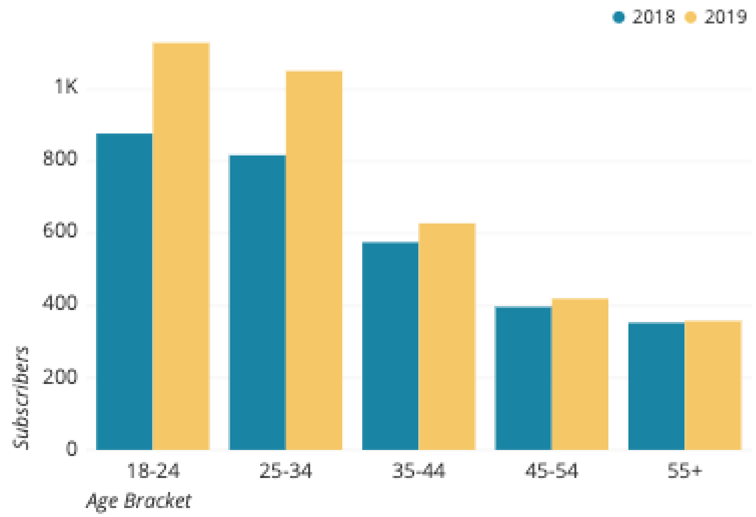
3) **No more than 4-5 items**

4) **Order the slices**

# Pie charts


2018 Subscribers by Age
29.0% 18-24
27.1% 25-34
19.0% 35-44
13.2% 45-54
11.7% 55+


2019 Subscribers by Age
31.5% 18-24
29.3% 25-34
17.5% 35-44
11.7% 45-54
10.0% 55+


2018
2019
1K
800
600
400
200
0
Subscribers
18-24
25-34
35-44
45-54
55+
Age Bracket

# Pie charts

1) **The parts of a pie chart must always be equal to 100%.**

2) **Do not use 3D:** Difficult to read the right values

3) **No more than 4-5 items**

4) **Order the slices**

5) **Do not use them for comparison**

# Pie charts

1) **The parts of a pie chart must always be equal to 100%.**

2) **Do not use 3D:** Difficult to read the right values

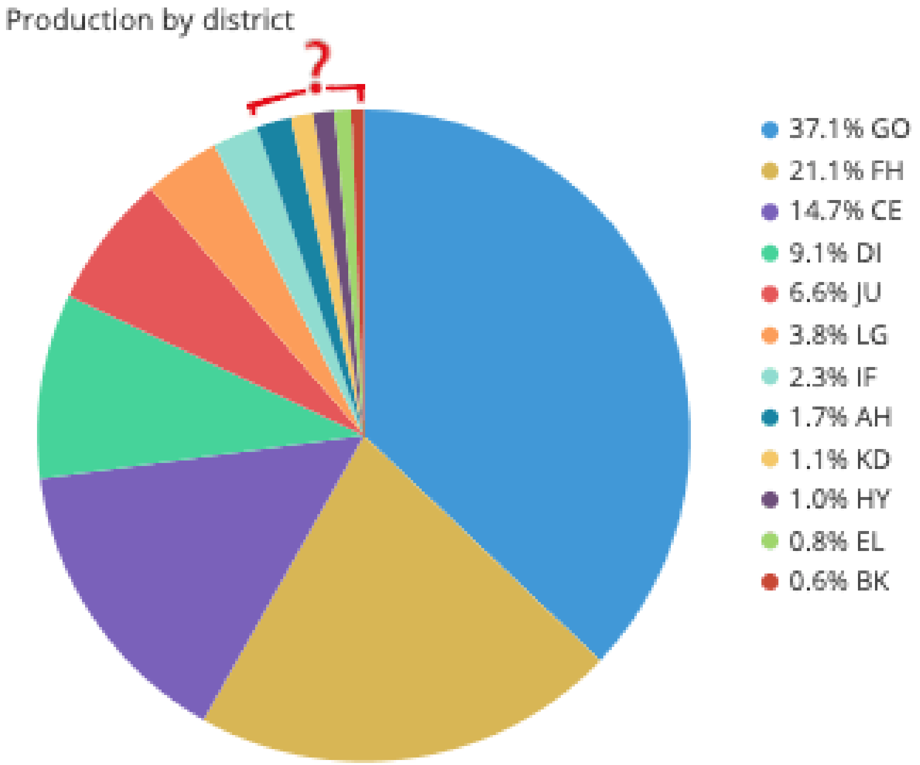
3) **No more than 4-5 items**

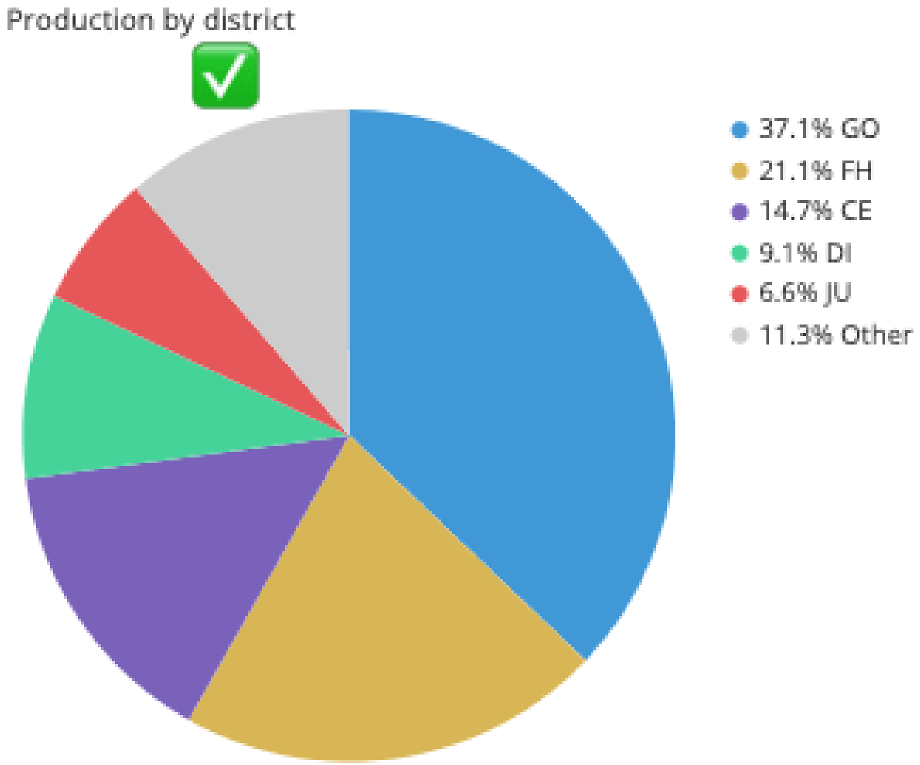
4) **Order the slices**

5) **Do not use them for comparison**

6) **Label values directly; don’t create an additional legend**

# Pie charts


Production by district
?
37.1% GO
21.1% FH
14.7% CE
9.1% DI
6.6% JU
3.8% LG
2.3% IF
1.7% AH
1.1% KD
1.0% HY
0.8% EL
0.6% BK


Production by district
37.1% GO
21.1% FH
14.7% CE
9.1% DI
6.6% JU
11.3% Other

# Pie charts

**1) The parts of a pie chart must always be equal to 100%.**

2) **Do not use 3D:** Difficult to read the right values

3) **No more than 4-5 items**

4) **Order the slices**

5) **Do not use them for comparison**

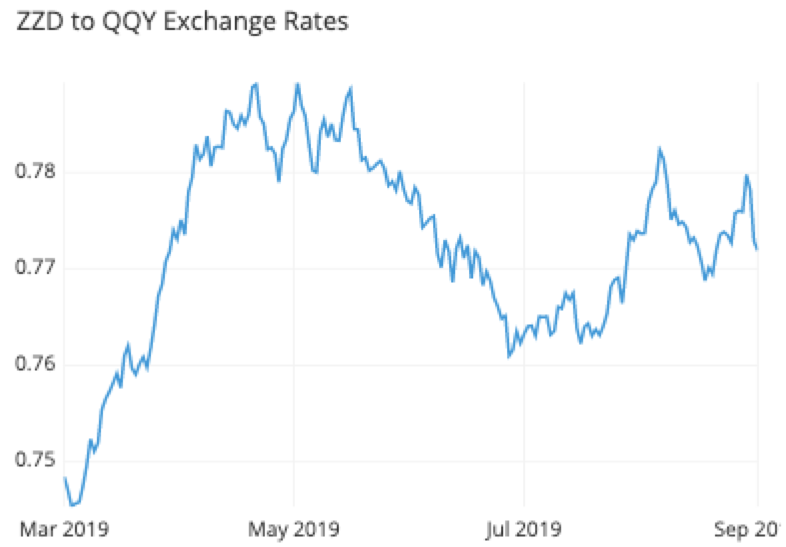
6) **Label values directly; don’t create an additional legend**

7) **Try to group small categories into ‘Other’**

# Line charts

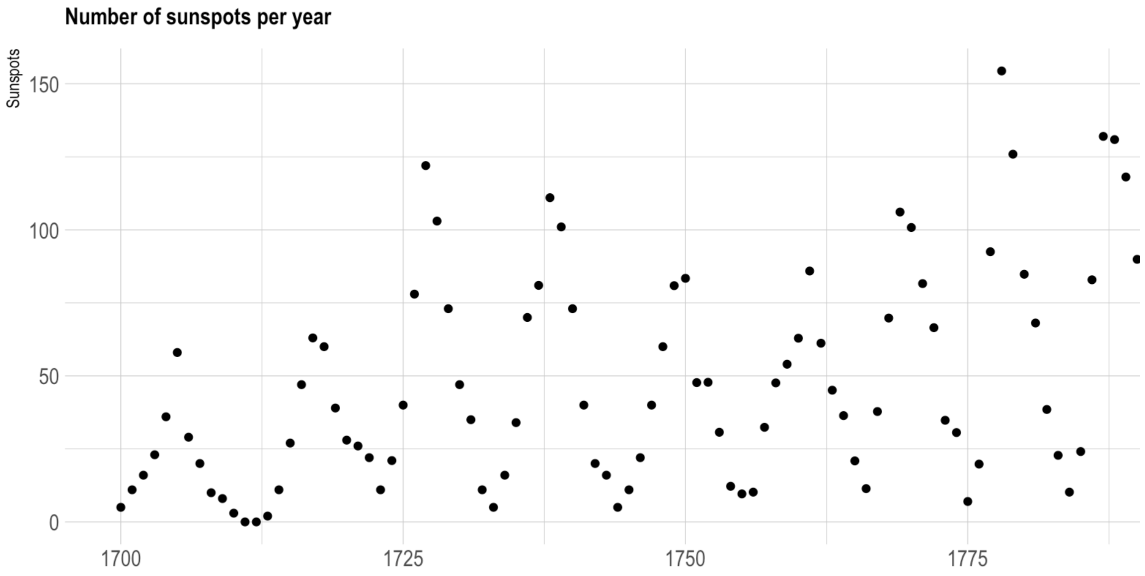
**A line chart uses points connected by line segments from left to right to demonstrate changes in value. The horizontal axis depicts a continuous progression, often that of time, while the vertical axis reports values for a metric of interest across that progression.**

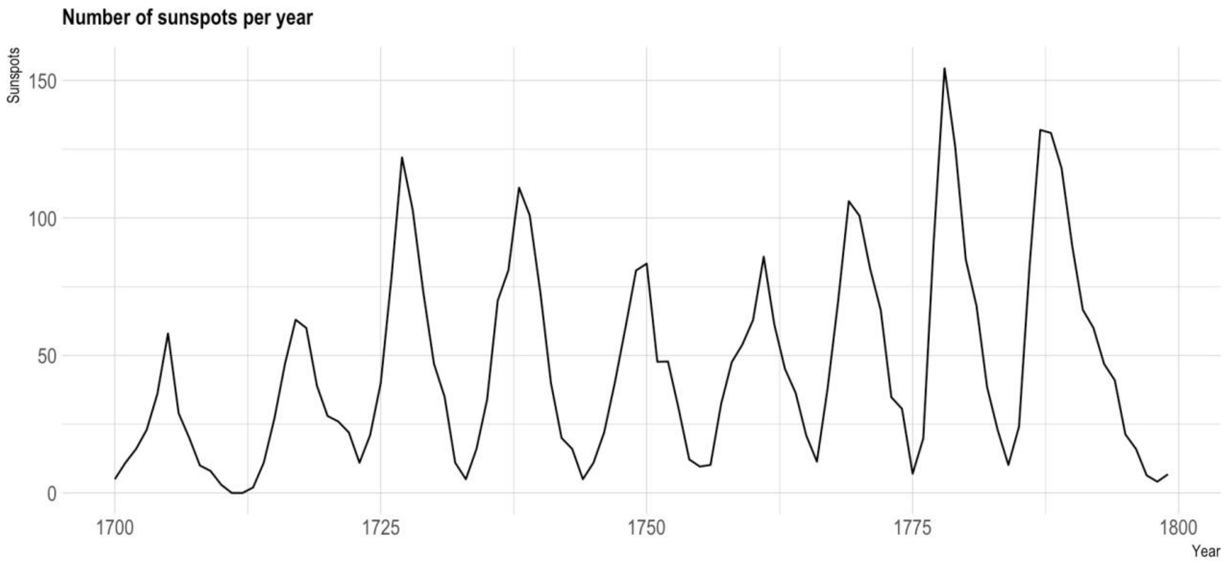
Your primary objective is to emphasize changes in values for one variable for **continuous values** of a second variable.



# Line charts

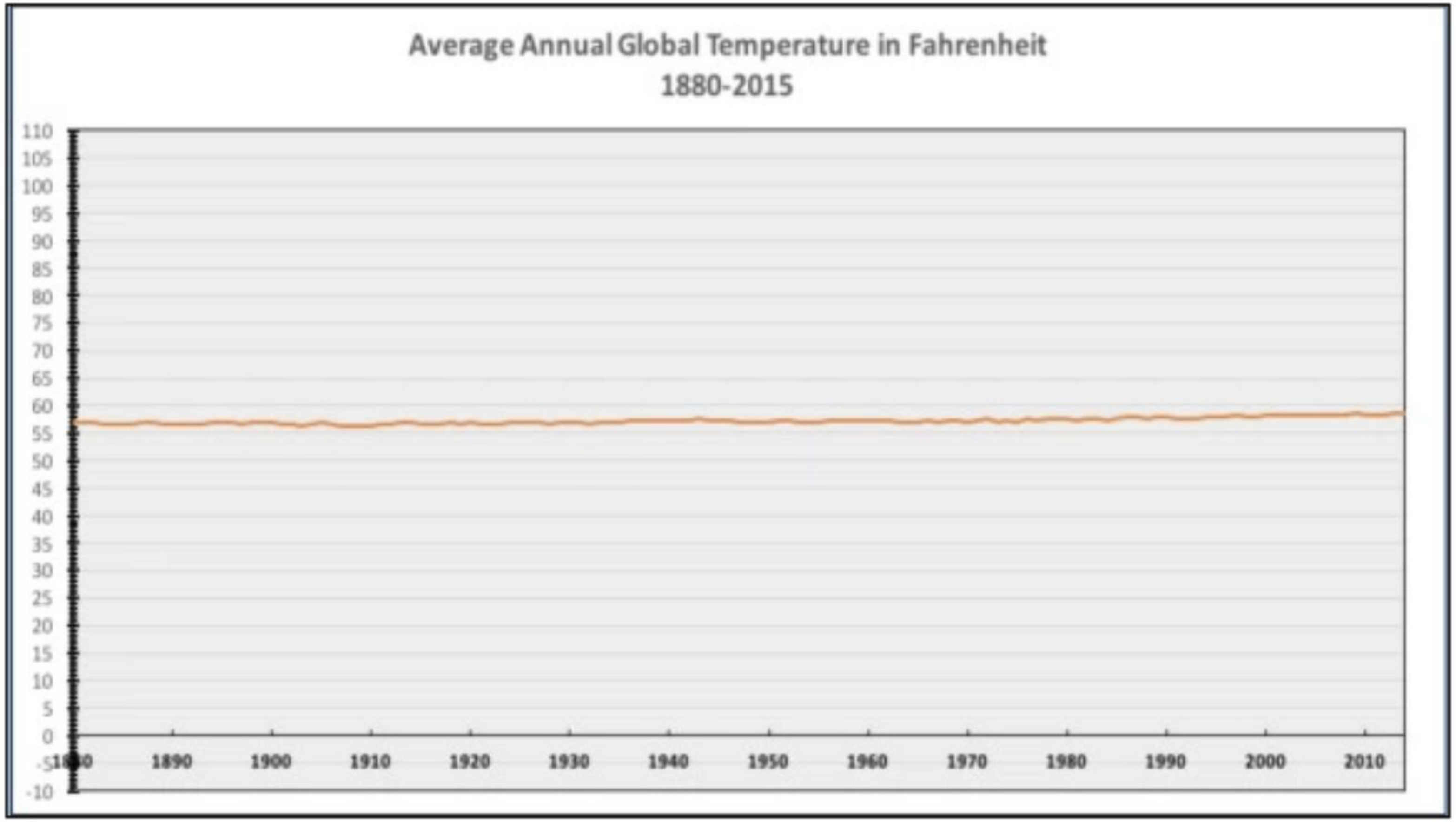
**CONNECT YOUR DOT!!**



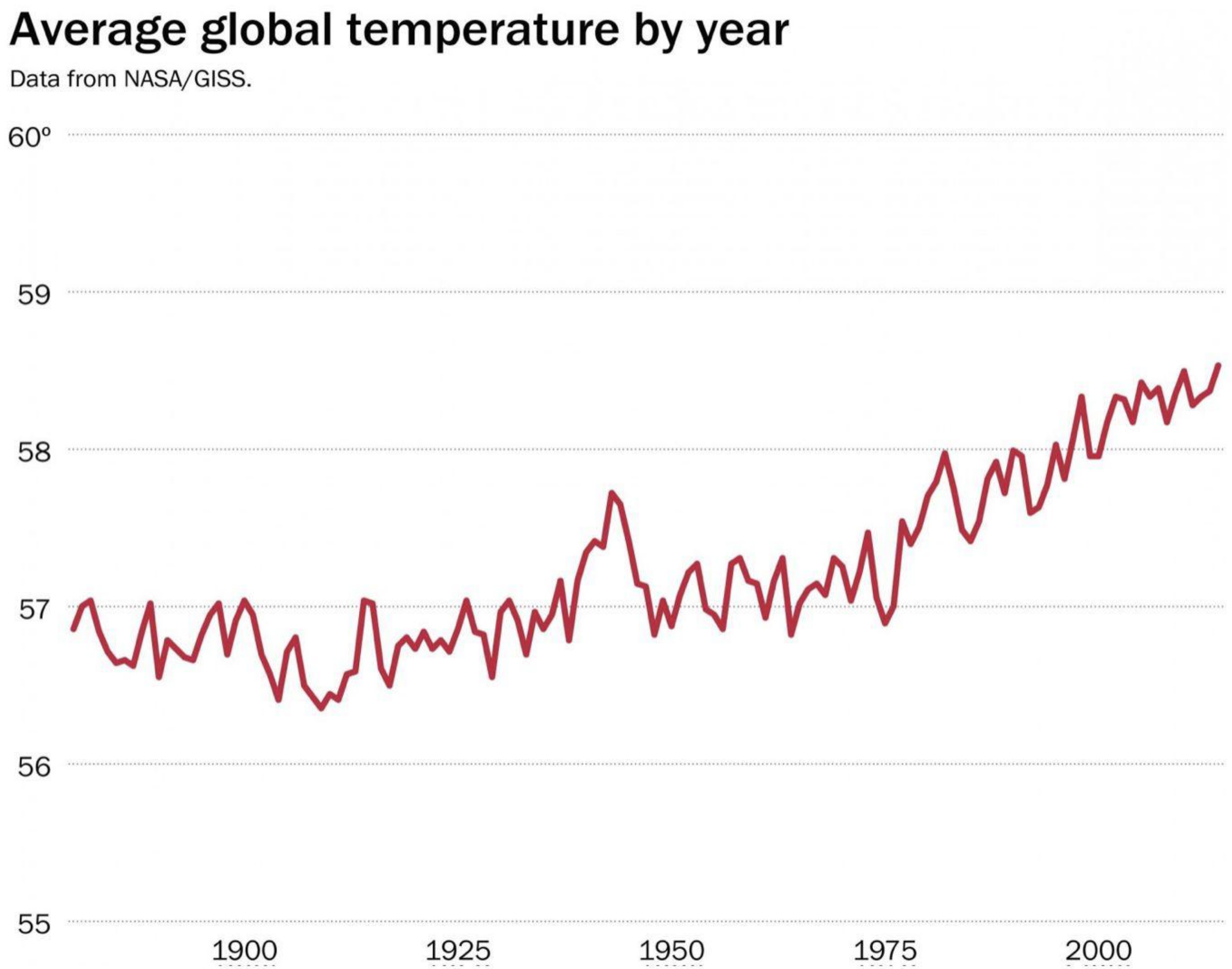
# Line charts

What do you conclude?



# Line charts

What do you conclude?

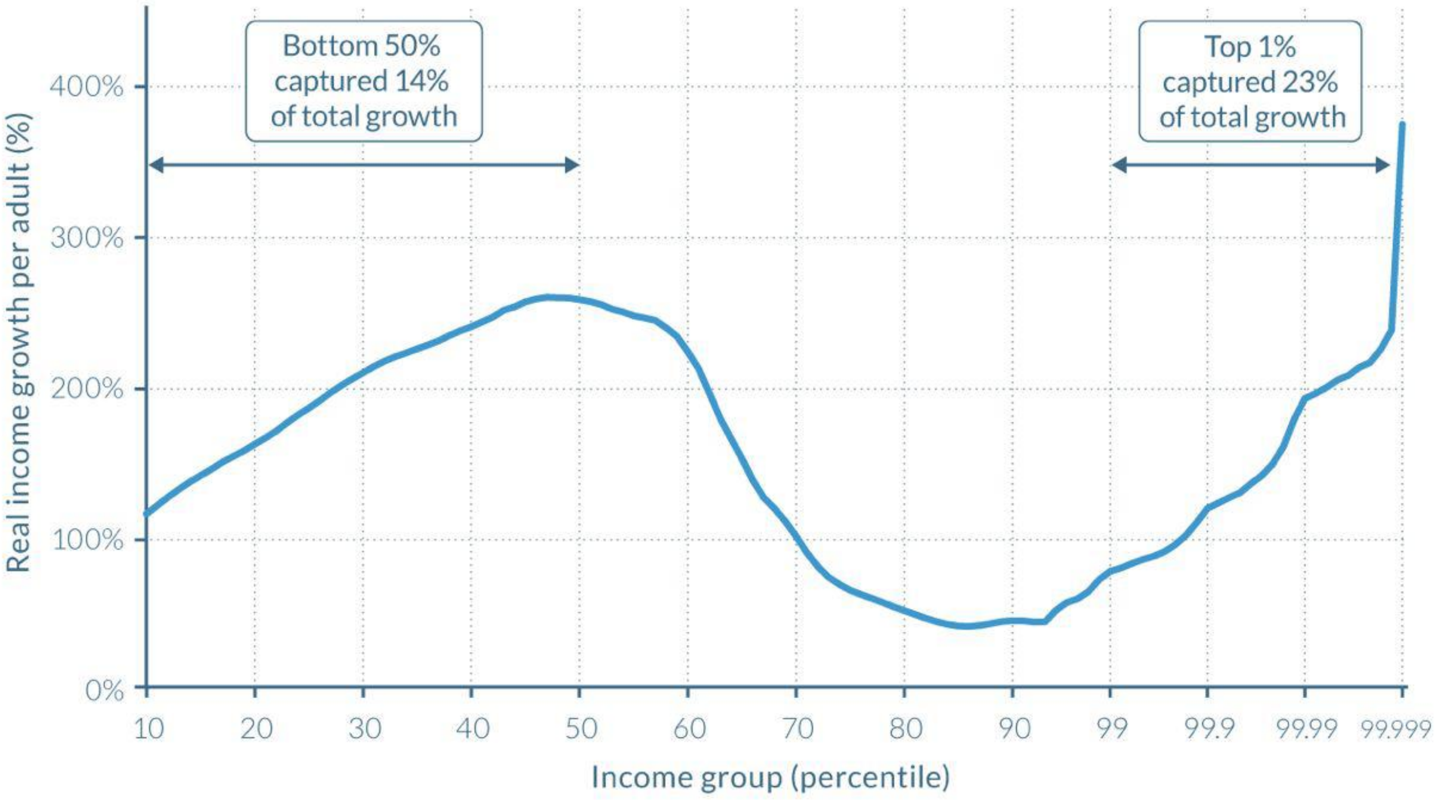


# Line charts

**1) no strict convention for starting the y axis at 0.** By starting the axis at 0 we can also **hide** information (same effect if we do not handle outliers appropriately in our data).

# Line charts

What do you see?




Source: WID.world (2017). See wir2018.wid.world/methodology.html for data series and notes.

On the horizontal axis, the world population is divided into a hundred groups of equal population size and sorted in ascending order from left to right, according to each group's income level. The Top 1% group is divided into ten groups, the richest of these groups is also divided into ten groups, and the very top group is again divided into ten groups of equal population size. The vertical axis shows the total income growth of an average individual in each group between 1980 and 2016. For percentile group p99p99.1 (the poorest 10% among the world's richest 1%), growth was 77% between 1980 and 2016. The Top 1% captured 23% of total growth over this period. Income estimates account for differences in the cost of living between countries. Values are net of inflation.

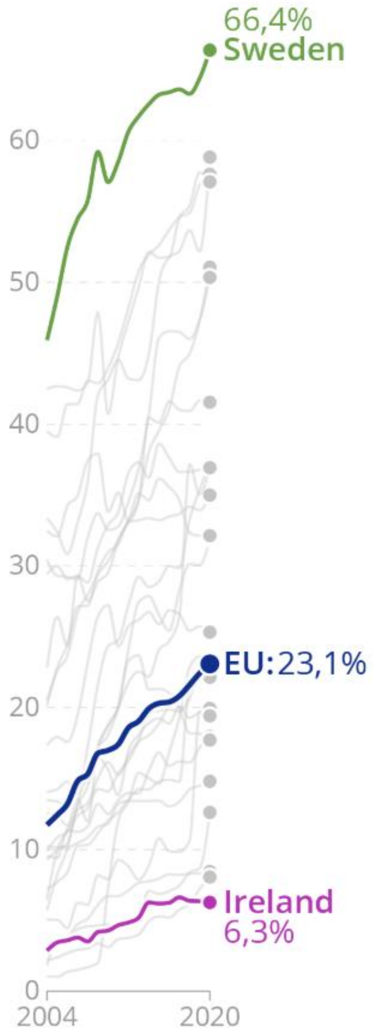
# Line charts

**1) no strict convention for starting the y axis at 0.** By starting the axis at 0 we can also **hide** information (same effect if we do not handle outliers appropriately in our data).

**2) scale** you choose on your axes must **remain consistent.**

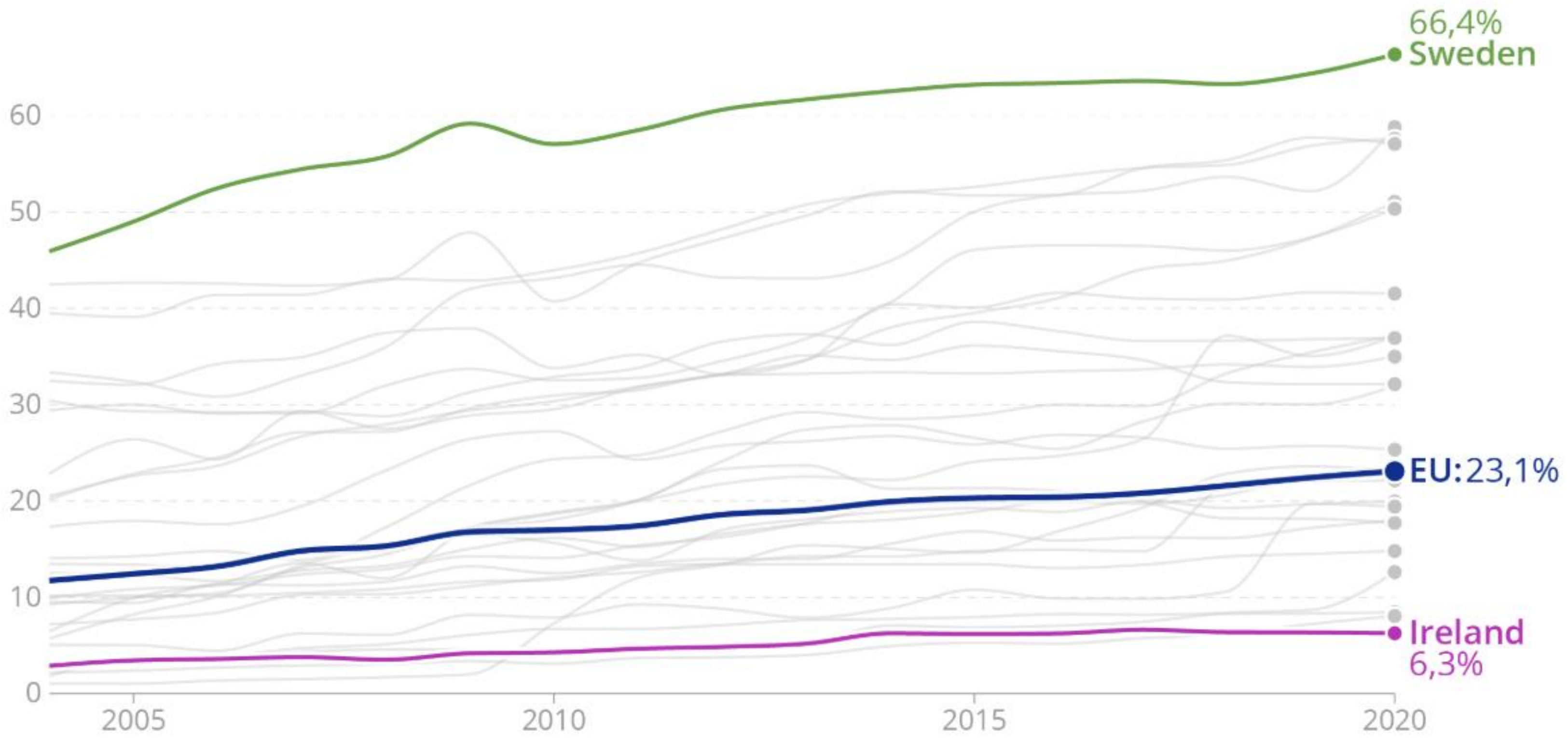
# Line charts

What do you conclude?

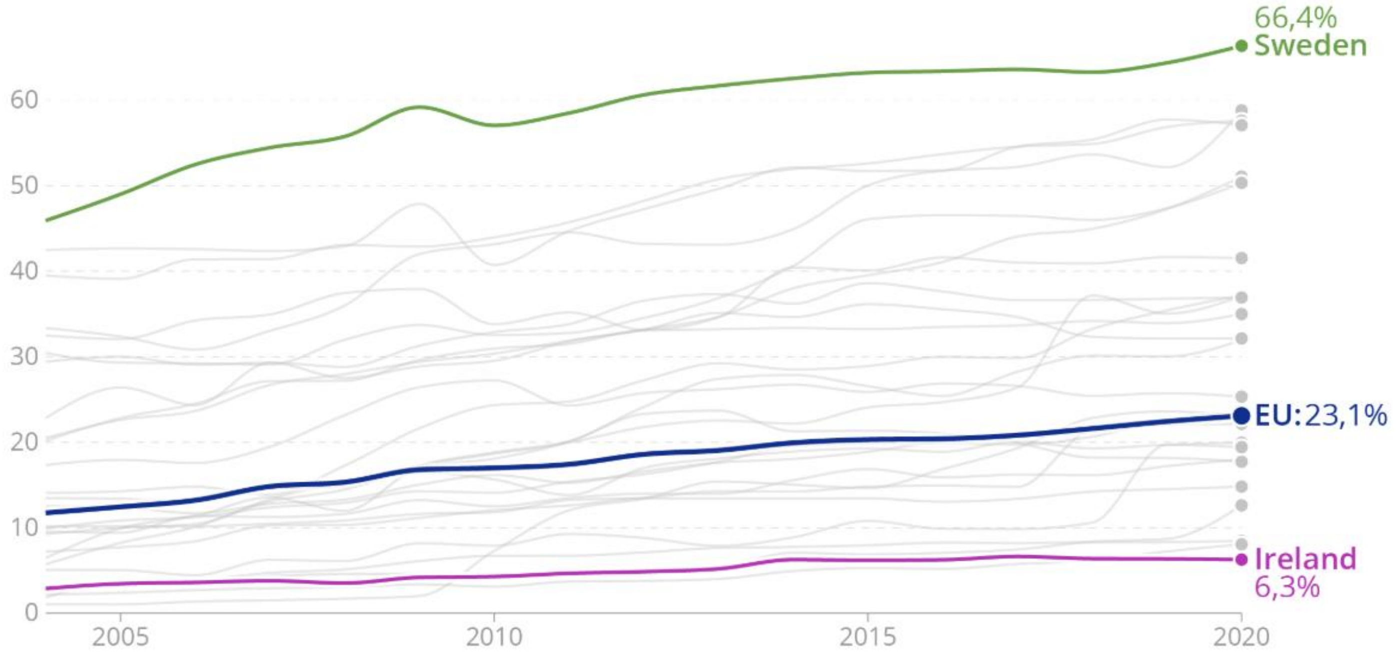


# Line charts

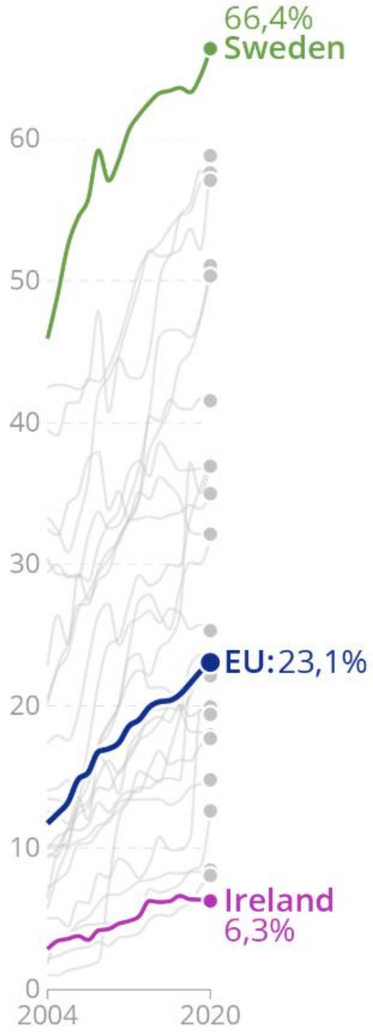
What do you conclude?



# Line charts




*A wide and flat layout suggest slow trends or even flat lines. Source: Maarten Lambrechts, CC BY 4.0*




*Narrow and high designs exaggerate trends. Source: Maarten Lambrechts, CC BY 4.0*

# Line charts

**1) no strict convention for starting the y axis at 0.** By starting the axis at 0 we can also **hide** information (same effect if we do not handle outliers appropriately in our data).

**2) scale** you choose on your axes must **remain consistent.**

**3) Respect the length/weight ratio of your graph**

# Line charts

**1) no strict convention for starting the y axis at 0.** By starting the axis at 0 we can also **hide** information (same effect if we do not handle outliers appropriately in our data).

**2) scale** you choose on your axes must **remain consistent.**

**3) Respect the length/weight ratio of your graph**

**4) Time goes from Left to Right**

# Line charts

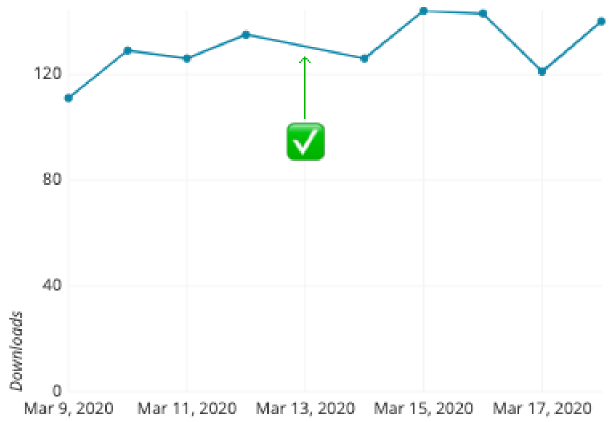
**1) no strict convention for starting the y axis at 0.** By starting the axis at 0 we can also **hide** information (same effect if we do not handle outliers appropriately in our data).

**2) scale** you choose on your axes must **remain consistent.**

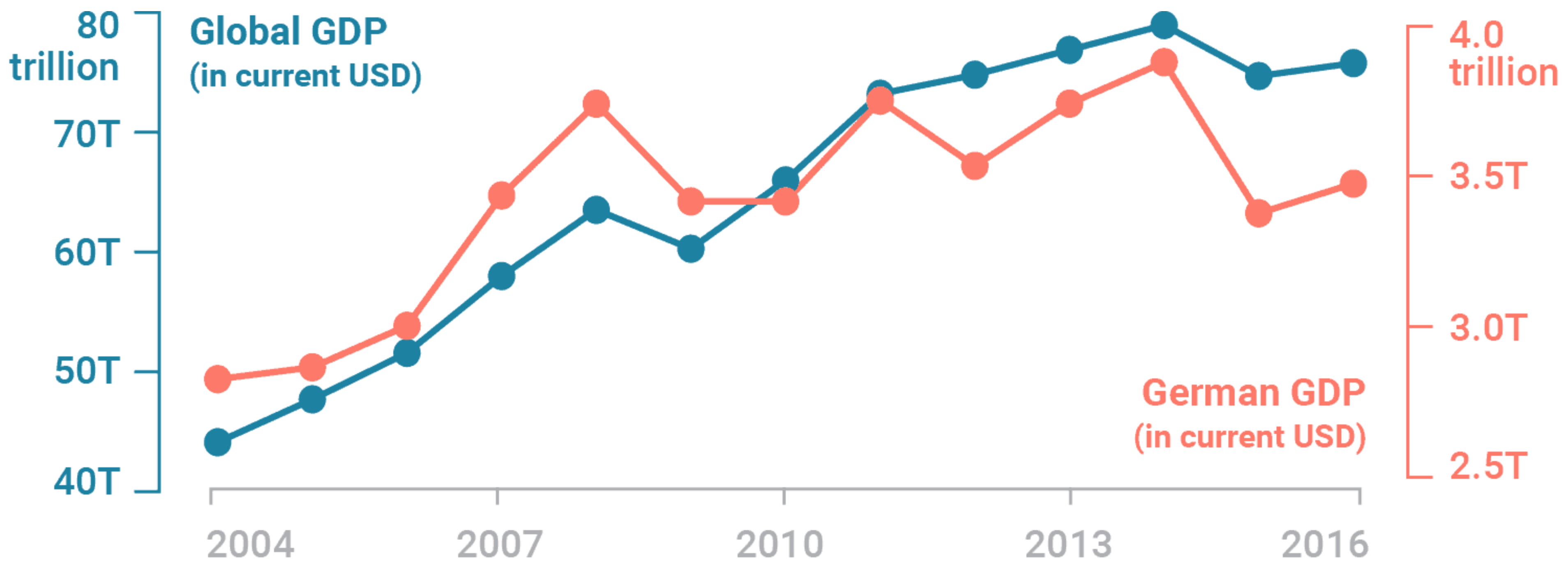
**3) Respect the length/weight ratio of your graph**

**4) Time goes from Left to Right**

**5) If you have missing data, make it clear from the chart or show dots**

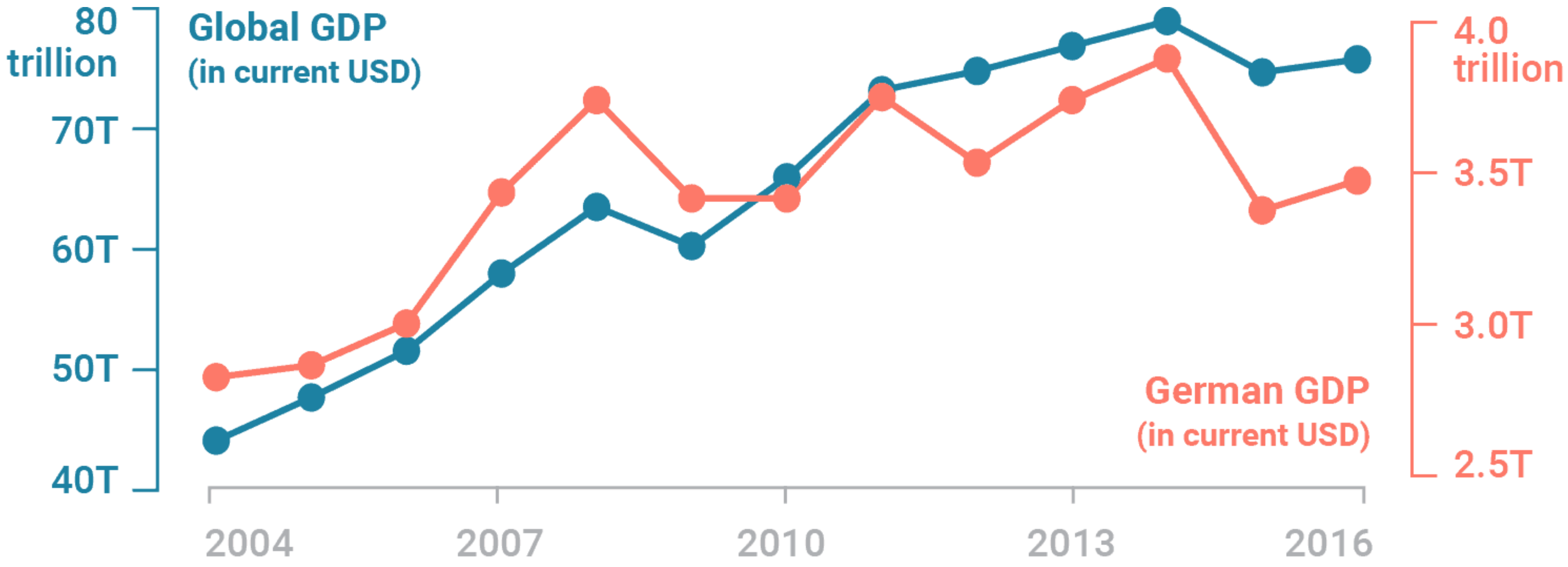
# Line charts



80
trillion
70T
60T
50T
40T
Global GDP
(in current USD)
4.0
trillion
3.5T
3.0T
2.5T
German GDP
(in current USD)
2004
2007
2010
2013
2016

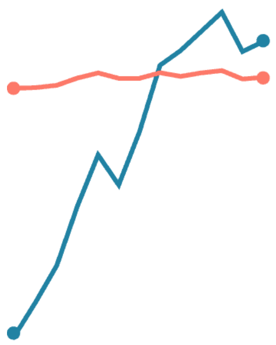
# Line charts

**AVOID DUAL SCALE: The scales of dual axis charts are arbitrary and can therefore (deliberately) mislead readers about the relationship between the two data series.**

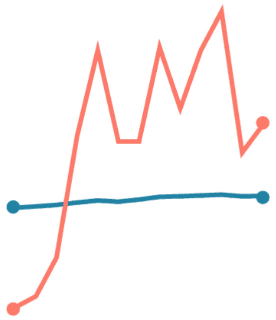



**the chart looks like** the German GDP and the global GDP go up at roughly the same rate (at least until 2014), they don't. The global GDP increased by 80% until 2014; the GDP of Germany by 40%

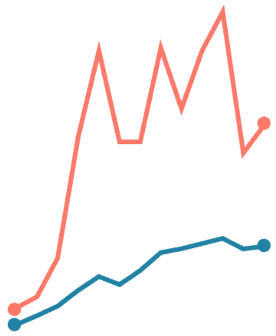
# Line charts



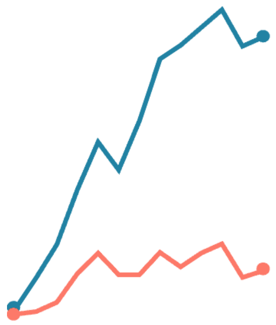
Orange steady,
Blue massively increasing.



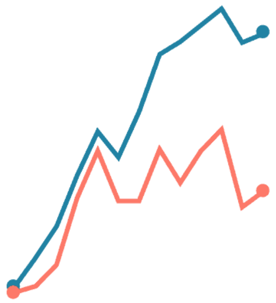
Blue steady,
Orange increasing.



Both started at the same level, but Orange increased far more than Blue.



Both started at the same level, but Blue increased far more than Orange.

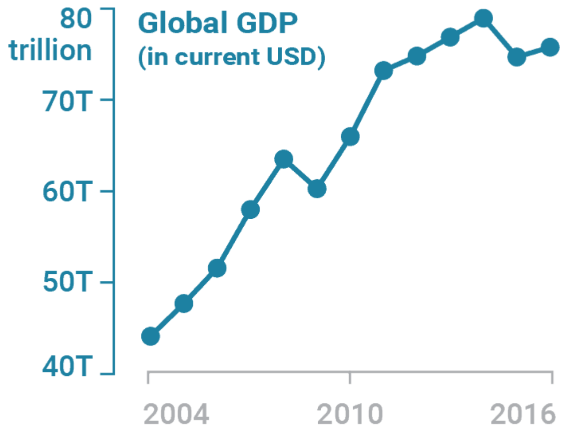


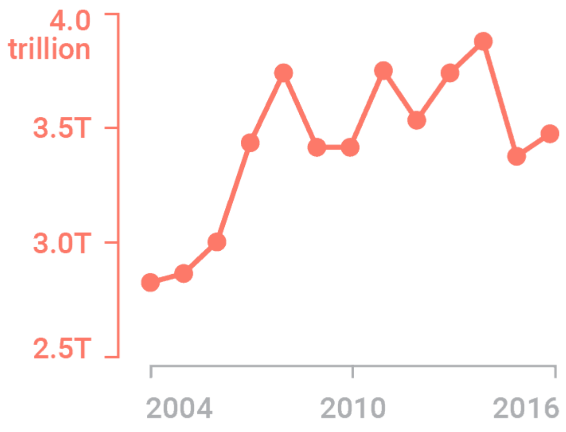
Both started with the same increase, then Blue raced to the top.

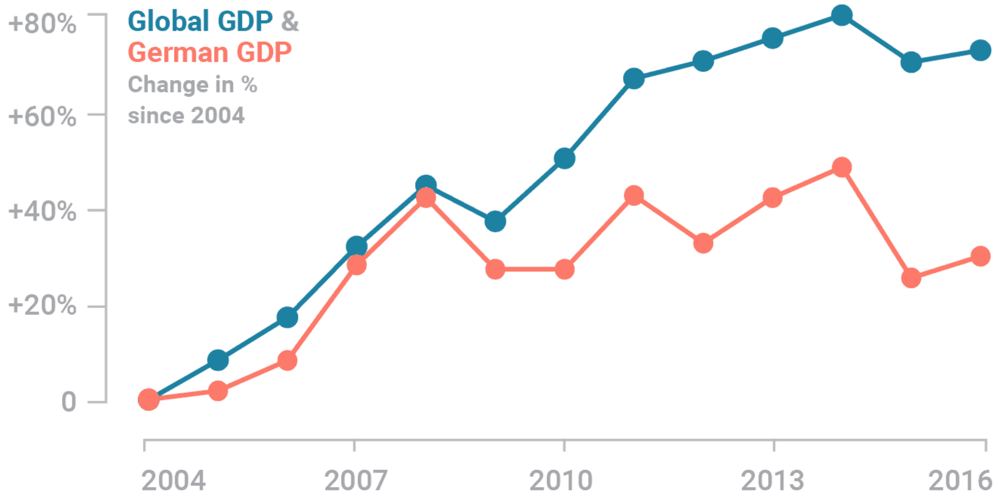
Both steady.

# Line charts

Solution



# Line charts

**1) no strict convention for starting the y axis at 0.** By starting the axis at 0 we can also **hide** information (same effect if we do not handle outliers appropriately in our data).

**2) scale** you choose on your axes must **remain consistent.**

**3) Respect the length/weight ratio of your graph**

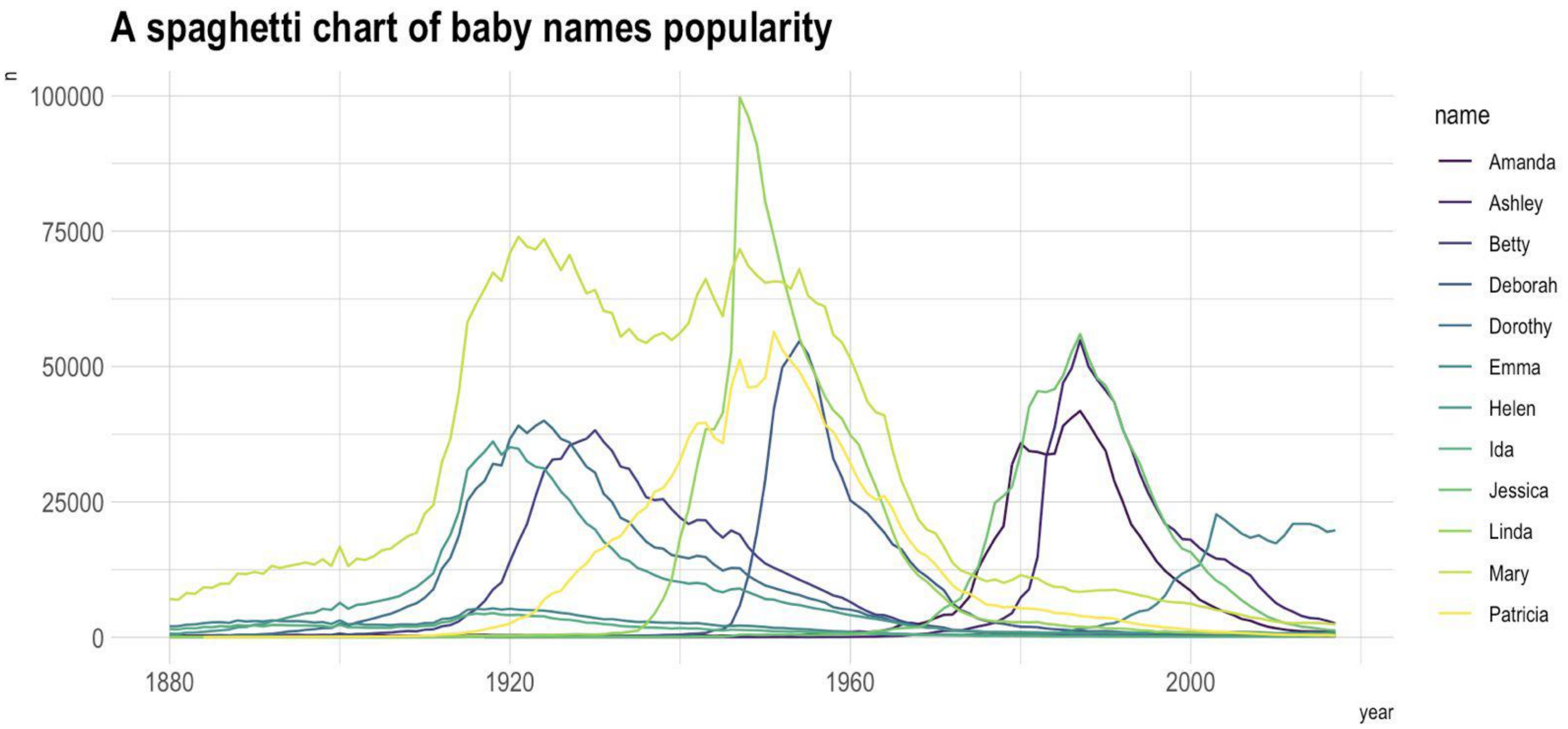
**4) Time goes from Left to Right**

**5) If you have missing data, make it clear from the chart or show dots**

**6) Avoid dual scales**

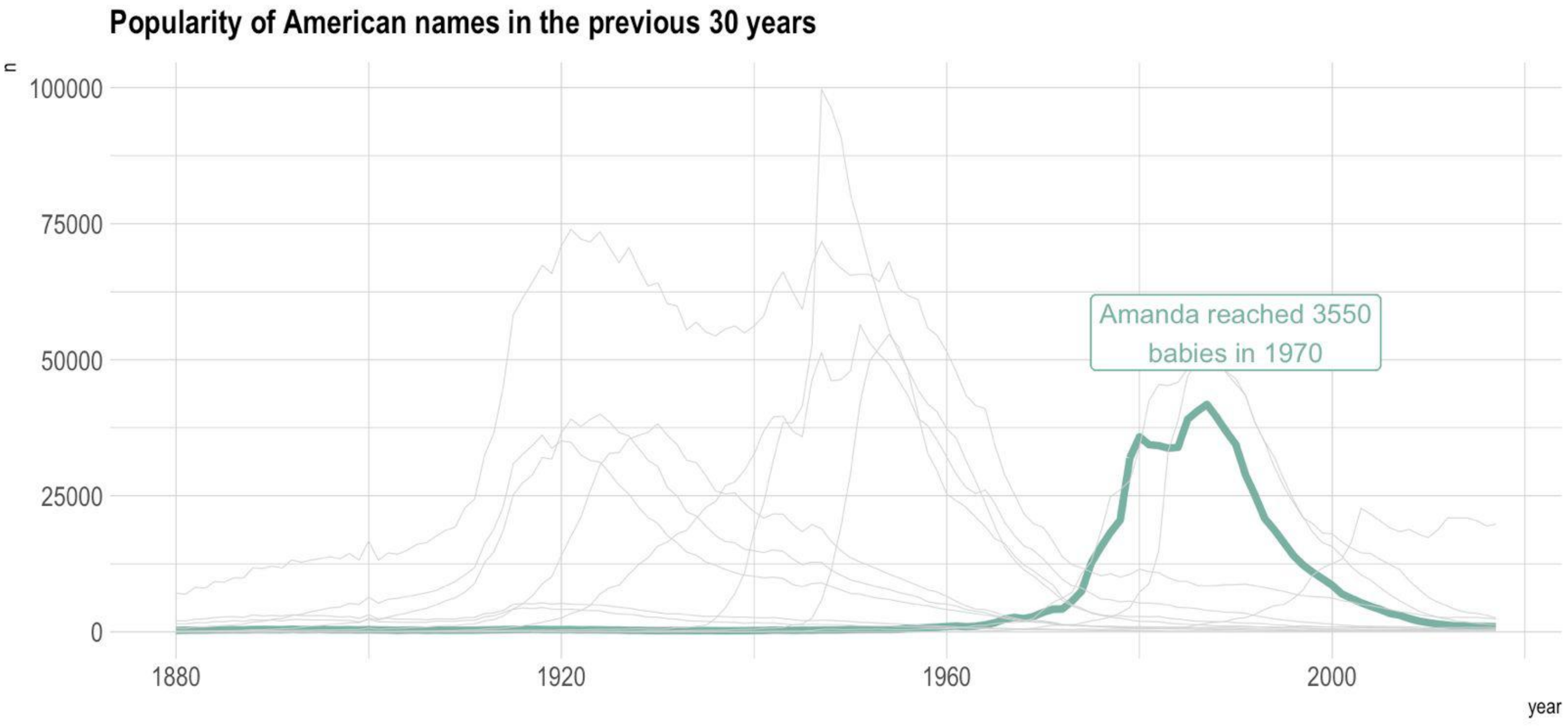
# Line charts

What do you see?



# Line charts

If you cannot reduce your categories, you can for example **gray** out all your curves, except the one you want to draw attention to, leaving it in color. This allows for a clear and readable narration.



# Line charts

**1) no strict convention for starting the y axis at 0.** By starting the axis at 0 we can also **hide** information (same effect if we do not handle outliers appropriately in our data).

**2) scale** you choose on your axes must **remain consistent.**

**3) Respect the length/weight ratio of your graph**

**4) Time goes from Left to Right**

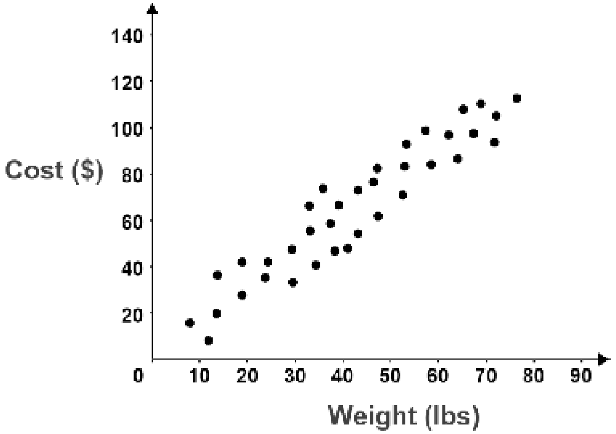
**5) If you have missing data, make it clear from the chart or show dots**

**6) Avoid dual scales**

**7) Limit the number of lines, highlight the one you want to draw attention**

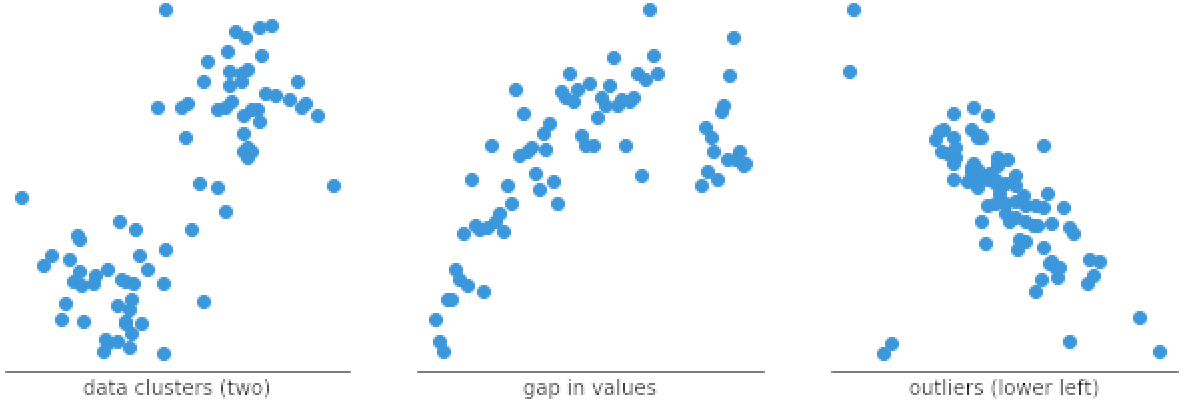
# Scatter charts

**A scatter plot (uses dots to represent values for two different numeric variables. The position of each dot on the horizontal and vertical axis indicates values for an individual data point. Scatter plots are used to observe relationships between variables.**



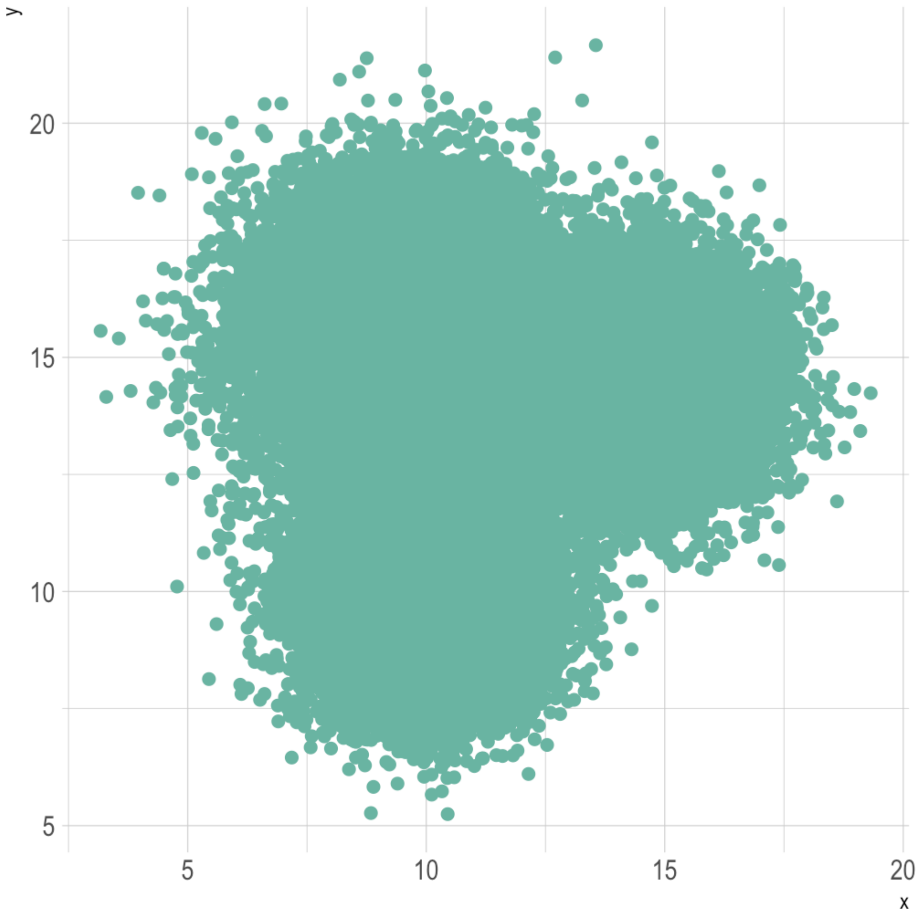

**Useful**

- to observe and show relationships between two numeric variables. Identification of correlational relationships are common with scatter plots.
- identifying other patterns in data

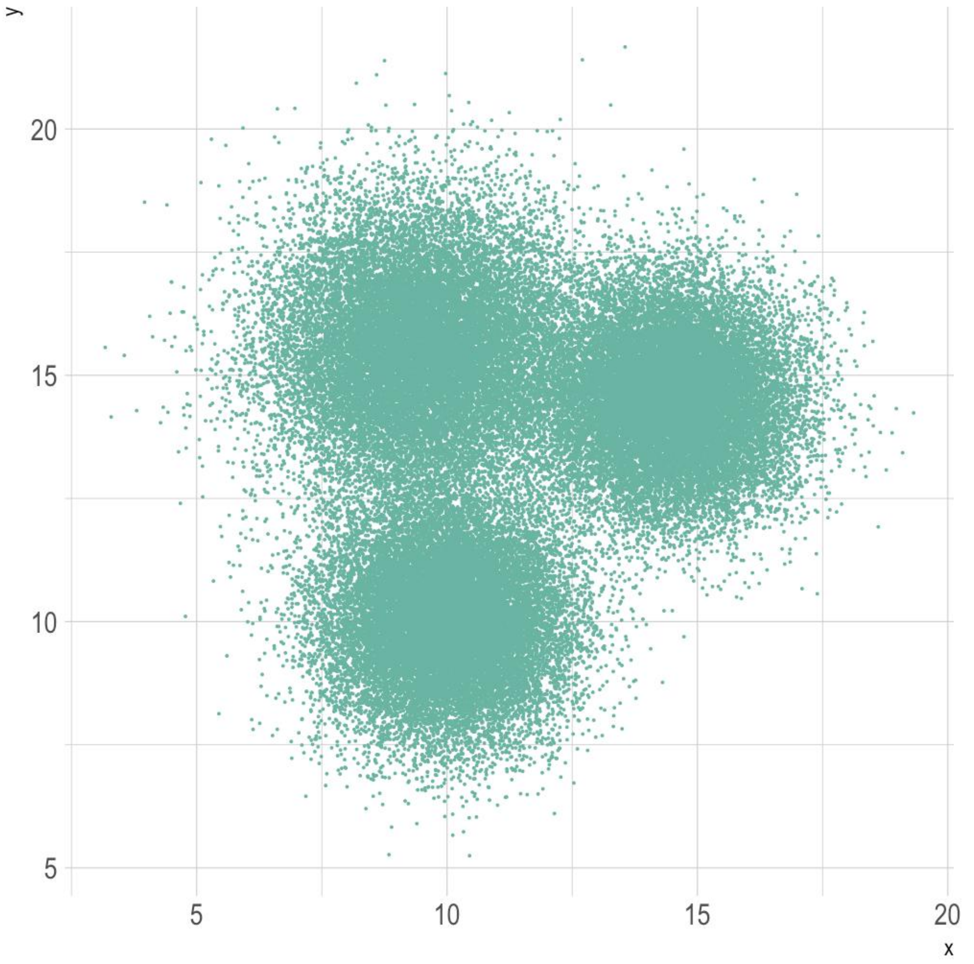


# Scatter charts

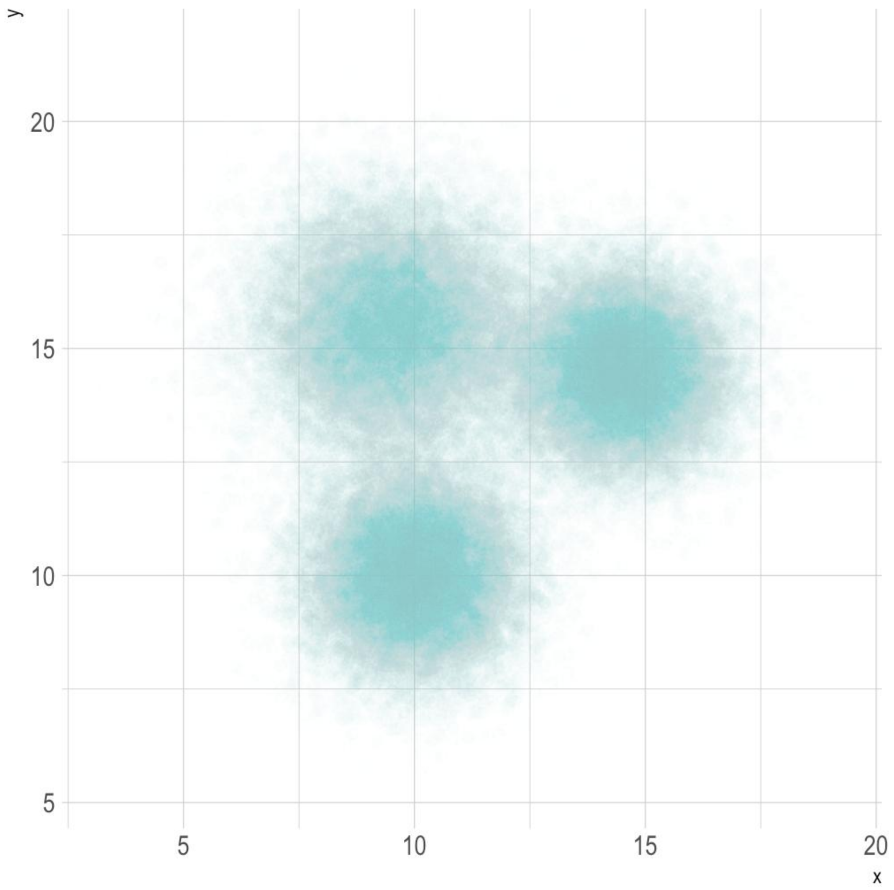
What to do?



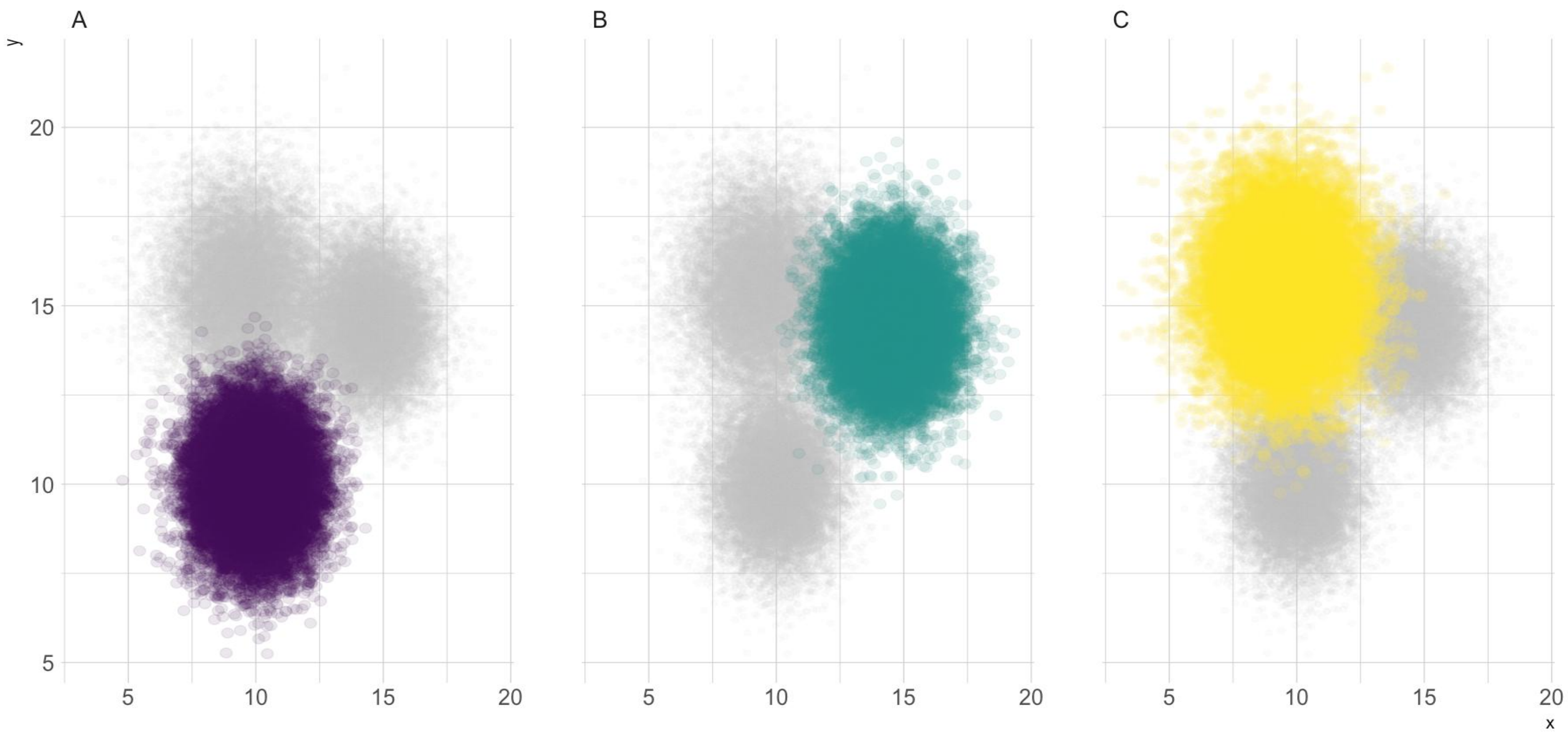
# Scatter charts



# Scatter charts

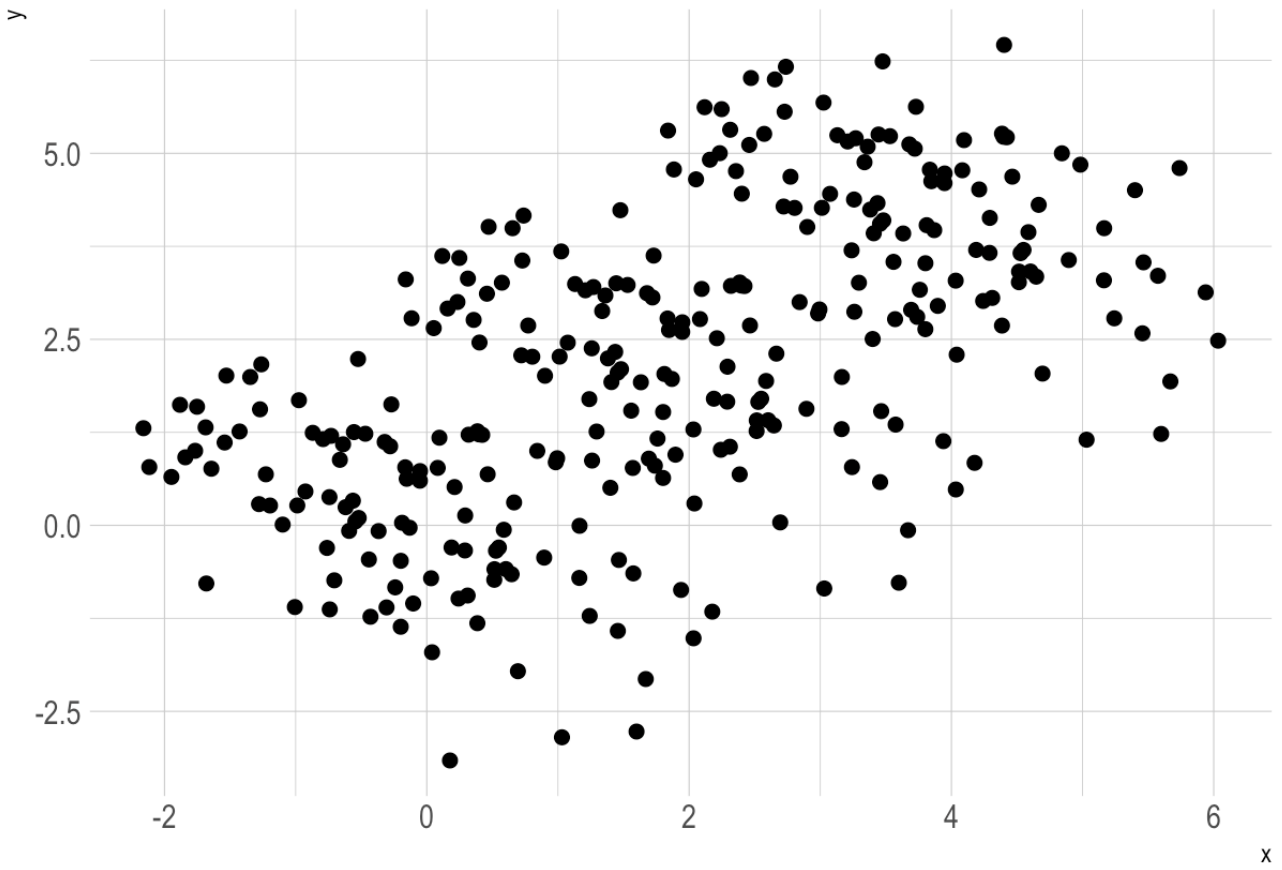


# Scatter charts



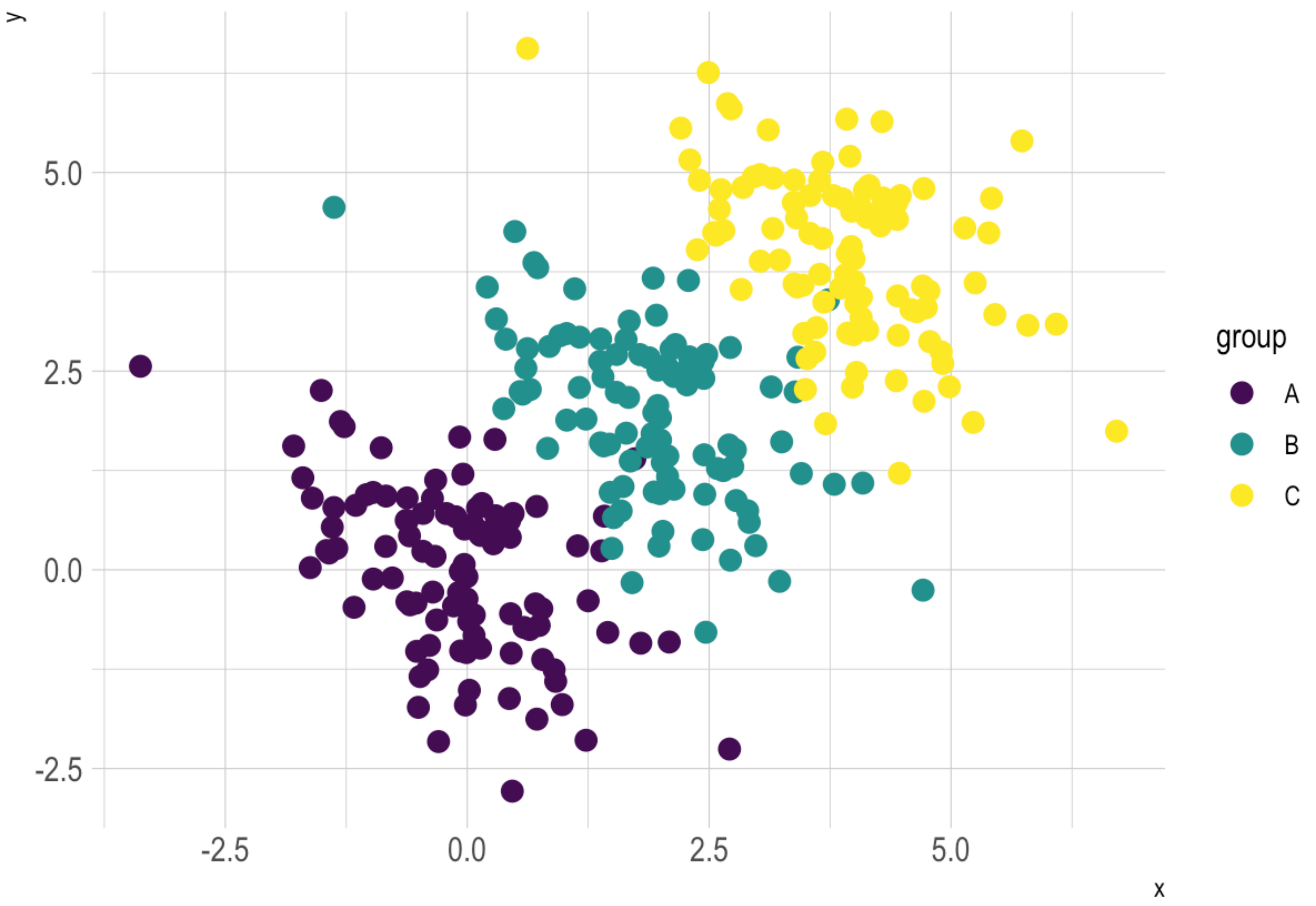
# Scatter charts

Do you see a correlation?



# Scatter charts

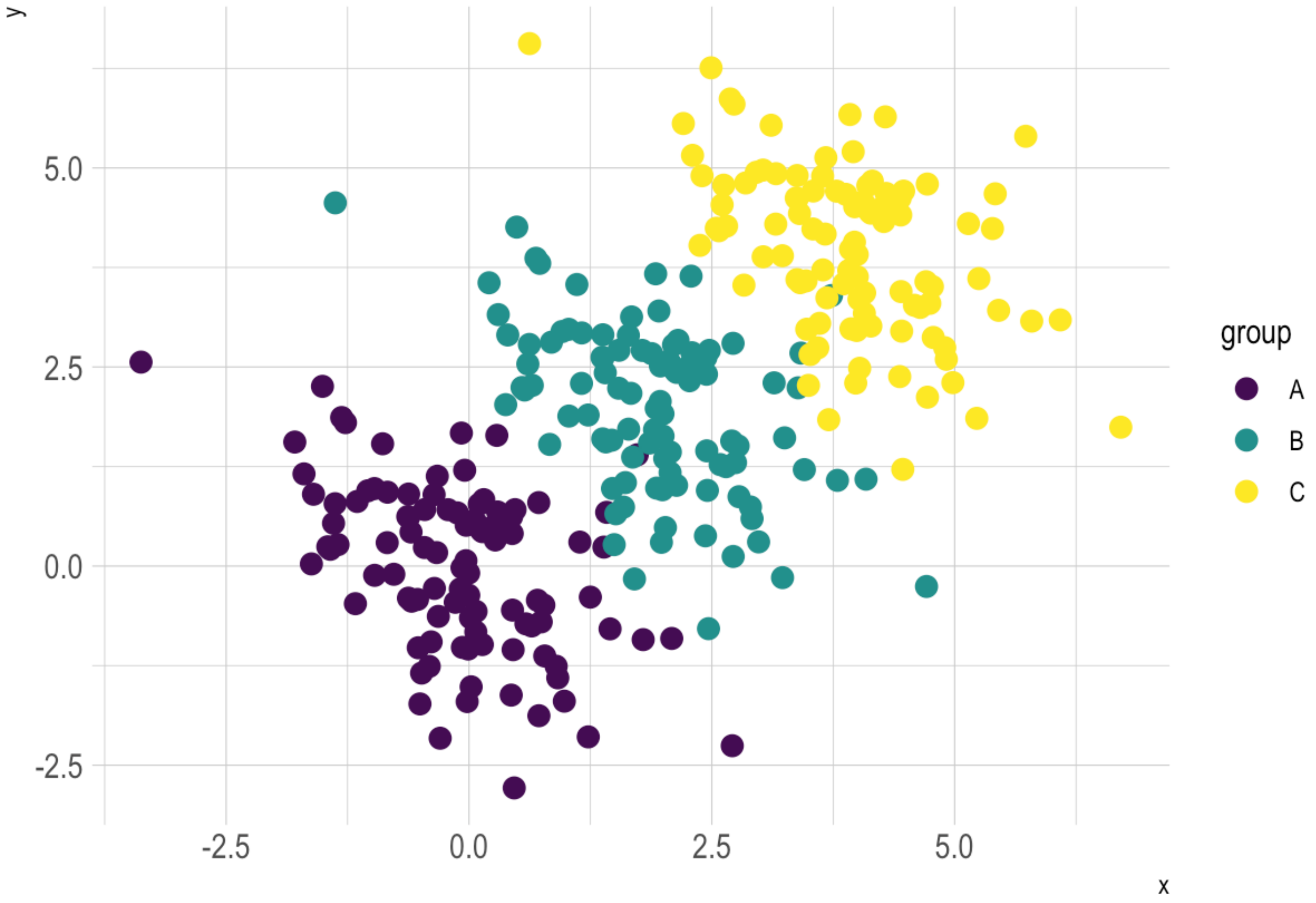
What happens if we group?



# Scatter charts

What happens if we group?

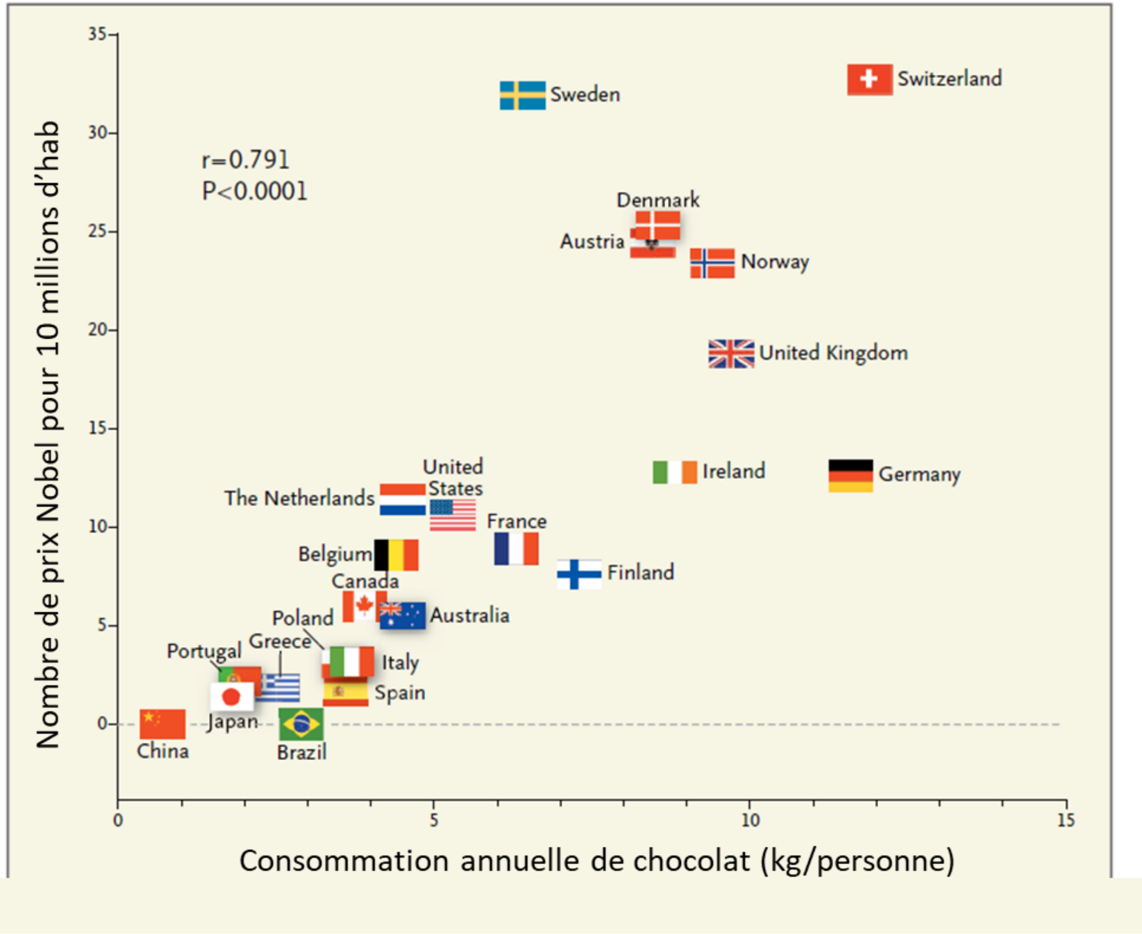
**Sympson's paradox: the trend between two different variables reverses when a third variable is included**



# Correlation != Causality

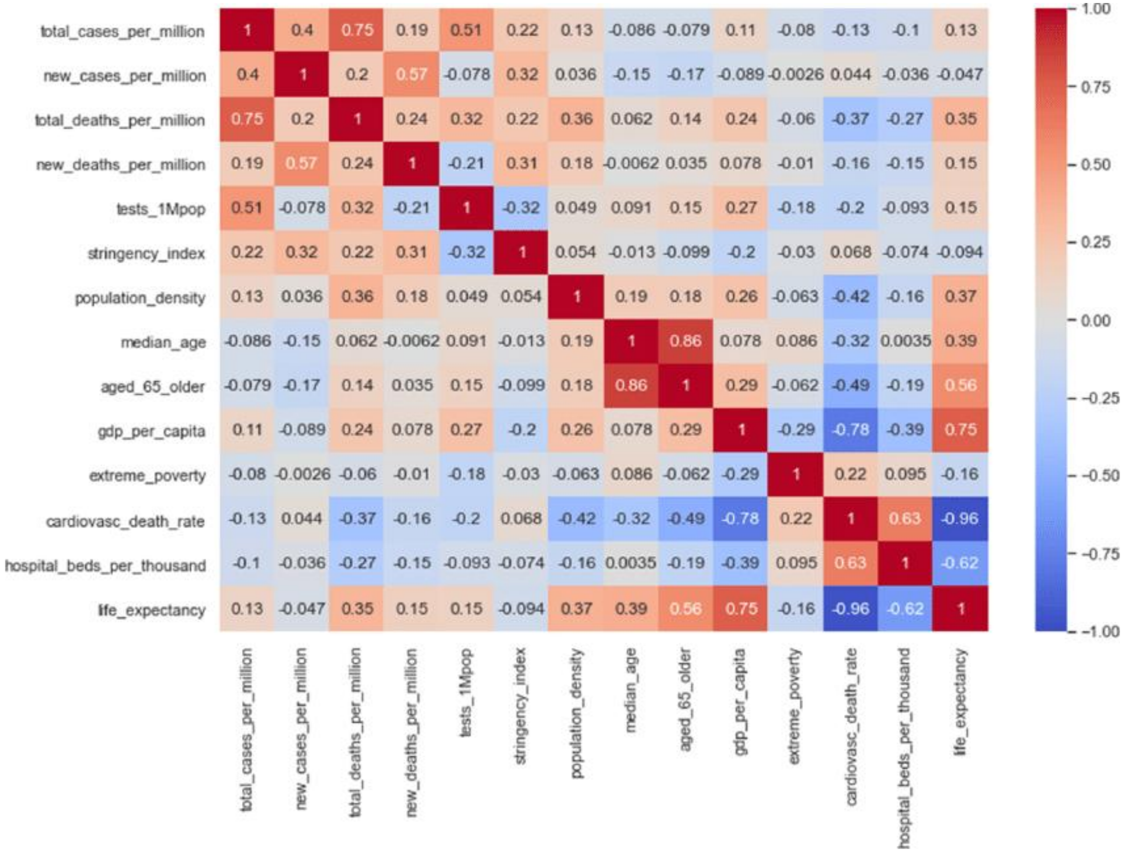
**Causality:** Relationship between two events where one event is affected by the other.

**Correlation:** Quantitative measure of joint variability between two or more variables. It is between -1 to 1.



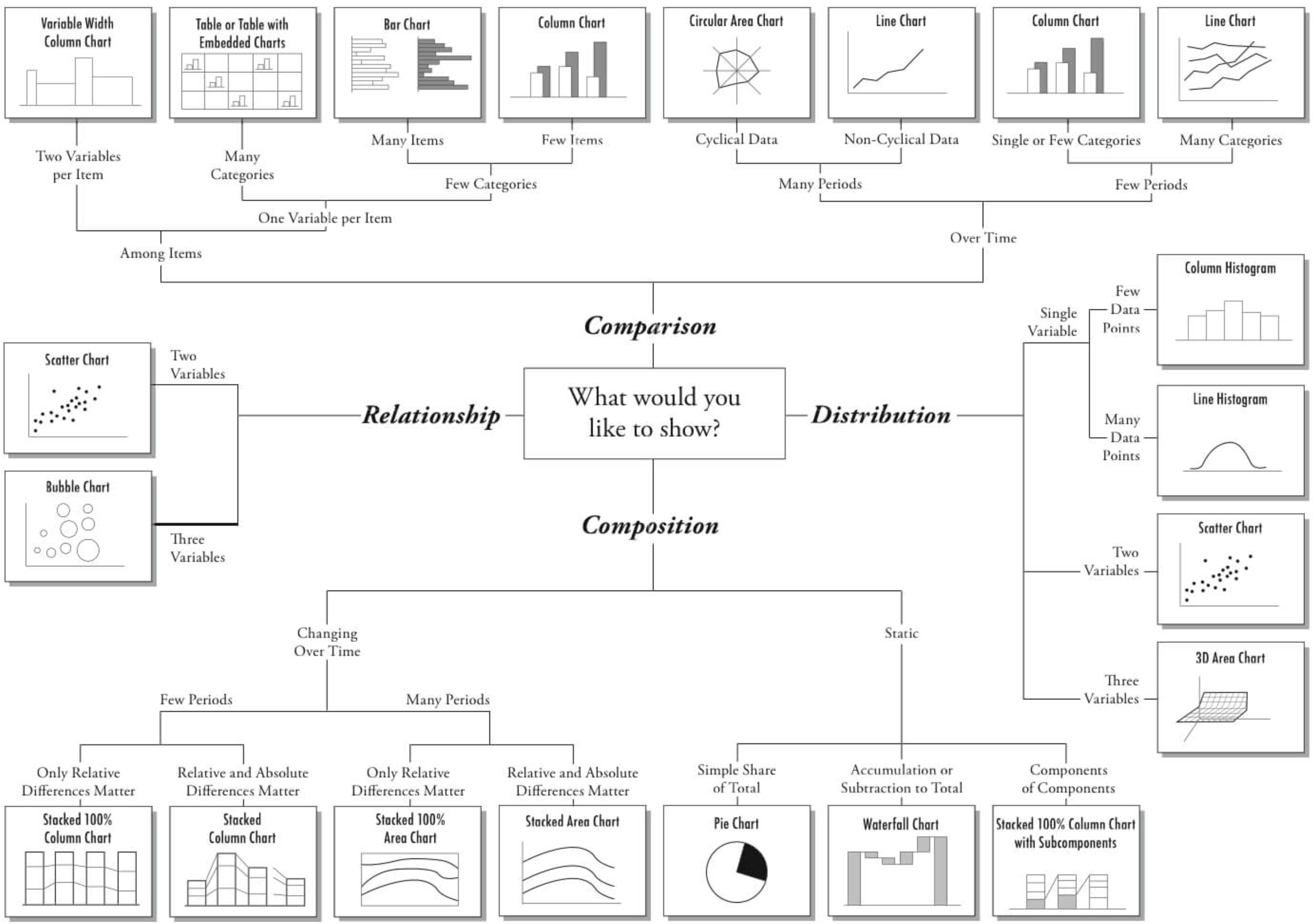
# Heatmap

**A heatmap displays data in a matrix format using color to represent values. It's ideal for spotting patterns, correlations and variations in large datasets.**



# Chart Suggestions—A Thought-Starter





# Selecting and Designing a Data Visualization

## Bar Chart

Shows: Comparison
Data: 2 variables - categorical & numerical

- Always start axis at zero.
- Turn the entire chart horizontal (like above) if categories are too long or there are too many. *Avoid the tilt & squeeze!**
- Bars should be the same color unless it is a clustered or stacked bar chart.
- Arrange by category order. No order? Arrange bars by size.
- To make a clustered or stacked bar chart: add a 3rd variable that is categorical.

## Line Chart

Shows: Trends
Data: 2 variables - chronological & numerical

- Line should show a trend across the same sample.
- Axis doesn't have to start at zero, but the line slope should be representative of the trend size.
- To make a chart with multiple lines: add a 3rd variable that is categorical.
- Lines should be solid and easily distinguishable.
- Label lines directly instead of providing a key/legend.
- Avoid more than 4 lines in one chart.

## Pie Chart

Shows: Percent to whole
Data: 2 variables - categorical & percentage

*Avoid using pie charts at all.
If a pie is absolutely needed:

- All slices must add up to 100% of a variable.
- Use for only 2-4 slices and when slices have very different sizes.
- Slices should start at the top and go clockwise largest to smallest, unless there are only two slices.
- Slices can be different colors, but should be directly labeled and percentages provided.

## Scatter Plot

Shows: Relationship
Data: 2 variables - numerical & numerical

*Great for 1st look at the data.

- All dots should be the same color, unless you add another variable that is categorical. Then you would distinguish categories by colors.
- Can also add another variable (ordinal or numerical) that is distinguished by dot size.



** the tilt = rotating any text, causing the viewer to tilt their head*
*the squeeze = trying to fit too many things in a constricted width*

## For All Vizzes

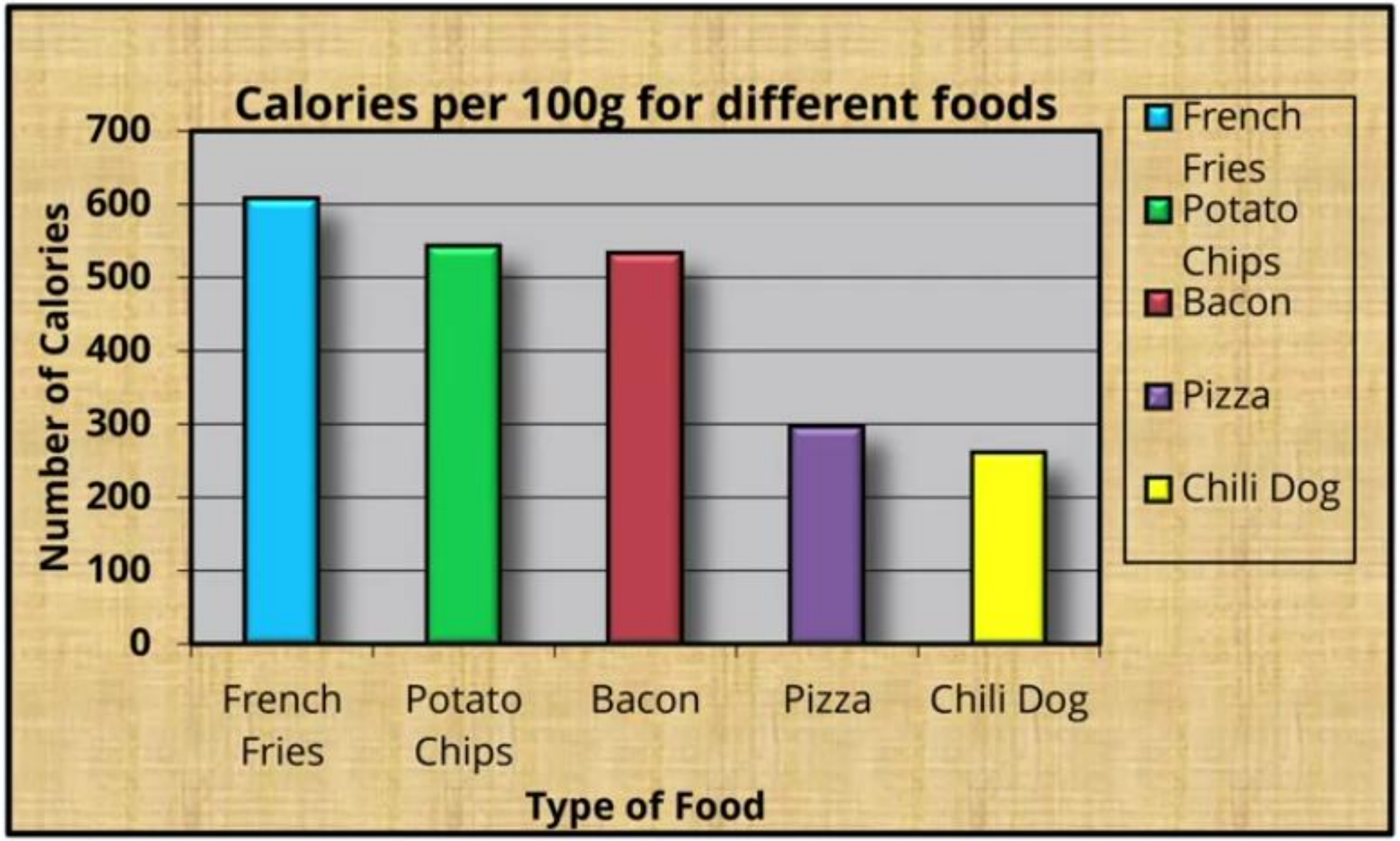
Only use a new type of visualization when there is a strong data-related purpose!

- **Be direct:** Directly label data and categories instead of using a legend.
- **Reduce clutter:** Avoid or minimize extra details, like gridlines, axes & tick marks.
- **Use Gestalt principles:** Use bright color to point something out and greys to push important, but supporting information to the back.
- **Make your point clear:** Use size, color, and borders to draw attention to a subsection of your visualization.
- **Be consistent** with sizes, fonts, and colors.
- **Align meaning:** Use colors that either align with the message and the variables being represented.
- **Be accessible:** Make sure of good contrast & large font. Avoid italics.

There are lots of other charts to choose from, such as area charts, treemaps, butterfly charts, bubble charts, and more! But, it is best to learn the 4 basic charts on the left before determining when to switch to another type.

# Data to Ink ratio

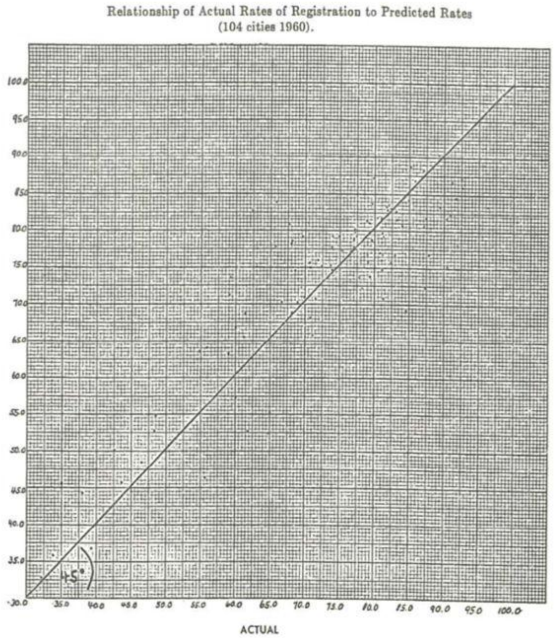
We want to compare calories in bacon with different food.

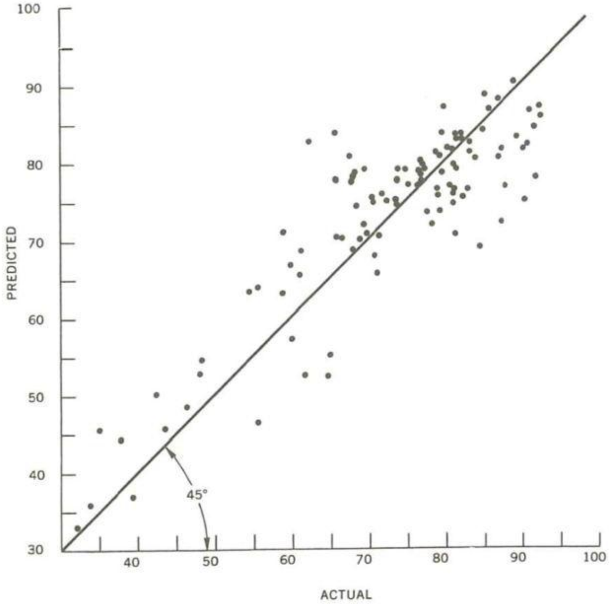


# Less is better

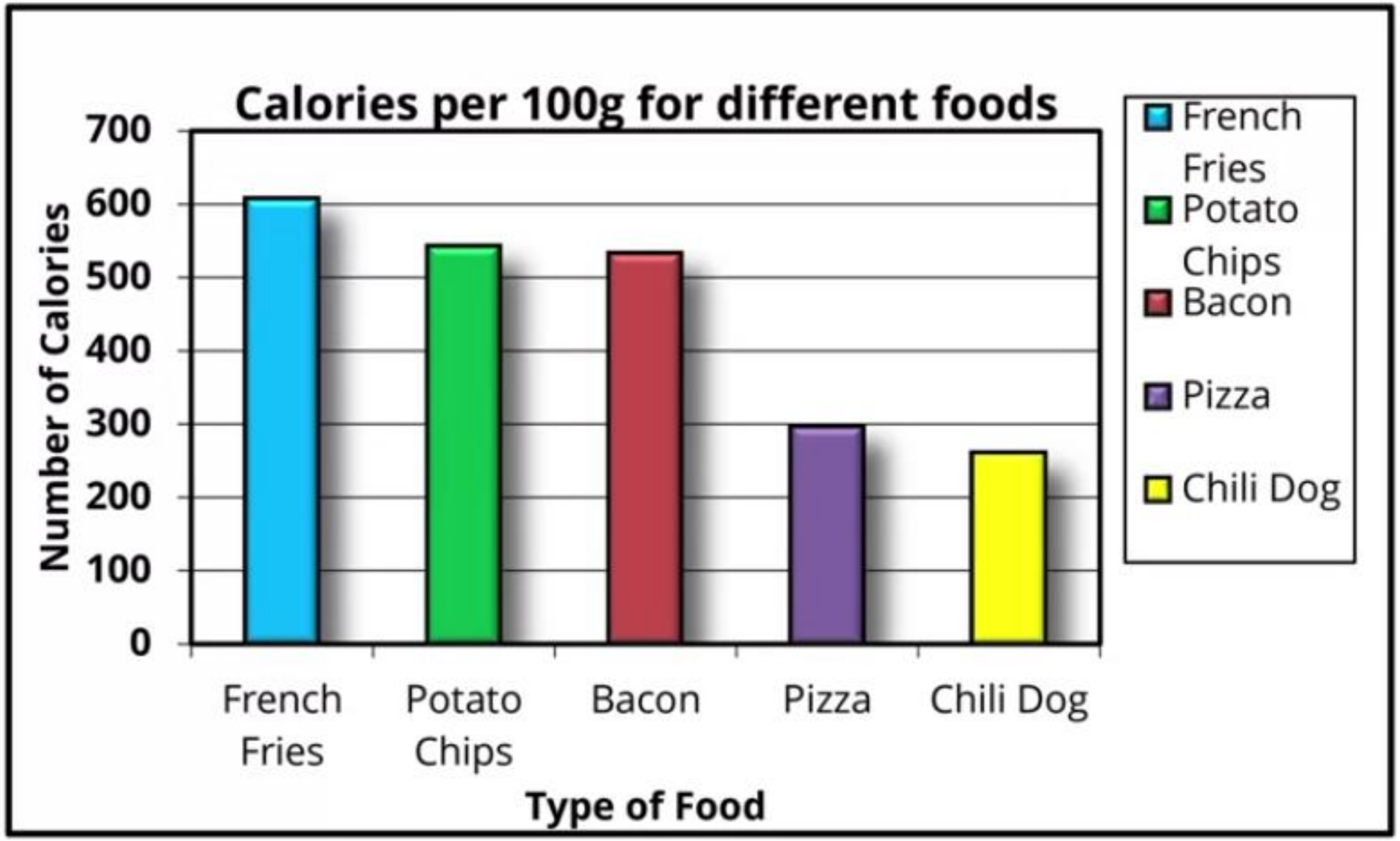
**Edward Tufte** is an author who invented the concept of **“data to ink ratio”**. Overall its concept leads us to identify the **quantity of ink used which really gives us information** in relation to the **total quantity of ink used.**

The overall idea is to **reduce redundant and non-essential information** so as not to have visual clutter when reading our graph.

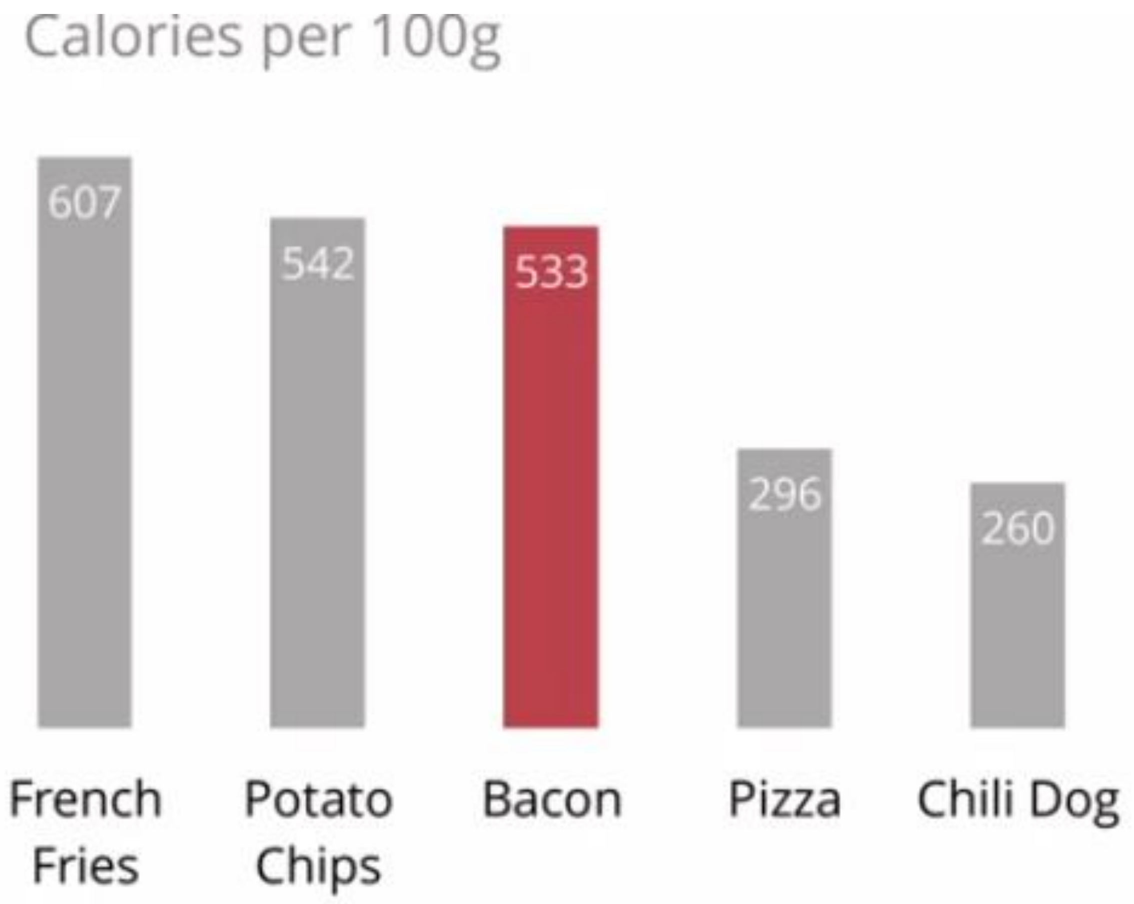


# Less is better

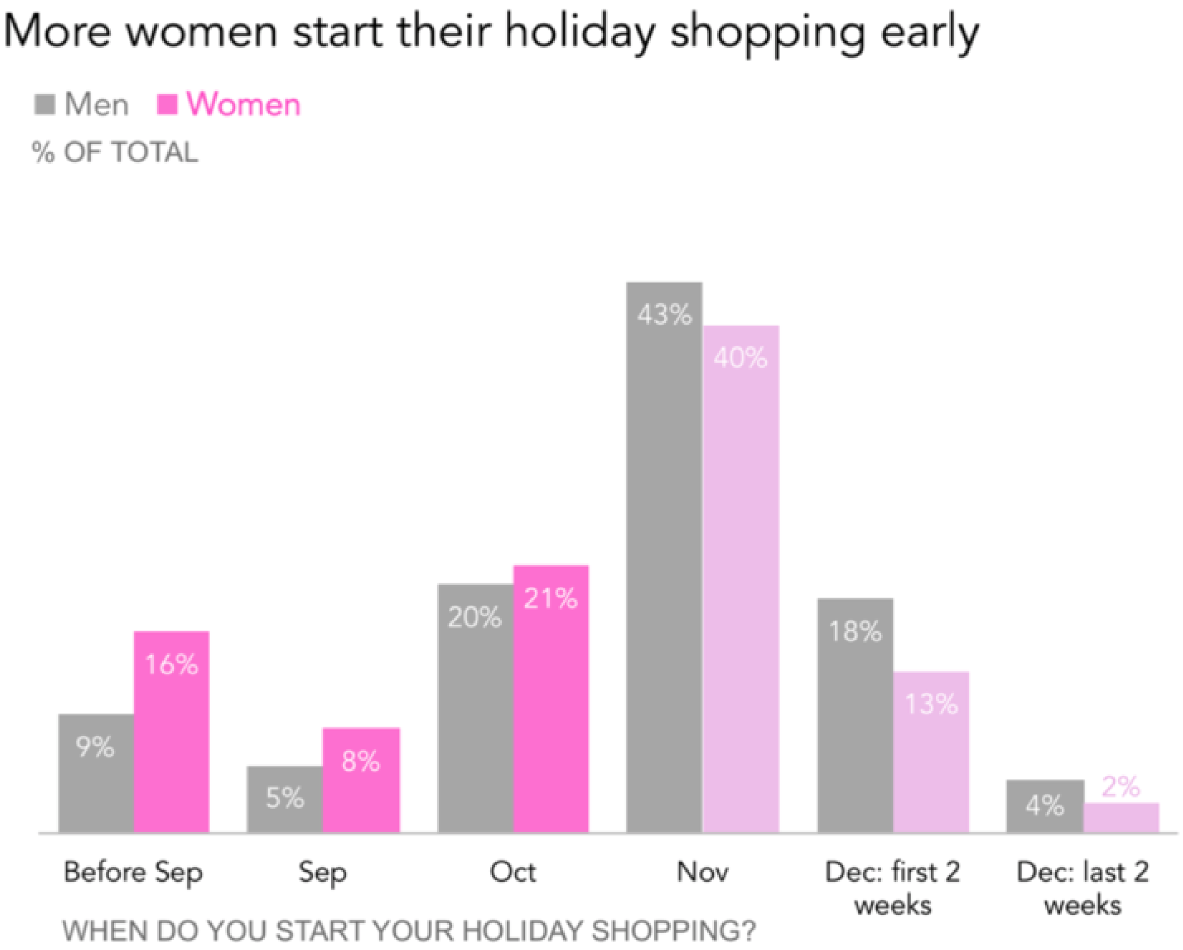


# Less is better

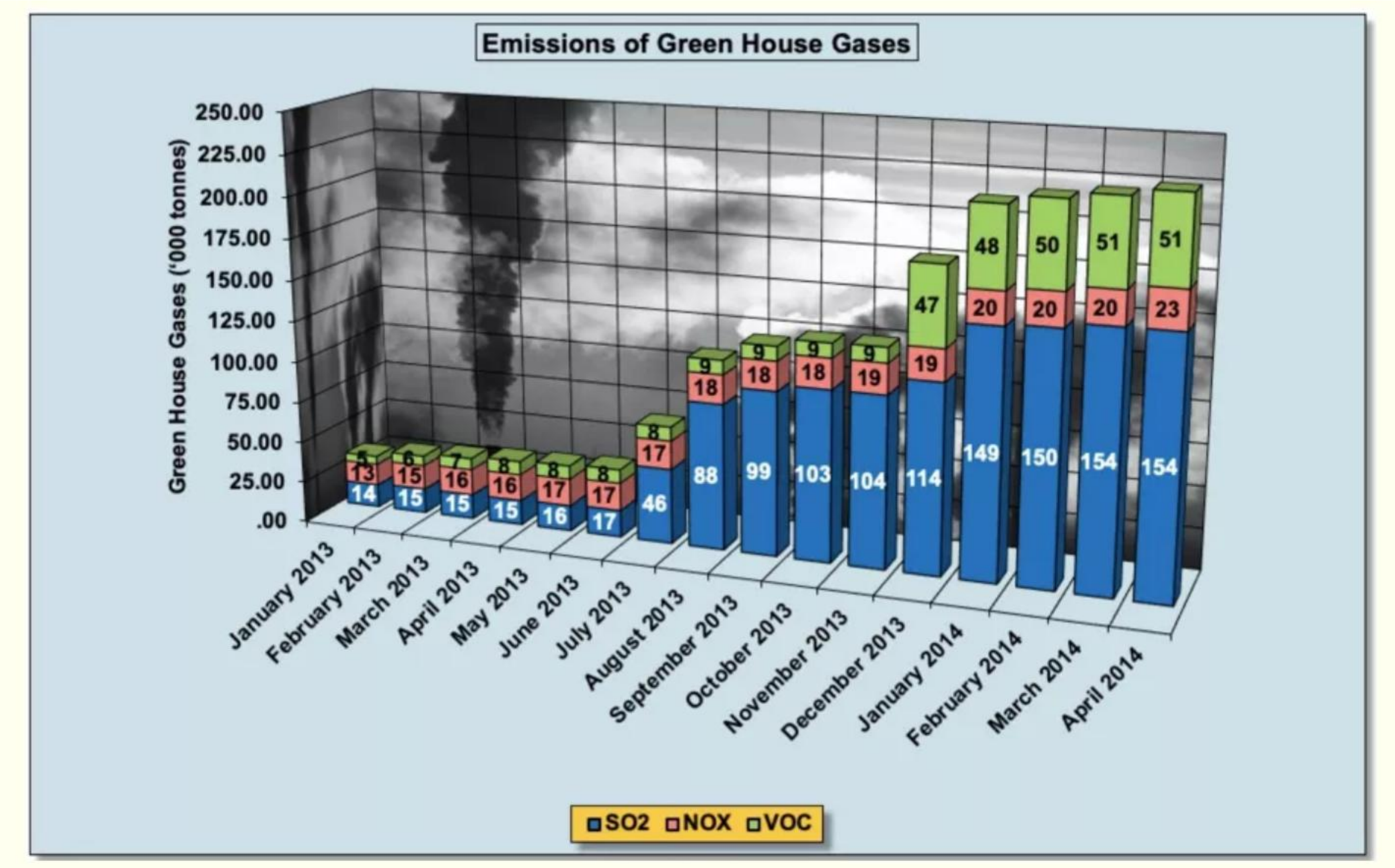


# Less is better

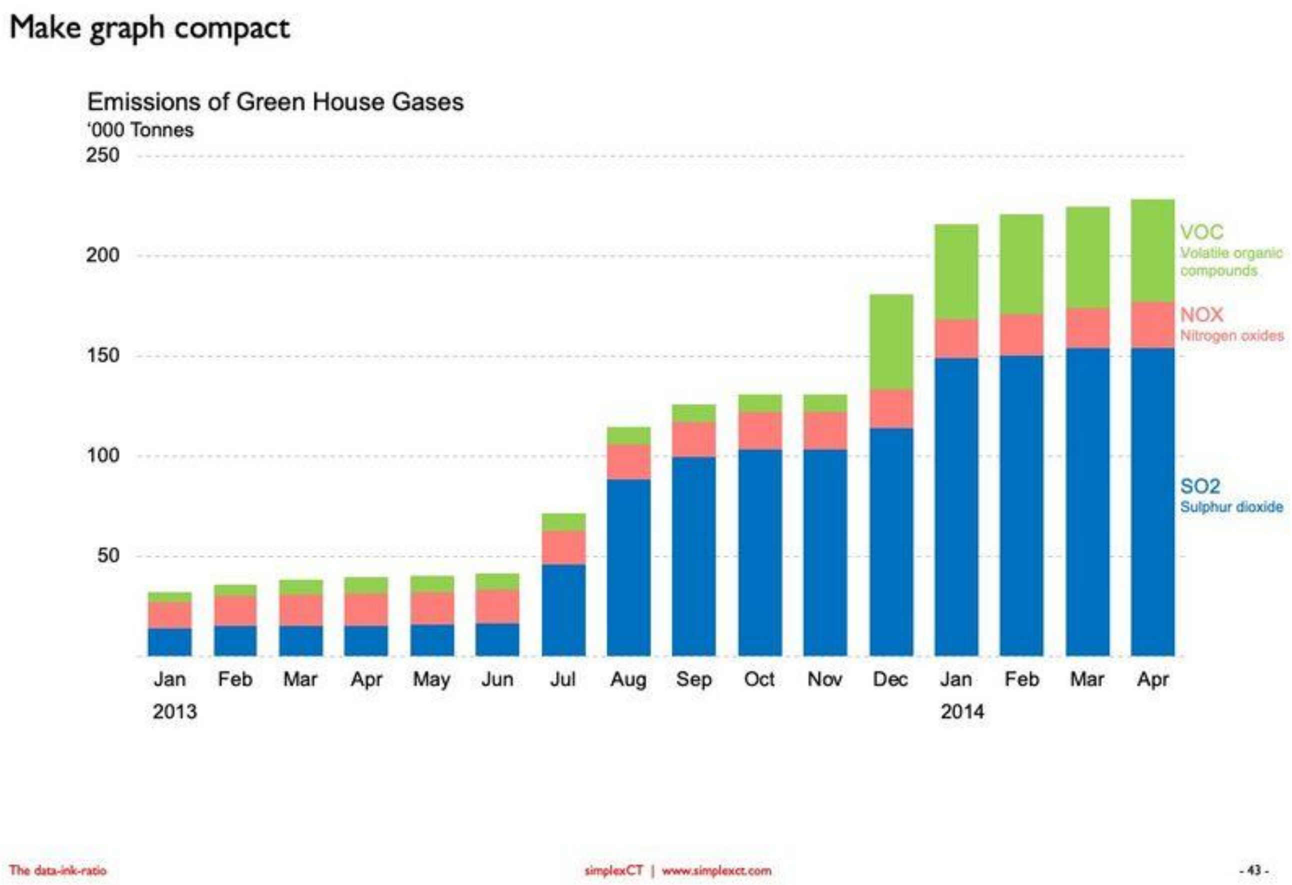
# Less is better



# Less is better



# Less is better



# Graph: text

Text in graphs should be chosen thoughtfully.

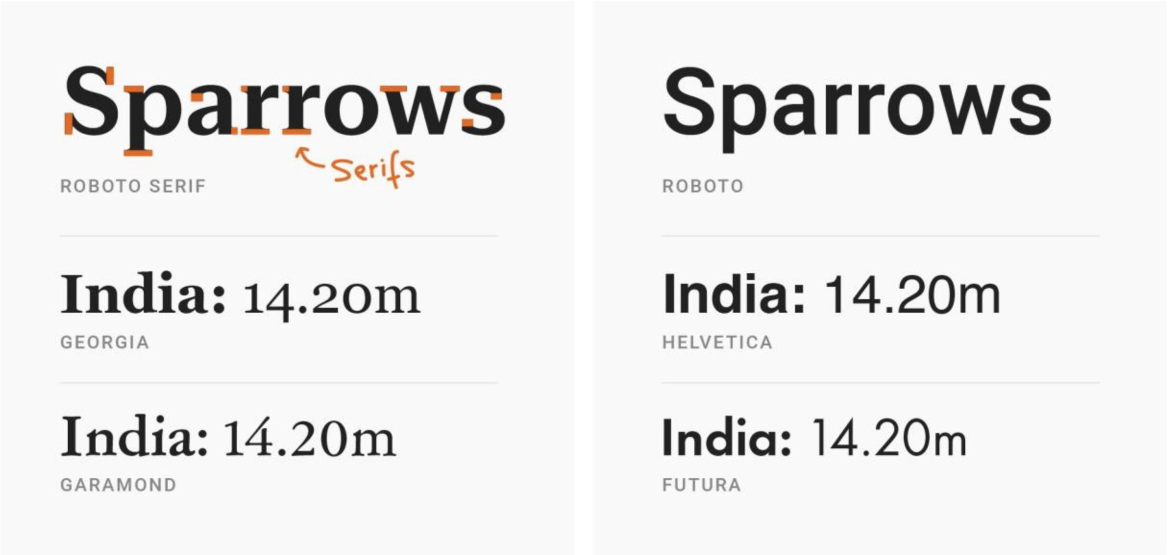
- The **title** should clearly state what the graph shows — **without interpreting or suggesting a trend**. It should reflect the data shown on the x- and y-axes in a **neutral and objective** manner.

- If you want to comment on patterns or trends, do so in a **subtitle** or in the accompanying **text or figure caption**, not in the main title.

- **legend of the axes and their scale/value is not an option**

# Graph: text

**Use sans-serif typefaces.**




SERIF FONTS SANS-SERIF FONTS

**Use a font with lining and tabular numbers.**


NOT IDEAL: OLDSTYLE BETTER: LINING

# Graph: text

**Use bold fonts only for emphasis.**


NOT IDEAL: EVERYTHING IS BOLD


BETTER: BOLD FOR EMPHASIS

# Graph: text

**Use uppercase text sparingly.**



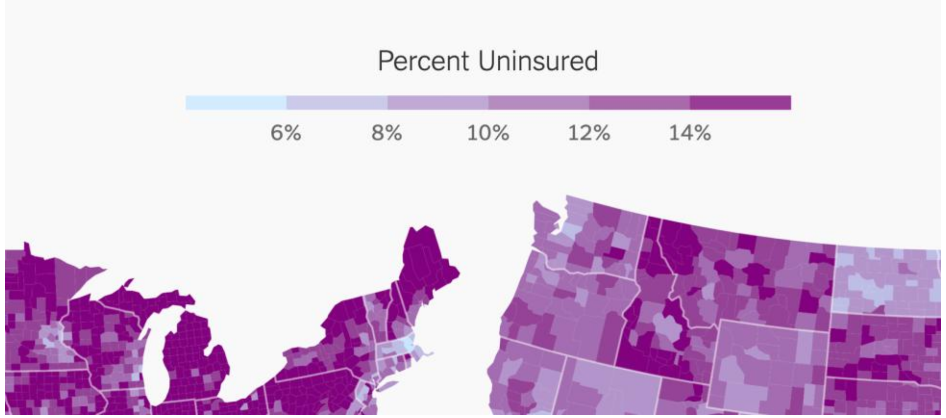
NOT IDEAL: UPPERCASE EVERYWHERE

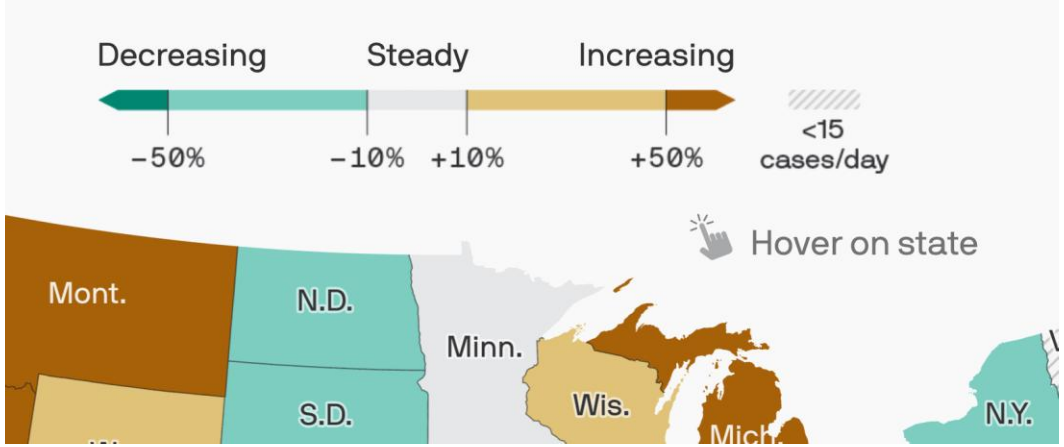
BETTER: UPPERCASE FOR FEW WORDS

# Graph: colors

**Sequential color** scales are gradients that go from bright to dark or the other way round. They're great for visualizing numbers that go from low to high, like income, temperature, or age.

**Diverging color** scales are the same as sequential color scales – but instead of just going from low to high, they have a bright middle value and then go darker to both ends of the scale in different hues.



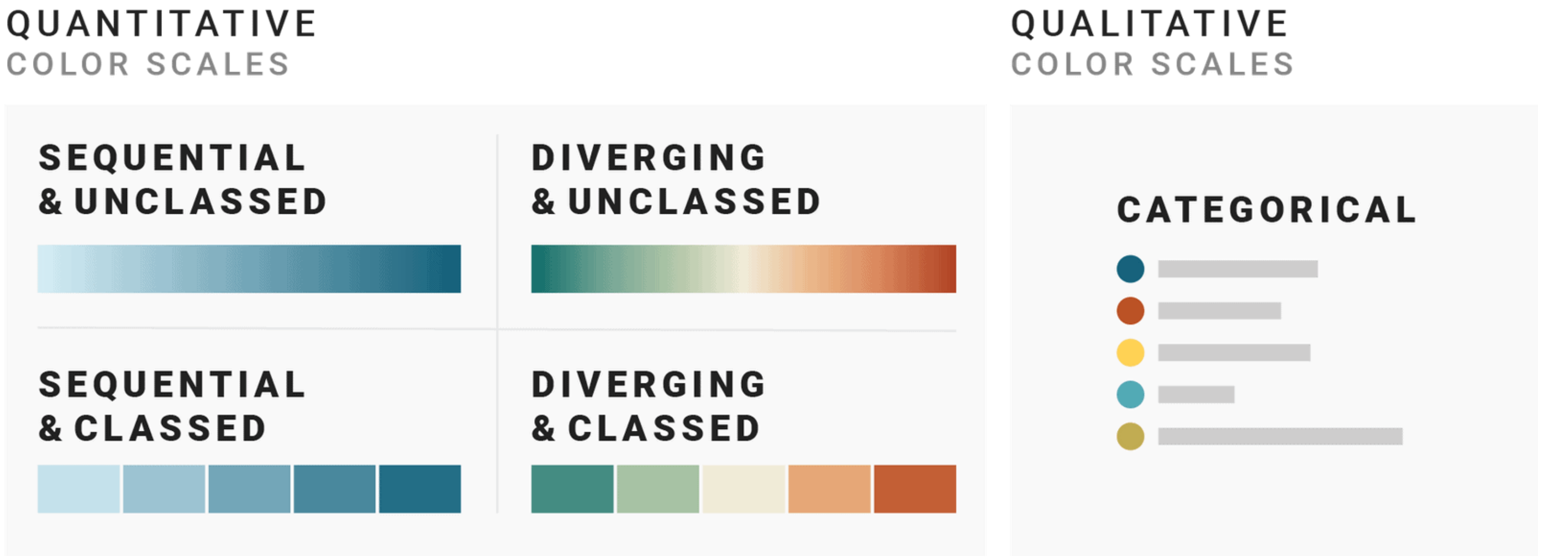




[Which color scale to use when visualizing data | Datawrapper Blog]

# Graph: colors

Color gradients can be imprecise and hard to read if not used correctly. The **type of your data** as well as **what you want to show** will help you choose the right gradient shades

- Use a diverging color scale if there’s a meaningful middle point
- Use a diverging color scale to emphasize the extremes
- Use a diverging color scale to let readers see more differences in the data
- Use a sequential color scale for a more intuitive reading

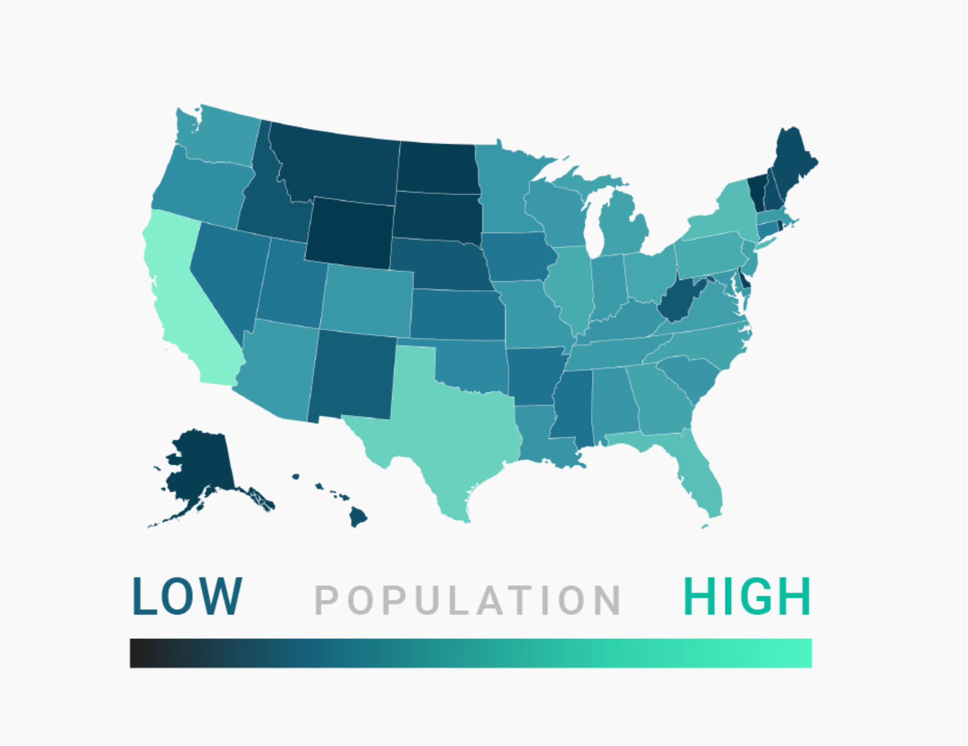



Which color scale to use when visualizing data | Datawrapper Blog

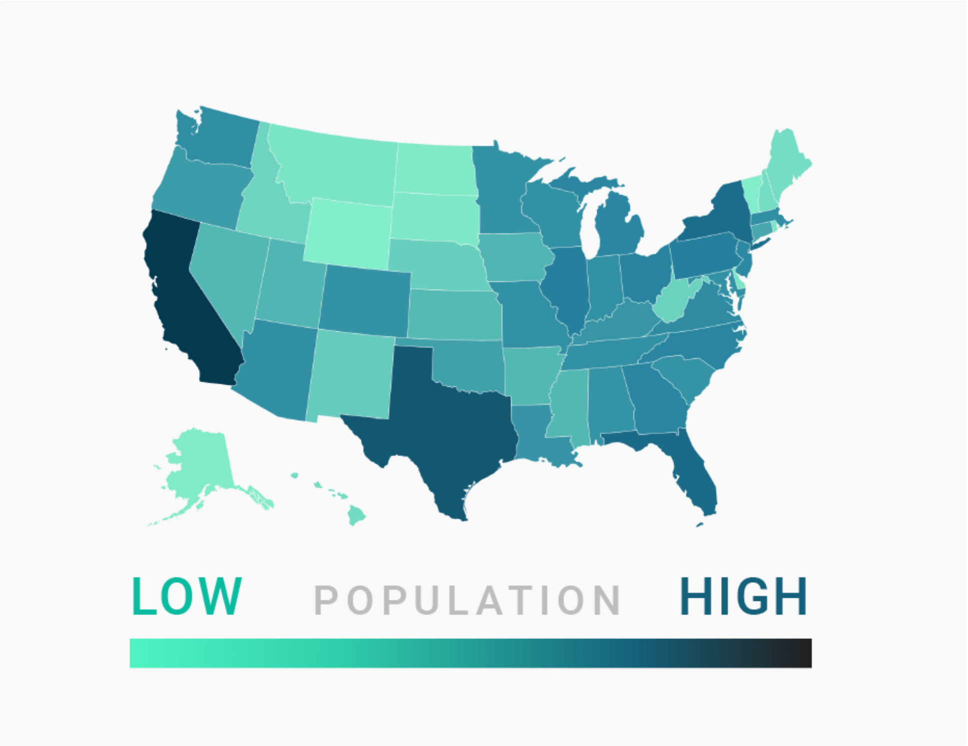
# Graph: colors

# Graph: colors

**Use light colors for low values and dark colors for high values**




NOT IDEAL




BETTER

# Graph: colors

**Avoid to use to much colours**


NOT IDEAL

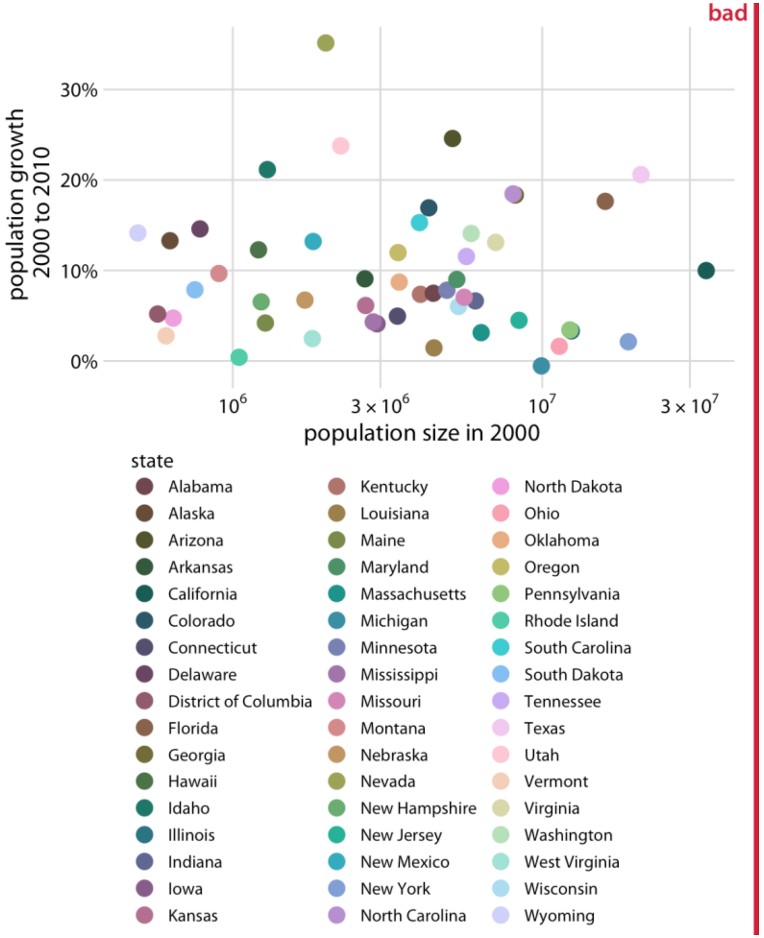

BETTER

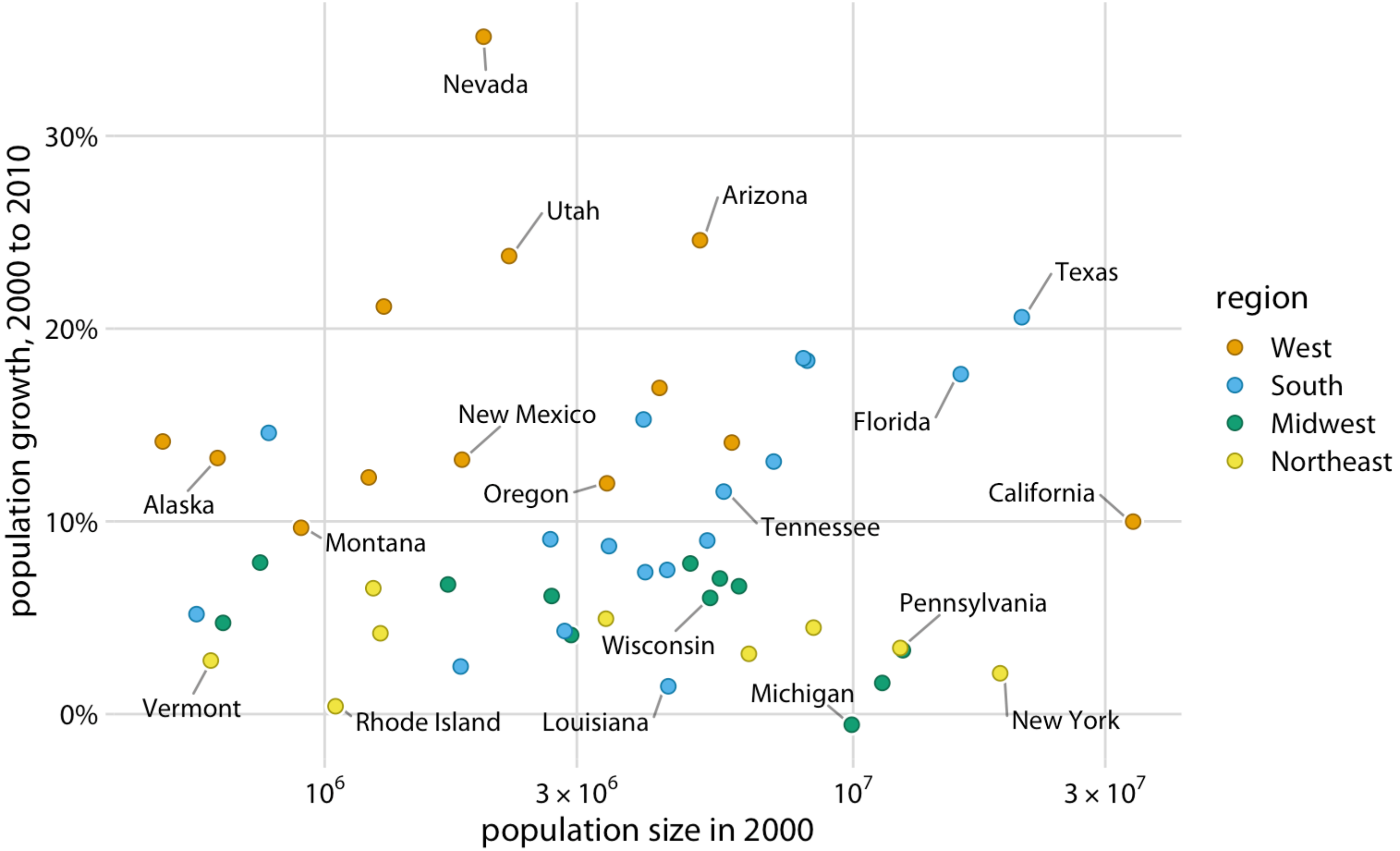
NOT IDEAL

BETTER

# Graph: colors

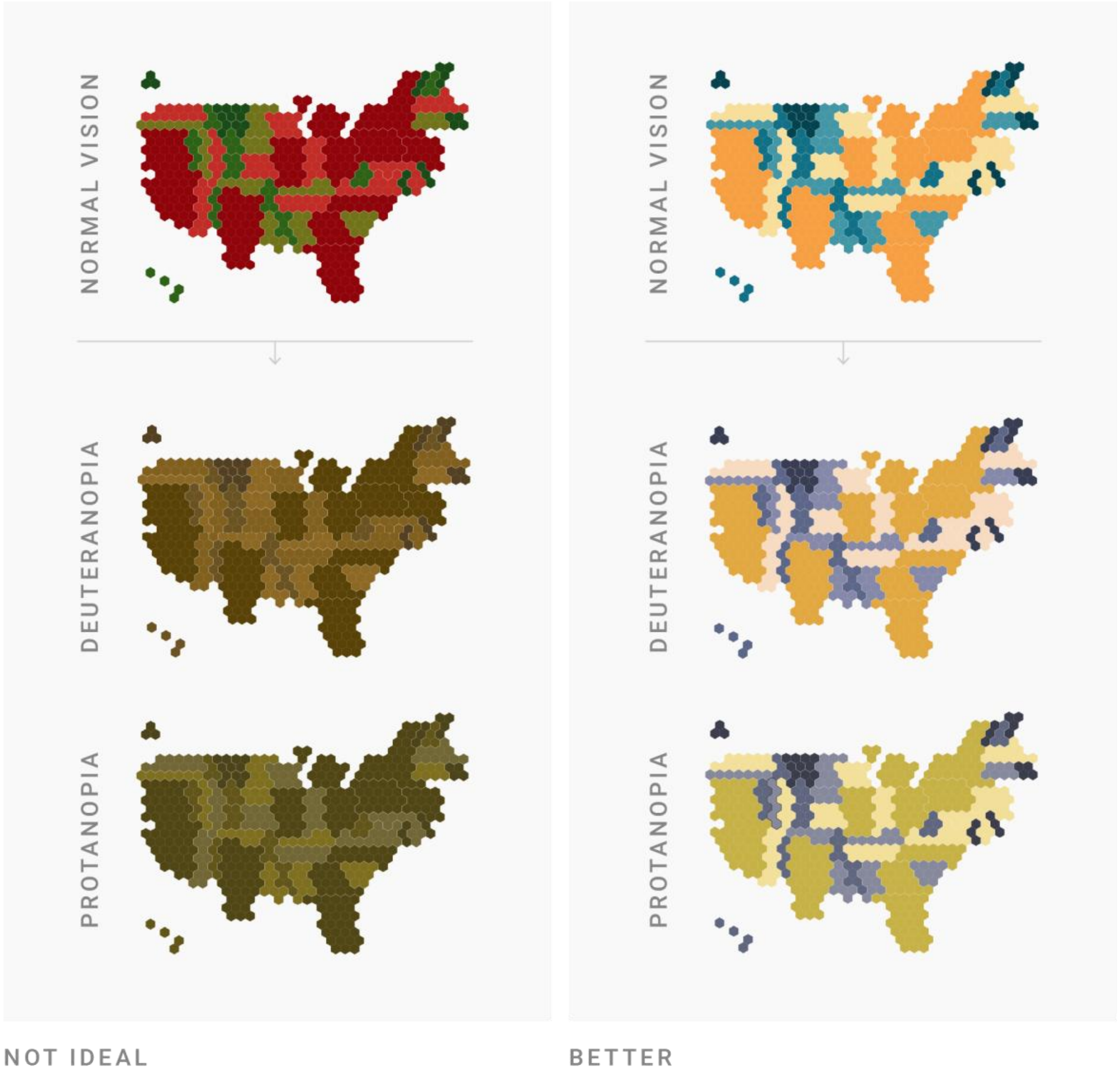
**Avoid to use to much colours**



# Graph: colors

**Consider color-blind people**



# Information hierarchy

It’s essential to consider how the audience will interpret the visuals we create. Knowing the visual hierarchy — which elements will catch attention first — allows us to guide viewers through the data in a logical and meaningful way. This helps ensure a clear message and reduces the risk of unintentional bias.

# Graph: Ethics

**Accuracy & Integrity**: Visuals must reflect the data truthfully, avoiding distortion or manipulation that could mislead the audience or push a hidden agenda.

**Clarity & Simplicity**: A good visualization communicates the message clearly, avoiding clutter or overly complex designs that confuse more than they inform.

**Objectivity & Fairness**: Visualizations should be unbiased and transparent. Clearly stating data sources, methods, and limitations supports fair and responsible interpretation.

**Privacy & Responsibility**: Always respect the confidentiality of personal or sensitive data. Follow data protection laws and ethical guidelines when sharing information.

**Accessibility & Inclusion**: Design with everyone in mind — use colorblind-friendly palettes, readable text, and alt text for screen readers. Be aware of cultural and contextual differences in interpretation.

# Python lab: Matplotlib

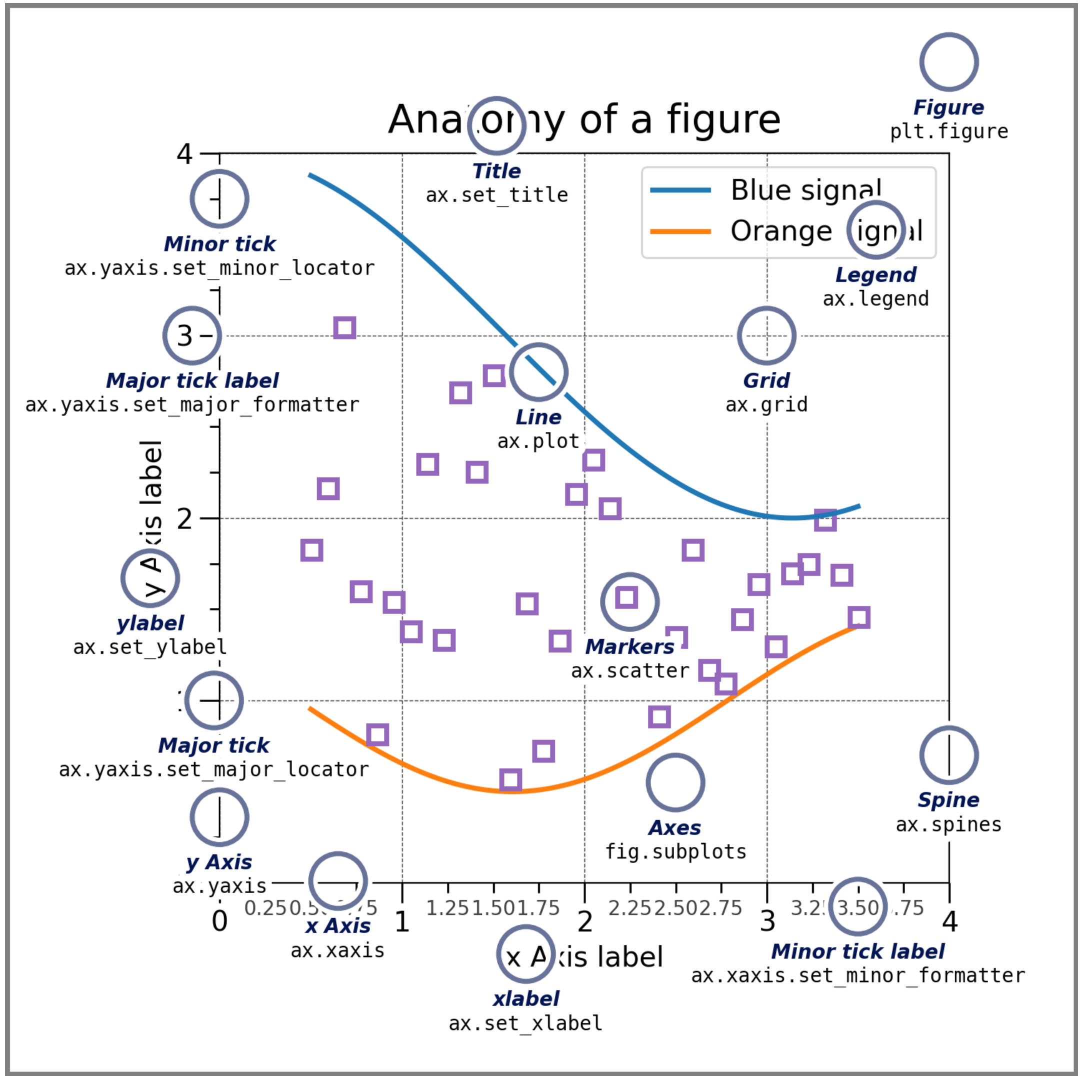
**matplotlib** primarily offers two levels of abstraction: the figure and the axes. The figure is essentially the “canvas” that contains one or more axes, where the charts are placed.

When you want to build a more complex graphs always use:

**fig, ax = plt.subplots()**

**Ex:**

```
ax.set_title("Monthly Sales", fontsize=16, color='navy')
ax.set_xlabel("Month", fontsize=14)
ax.set_ylabel("Revenue (€)", fontsize=14)
ax.legend(["Product A"], loc="upper left")
```




[Quick start guide — Matplotlib 3.10.5 documentation](#)